Test Case Register

Requirement-linked cases and recorded execution status

Execution snapshot

Test branch: feature/TEST-042-final-testing-deliverables. PDF generation source SHA: 738db9d.

Register rows: 663 | Passed: 117 | Failed: 0 | Skipped: 0 | Not Run: 546 | Not Mapped: 0.

Evidence metadata: run=40180387881; test SHA=738db9d; backend run=40180383416; backend SHA=8cb633c2c12868cb3239a8940e24cad1abd42746. Historical local JUnit results without a captured source SHA are not attributed to the current commit.

Actual output is represented by the recorded execution status and JUnit report reference. Not Run rows are not treated as passes. Detailed case source, preconditions, and expected behavior remain in the Markdown catalogs.

Case-by-case register

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-COURSE-022	Teacher portal dashboard is available to teachers	Read dashboard	TeacherPortalAuthorizationIT#shouldAllowTeacherToReadDashboardAndSearch	integration / high	Course Service and teacher credentials	Expected: 200 success response with data | Actual: Not executed in this run | Status: Not Run
TC-COURSE-023	Teacher portal search is available to teachers	Search portal records	TeacherPortalAuthorizationIT#shouldAllowTeacherToReadDashboardAndSearch	integration / high	Course Service and teacher credentials	Expected: 200 success response with data | Actual: Not executed in this run | Status: Not Run
TC-COURSE-024	Teacher portal protected reads require authentication	Unauthenticated dashboard and search	TeacherPortalAuthorizationIT#shouldRejectUnauthenticatedDashboardAndSearch	security / high	Course Service	Expected: 401 Unauthorized | Actual: Not executed in this run | Status: Not Run
TC-COURSE-025	Teacher portal reads reject student role	Student denied on teacher routes	TeacherPortalAuthorizationIT#shouldRejectStudentFromTeacherPortalReads	security / high	Course Service and student credentials	Expected: 403 Forbidden | Actual: Not executed in this run | Status: Not Run
TC-COURSE-026	Teacher portal client rejects missing required inputs	Validate portal identifiers and query	CourseClientTest#shouldRejectBlankCourseIdForLookupUpdateAndDelete	unit / medium	None	Expected: IllegalArgumentException for blank values | Actual: Passed; see JUnit report | Status: Passed
TC-COURSE-027	Students cannot write course content	Student denied course content mutations	CourseServiceAuthorizationMatrixIT#shouldDenyStudentAllCourseContentMutations	security / high	Course Service and student credentials	Expected: 403 Forbidden for all tested mutation routes | Actual: Not executed in this run | Status: Not Run
TC-COURSE-028	Grade administration is restricted to admins	Teacher denied grade and monthly fee administration	CourseServiceAuthorizationMatrixIT#shouldRestrictGradeAdministrationToAdministrators	security / high	Course Service and teacher credentials	Expected: 403 Forbidden for all grade administration routes | Actual: Not executed in this run | Status: Not Run
TC-COURSE-029	Course content and grade routes require authentication	Unauthenticated protected-route checks	CourseServiceAuthorizationMatrixIT#shouldRequireAuthenticationForCourseContentAndGradeAdministration	security / high	Course Service	Expected: 401 Unauthorized | Actual: Not executed in this run | Status: Not Run
TC-COURSE-030	Announcement and progress routes follow student-role rules	Student success and teacher denial	CourseServiceAuthorizationMatrixIT#shouldKeepStudentProgressAndAnnouncementsStudentOnly	security / high	Course Service and student/teacher credentials	Expected: Student succeeds and teacher is denied | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-015-09	Completed attendance sessions reject additional marks	Mark after completing a session	AttendanceSessionWorkflowIT#teacherCanCreateListMarkAndCompleteSession	integration / high	Attendance Service plus enrolled-student fixture	Expected: 409 Conflict and no state change | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-10	Cancelled attendance sessions reject completion and marks	Attempt transitions after cancellation	AttendanceSessionWorkflowIT#teacherCanCancelOpenSessionAndCancelledSessionCannotBeCompleted	integration / high	Attendance Service plus enrolled-student fixture	Expected: 409 Conflict | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-11	Incomplete attendance sessions cannot be completed	Complete while records remain NOT_MARKED	AttendanceSessionWorkflowIT#teacherCanCreateListMarkAndCompleteSession	integration / high	Attendance Service plus course containing enrolled students	Expected: 409 Conflict and session remains OPEN | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-12	Attendance batch rejects unknown or duplicate student records	Mark out-of-session student and duplicate batch	AttendanceSessionWorkflowIT#teacherCanCreateListMarkAndCompleteSession	integration / high	Attendance Service plus enrolled-student fixture	Expected: 404 for unknown student and 400 for duplicate student batch | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-13	Attendance batch rejects NOT_MARKED as a submitted status	Attempt to submit the initial-only NOT_MARKED value	AttendanceSessionWorkflowIT#teacherCanCreateListMarkAndCompleteSession	integration / high	Attendance Service plus enrolled-student fixture	Expected: 400 Bad Request | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-14	Student summary counts and percentage are internally consistent	Validate student summary aggregates	AttendanceSummaryIT#studentCanReadOwnCourseSummary	integration / high	Attendance Service and course fixture	Expected: Counts match and percentage is correctly rounded | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-15	Teacher student-summary counts and percentage are internally consistent	Validate teacher summary aggregates	AttendanceSummaryIT#teacherCanReadStudentCourseSummary	integration / high	Attendance Service plus teacher student and course fixtures	Expected: Requested student/course and aggregate arithmetic match | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-16	Teacher cannot list sessions for another teacher course	Reject cross-teacher course access	AttendanceAuthenticationIT#teacherCannotListSessionsForAnotherTeachersCourse	security / high	Attendance Service; TEST_ATTENDANCE_COURSE owned by primary teacher and distinct secondary teacher credentials	Expected: 403 Forbidden | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-19	Assignment partial update preserves omitted fields	PATCH assignment title while draft	AssignmentWorkflowIT#teacherCanCreateReadListUpdatePublishCloseAndDeleteAssignment	integration / high	Assignment Service with write fixtures enabled	Expected: Changed field persists and omitted fields are unchanged | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-20	Student reads only eligible assignments in an enrolled course	Read paged student assignment list filtered by course	StudentAssignmentIT#studentCanReadFilteredPagedAssignmentsForAnEnrolledCourse	integration / high	Assignment and Enrollment Services plus matching student/course fixtures	Expected: 200 with valid page metadata and no DRAFT entries | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-21	Student assignment query rejects invalid page/date ranges	Invalid page size and reversed due-date range	StudentAssignmentIT#studentAssignmentFiltersAndPagingRejectInvalidRanges	validation / medium	Assignment Service and student credentials	Expected: 400 Bad Request | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-011-22	Empty assignment PATCH is rejected	PATCH with no fields	AssignmentAuthenticationIT#rejectsInvalidAssignmentPayloadBeforeCreation	validation / medium	Assignment Service and teacher credentials	Expected: 400 Bad Request | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-23	Assignment attempt penalty and rubric DTO boundaries	Reject invalid attempts penalty and rubric shapes	AssignmentAuthenticationIT#rejectsAssignmentAttemptPenaltyAndRubricValidationBoundaries	validation / high	Assignment Service and teacher credentials	Expected: 400 for attempts below 1 penalty above 100 blank rubric criterion and more than 50 rubric items | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-24	Assignment rubric persistence and partial update	Preserve rubric on update and PATCH	AssignmentWorkflowIT#teacherCanCreateReadListUpdatePublishCloseAndDeleteAssignment	integration / high	TEST_ASSIGNMENT_WRITE_ENABLED plus course/module fixtures	Expected: Create returns ordered rubric criteria and marks; update and PATCH without rubric retain it | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-25	Assignment grade DTO marks boundaries	Reject missing negative and over-maximum marks	AssignmentAuthenticationIT#rejectsMissingNegativeAndOverMaximumGradeMarksBeforeLookup	validation / high	Assignment Service and teacher credentials	Expected: 400 for missing marks and values below 0 or above 100 before submission lookup | Actual: Not executed in this run | Status: Not Run
TC-TEST-012-13	Assignment multipart upload workflow and cleanup	Upload assignment attachment through Material Service	AssignmentAttachmentWorkflowIT#teacherCanUploadAndCleanUpSyntheticAssignmentAttachment	integration / high	TEST_ATTACHMENT_WRITE_ENABLED plus assignment/course fixtures and Material Service	Expected: 201 upload; list and download metadata verified; attachment and test-created Material are deleted | Actual: Not executed in this run | Status: Not Run
TC-TEST-012-14	Submission multipart upload workflow and cleanup	Upload submission file through Material Service	SubmissionFileWorkflowIT#studentCanUploadAndCleanUpSyntheticSubmissionFile	integration / high	TEST_ATTACHMENT_WRITE_ENABLED plus owned draft submission/course fixtures and Material Service	Expected: 201 upload; list and download metadata verified; student deletes file and admin removes test-created Material | Actual: Not executed in this run | Status: Not Run
TC-TEST-012-15	Multipart attachment routes enforce role access	Reject unauthorized multipart uploads	AssignmentAttachmentAuthenticationIT#multipartUploadRoutesEnforceAuthenticationAndRoles	security / high	Assignment and Material Services	Expected: Anonymous requests return 401; disallowed student/teacher roles return 403 | Actual: Not executed in this run | Status: Not Run
TC-TEST-012-16	Submission-file cross-student ownership	Prevent another student reading or deleting submission files	SubmissionFileWorkflowIT#anotherStudentCannotReadOrDeleteSubmissionFiles	integration / high	TEST_ATTACHMENT_WRITE_ENABLED plus owned draft submission fixture and distinct secondary student credentials	Expected: Other student receives 403 for list download and delete; owner cleans up synthetic file and Material | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-10	Student quiz projection hides correct-answer fields	Read published quiz as student and inspect option fields	QuizAttemptWorkflowIT#enrolledStudentCanAccessPublishedQuizAndCompleteAttempt	integration / high	Assignment Service and enrolled student fixture	Expected: Student option response omits correctness flags | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-014-11	Quiz attempt starts in progress and records attempt number	Start first attempt	QuizAttemptWorkflowIT#enrolledStudentCanAccessPublishedQuizAndCompleteAttempt	integration / high	Assignment Service and enrolled student fixture	Expected: 201 with IN_PROGRESS and attempt number one | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-13	Quiz submission calculates score percentage and pass state	Answer correctly and submit attempt	QuizAttemptWorkflowIT#enrolledStudentCanAccessPublishedQuizAndCompleteAttempt	integration / high	Assignment Service and enrolled student fixture	Expected: Score and percentage match answer key and pass threshold | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-14	Quiz attempt limit is enforced	Start beyond configured maximum attempts	QuizAttemptWorkflowIT#enrolledStudentCanAccessPublishedQuizAndCompleteAttempt	integration / high	Assignment Service and enrolled student fixture	Expected: 409 Conflict | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-31	Quiz prevents concurrent active attempts	Start another attempt before submitting the first	QuizAttemptWorkflowIT#enrolledStudentCanAccessPublishedQuizAndCompleteAttempt	integration / high	Assignment Service and enrolled student fixture	Expected: 409 Conflict | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-32	Submitted quiz attempts cannot be resubmitted or changed	Submit twice and try to change a saved answer	QuizAttemptWorkflowIT#enrolledStudentCanAccessPublishedQuizAndCompleteAttempt	integration / high	Assignment Service and enrolled student fixture	Expected: 409 Conflict | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-33	Unanswered quiz attempt is graded as zero	Submit a second attempt without answers	QuizAttemptWorkflowIT#enrolledStudentCanAccessPublishedQuizAndCompleteAttempt	integration / medium	Assignment Service and enrolled student fixture	Expected: Zero marks and failed pass state | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-34	Teacher quiz results return each student's latest submitted attempt	Read quiz results after repeated student attempts	QuizAttemptWorkflowIT#enrolledStudentCanAccessPublishedQuizAndCompleteAttempt	integration / high	TEST_QUIZ_WRITE_ENABLED plus Assignment Course Enrollment and User Services	Expected: Results contain one latest record per student with score percentage pass state name and submittedAt | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-35	Teacher course gradebook lists closed quizzes and results	Read closed quiz gradebook results	QuizManagementIT#teacherCanReadClosedQuizResultsInGradebook	integration / high	TEST_QUIZ_GRADEBOOK_CC _ID and TEST_QUIZ_CLOSED_ID with assigned teacher credentials	Expected: Closed quiz appears with title and result list | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-36	Unauthenticated quiz endpoints reject access	Probe protected quiz routes without credentials	QuizAuthenticationIT#protectedQuizRequestsRejectMissingAuthentication	security / high	Assignment Service and API Gateway	Expected: Protected quiz endpoints return 401 | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-38	Students cannot use teacher quiz management routes	Probe teacher management routes with student credentials	QuizAuthenticationIT#studentCannotUseTeacherQuizManagementEndpoints	security / high	Assignment Service and student credentials	Expected: Teacher management routes return 403 | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-39	Teacher gradebook quiz route validates course identifier	Probe missing and malformed gradebook course IDs	QuizAuthenticationIT#teacherGradebookQuizRouteValidatesRequiredCourseId	validation / medium	Assignment Service and teacher credentials	Expected: Missing and malformed courseId return 400 | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-014-40	Quiz management is restricted to its owning teacher	Reject another teacher reading updating or deleting quiz	QuizManagementIT#anotherTeacherCannotReadOrManag QuizOwnedByTeacher	security / high	Assignment Service and Course Service; TEST_QUIZ_ WRITE_ENABLED=true; TEST_QUIZ_COURSE_ID; distinct secondary teacher credentials	Expected: 403 for read update and delete; owner deletes synthetic quiz | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-37	AI quiz generation validates inputs before provider use	Reject invalid generation settings	QuizValidationIT#aiQuizGenerationRejectsInvalidSe ttingsBeforeCallingProvider	validation / medium	Assignment Service and synthetic multipart file	Expected: Zero question count and unsupported question type return 400 without AI provider invocation | Actual: Not executed in this run | Status: Not Run
TC-TEST-019-02	Notification pages expose valid bounded page metadata	Read defaults unread page and later page	NotificationWorkflowIT#userCanRetrievePaginatedAn dUnreadNotifications	integration / high	Notification Service and synthetic user credentials	Expected: Response metadata and item bounds match request | Actual: Not executed in this run | Status: Not Run
TC-TEST-019-05	Notification endpoint rejects out-of-range pagination	Send invalid page and size directly to HTTP endpoint	NotificationValidationIT#httpEndpointRejectsInval idPaginationRanges	validation / high	Notification Service and authenticated user	Expected: 400 Bad Request for invalid page/size | Actual: Not executed in this run | Status: Not Run
TC-TEST-019-06	Unread notification results contain only unread records	Verify unreadOnly records have null readAt	NotificationWorkflowIT#userCanRetrievePaginatedAn dUnreadNotifications	integration / high	Notification Service and synthetic user credentials	Expected: Every returned item is unread | Actual: Not executed in this run | Status: Not Run
TC-TEST-019-08	Notification mark-read is idempotent	Mark same notification read twice	NotificationWorkflowIT#ownerCanMarkReadMarkAllRea dAndDeleteApprovedFixture	integration / medium	One-time disposable notification owned by configured student	Expected: 200 and stable non-null readAt | Actual: Not executed in this run | Status: Not Run
TC-TEST-019-12	Notification mutations are scoped to the recipient	Teacher attempts to mutate student-owned notification	NotificationOwnershipIT#anotherUserCannotMarkOrDe leteAStudentOwnedNotification	security / high	Notification Service and student-owned notification fixture	Expected: 404 for cross-user mark/delete attempts | Actual: Not executed in this run | Status: Not Run
TC-TEST-019-14	Malformed notification identifier returns bad request	Use invalid UUID in mark-read route	NotificationValidationIT#unknownAndMalformedNotif icationIdsReturnDocumentedClientErrors	validation / medium	Notification Service and authenticated user	Expected: 400 Bad Request | Actual: Not executed in this run | Status: Not Run
TC-TEST-019-15	Unknown notification identifier returns not found	Mark/delete unknown notification	NotificationValidationIT#unknownAndMalformedNotif icationIdsReturnDocumentedClientErrors	validation / medium	Notification Service and authenticated user	Expected: 404 Not Found | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-02	Payment initiation returns the documented pending response	Start a course payment and inspect safe response data	PaymentStudentIT#studentCanStartAndReadOwnCourseP ayment	integration / high	Test Stripe mode and approved payable course/student fixture	Expected: 201 with positive amount three-letter currency PENDING and client secret | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-03	Student payment history contains the initiated payment	Read own payment list after initiation	PaymentStudentIT#studentCanStartAndReadOwnCourseP ayment	integration / high	Payment Service and test-mode PaymentIntent	Expected: 200 and payment ID appears in own list | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-018-04	Student payment detail matches payment initiation	Read own payment details	PaymentStudentIT#studentCanStartAndReadOwnCoursePayment	integration / high	Payment Service and test-mode payment	Expected: Payment/course/status/provider fields match | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-05	Student payment status agrees with payment detail	Read status endpoint for initiated payment	PaymentStudentIT#studentCanStartAndReadOwnCoursePayment	integration / high	Payment Service and test-mode payment	Expected: Payment ID/status match detail resource | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-06	Payment detail is private to its owning student	Second student attempts to read another student's payment	PaymentStudentIT#studentCanStartAndReadOwnCoursePayment	security / high	Second student credentials and matching ID fixture optional	Expected: Other user's list omits payment and detail/status return 404 | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-07	Admin payment listing returns valid page metadata	Read admin payment page	PaymentAdminIT#administratorCanListFilterAndInspectPayments	integration / high	Payment Service and admin credentials	Expected: Page number/size/total count and content bounds are valid | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-08	Admin payment filters return matching records	Filter by student course status date and payment reference	PaymentAdminIT#administratorCanListFilterAndInspectPayments	integration / high	Payment Service and approved payment fixtures	Expected: Every returned record matches requested filter values | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-12	Admin refund changes a dedicated successful payment to refunded	Refund the configured one-time payment fixture	PaymentAdminIT#administratorCanInspectAndSafelyRefundAnApprovedTestPayment	integration / high	TEST_PAYMENT_WRITE_ENA=true and TEST_PAYMENT_REFUND_ID with test-mode Stripe refund support	Expected: 200 REFUNDED and provider refund metadata persists | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-14	Overlong refund reason returns bad request	Submit a reason exceeding 500 characters	PaymentValidationIT#refundReasonOverLimitReturnsBadRequest	validation / medium	Payment Service and admin credentials	Expected: 400 Bad Request | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-18	Invalid payment IDs and input return exact client errors	Submit malformed/missing IDs and unknown payment IDs	PaymentValidationIT#invalidPaymentRequestsReturnExactClientErrorsOrNotFoundResponses	validation / high	Payment Service and student credentials	Expected: 400 for invalid input and 404 for unknown payment | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-27	Invalid admin payment filters return bad request	Reverse dates and submit invalid status/student UUID	PaymentValidationIT#invalidAdminFilterValuesReturnBadRequest	validation / high	Payment Service and admin credentials	Expected: 400 Bad Request | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-29	Unknown admin payment detail/refund IDs return not found	Request unknown admin detail and refund	PaymentValidationIT#invalidAdminFilterValuesReturnBadRequest	validation / medium	Payment Service and admin credentials	Expected: 404 Not Found | Actual: Not executed in this run | Status: Not Run
TC-AUTH-RECOVERY-001	Unknown password reset account is not disclosed	Generic reset request	AuthenticationRecoveryIT#shouldReturnGenericResultForUnknownPasswordResetAccount	security / high	Auth Service and Gateway	Expected: 200 generic response without supplied account value | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-AUTH-RECOVERY-002	Invalid password reset OTP/token is rejected	Reject invalid recovery credentials	AuthenticationRecoveryIT#shouldRejectInvalidPasswordResetOtpAndToken	validation / high	Auth Service and Gateway	Expected: Client error for invalid OTP and token | Actual: Not executed in this run | Status: Not Run
TC-AUTH-JWKS-001	Public JWKS endpoint returns signing keys	Read signing-key document	AuthenticationRecoveryIT#shouldExposePublicJwksDocument	integration / high	Auth Service and Gateway	Expected: 200 response with keys array | Actual: Not executed in this run | Status: Not Run
TC-COURSE-031	Teacher portal data endpoints return configured course data	Read directory summary gradebook and analytics	TeacherPortalDataIT#shouldReturnDirectorySummaryGradebookAndAnalyticsForConfiguredCourse	integration / high	Course Service teacher-owned course enrolled student	Expected: 200 successful response with data | Actual: Not executed in this run | Status: Not Run
TC-CAL-017	Venue listing requires authentication	Read venue list	CalendarVenueAuthorizationIT#shouldAllowAuthenticatedUsersToListVenues	security / high	Calendar Service and student credentials	Expected: 200 with data; unauthenticated 401 checked separately | Actual: Not executed in this run | Status: Not Run
TC-CAL-018	Only administrators can create venues	Deny non-admin venue creation	CalendarVenueAuthorizationIT#shouldRestrictVenueCreationToAdministrators	security / high	Calendar Service and student/teacher credentials	Expected: 401/403 without creating data | Actual: Not executed in this run | Status: Not Run
TC-CAL-019	Venue input is validated	Reject empty venue fields	CalendarVenueAuthorizationIT#shouldRejectInvalidVenueDataForAdministrator	validation / medium	Calendar Service and admin credentials	Expected: 400 Bad Request | Actual: Not executed in this run | Status: Not Run
TC-CAL-020	Teacher can read updated event details	Create update and retrieve event	CalendarEventWorkflowIT#shouldAllowTeacherToCreateUpdateAndReadAnEvent	integration / high	Calendar Service and teacher credentials; disposable event created by test	Expected: Read-after-write returns matching event ID and updated title; event cancelled in cleanup | Actual: Not executed in this run | Status: Not Run
TC-CAL-021	Unknown event detail returns not found	Get unknown event details	CalendarEventWorkflowIT#shouldReturnNotFoundForUnknownEvent	validation / medium	Calendar Service and teacher credentials	Expected: Unknown event detail and update return 404 | Actual: Not executed in this run | Status: Not Run
TC-CAL-022	Calendar event search requires date range	Reject event search without required date parameters	CalendarEventWorkflowIT#shouldRejectEventSearchWithoutRequiredDateRange	integration / medium	Calendar Service and teacher credentials	Expected: Missing required start/end returns 400; valid bounded search is asserted by TC-CAL-008 | Actual: Not executed in this run | Status: Not Run
TC-CAL-023	Calendar event detail requires authentication	Reject anonymous event detail requests	CalendarAuthenticationIT#shouldRejectEventDetailsWithoutAuthentication	security / high	Calendar Service and API Gateway	Expected: 401 Unauthorized | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-CAL-024	Calendar event access requires creator or assigned course teacher	Reject another teacher reading updating or cancelling an unassigned course event	CalendarEventWorkflowIT#anotherTeacherCannotUpdateOrCancelAnUnassignedCourseEvent	security / high	Calendar and Course Services; TEST_CALENDAR_COURSE_I assigned to primary teacher only; distinct secondary teacher credentials	Expected: 404 for read and 403 for update/cancel; owner cleans up event | Actual: Not executed in this run | Status: Not Run
TC-CAL-008	Student event search supplies required date range	Read calendar events in bounded date range	CalendarEventWorkflowIT#shouldAllowStudentToRetrieveCalendarEvents	integration / high	Calendar Service and student credentials	Expected: GET includes required start/end timestamps and returns 200 with successful response data | Actual: Not executed in this run | Status: Not Run
TC-COURSE-032	Teacher can manage course modules and lessons	Module and lesson CRUD lifecycle	CourseContentWorkflowIT#shouldCreateReadUpdateAndDeleteModuleAndLesson	integration / high	Course Service and teacher-owned course	Expected: Created lesson and module are read updated and cleaned up | Actual: Not executed in this run | Status: Not Run
TC-COURSE-033	Teacher can manage course learning-material references	Learning material reference CRUD lifecycle	CourseContentWorkflowIT#shouldCreateReadUpdateAndDeleteLearningMaterialReferences	integration / high	Course Service and teacher-owned course	Expected: Created material lesson and module are read updated and cleaned up | Actual: Not executed in this run | Status: Not Run
TC-COURSE-034	Administrator can manage grades and monthly fees	Grade and monthly fee CRUD lifecycle	CourseGradeWorkflowIT#shouldManageGradeAndMonthlyFeeThenDeleteCreatedRecords	integration / high	Course Service and TEST_COURSE_GRADE_WRITE_ENABLrue	Expected: Created grade and fee are cleaned up | Actual: Not executed in this run | Status: Not Run
TC-COURSE-035	Students can read their announcements and progress	Student self-service routes and role validation	CourseAnnouncementsProgressIT#shouldAllowStudentsToReadOwnAnnouncementsAndProgress	integration / high	Course Service and student credentials	Expected: 200 successful data response | Actual: Not executed in this run | Status: Not Run
TC-COURSE-036	Enrolled students can read course progress	Course-specific progress read	CourseAnnouncementsProgressIT#shouldReadProgressForAnEnrolledStudentCourseFixture	integration / high	Course Service and TEST_STUDENT_PROGRESS_COURSE	Expected: 200 with matching course ID and progress data | Actual: Not executed in this run | Status: Not Run
TC-COURSE-037	Teacher creates course announcement	Announcement create success	CourseAnnouncementsProgressIT#shouldCreateAnnouncementForApprovedTeacherCourseFixture	integration / medium	TEST_COURSE_ANNOUNCEMRITE_ENABLED and TEST_TEAHER_PORTAL_COURSE_ID	Expected: 201 with course and title; persistent synthetic record | Actual: Not executed in this run | Status: Not Run
TC-COURSE-038	Student updates lesson completion	Student lesson progress update and restore	CourseAnnouncementsProgressIT#shouldUpdateAndRestoreLessonProgressForApprovedStudentFixture	integration / medium	TEST_STUDENT_PROGRESS_E_ENABLED plus enrolled course and lesson fixture IDs	Expected: 200 with lesson/course and progress; prior completion restored | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-COURSE-039	Teacher uploads module and lesson materials	Cross-service course content file upload	CourseContentWorkflowIT#shouldUploadModuleAndLessonMaterialsAndAssociateThemAcrossServices	integration / high	TEST_COURSE_CONTENT_UI _WRITE_ENABLED plus teacher-owned course and Course/Material services	Expected: Uploaded files are associated with module/lesson/course and synthetic records are cleaned up | Actual: Not executed in this run | Status: Not Run
TC-TEST-010-01	Student self-enrollment creates pending enrollment	Self-enroll with free course	EnrollmentAuthenticationIT#shouldSupportStudentSelfEnrollmentWhenWriteFixturesAreExplicitlyEnabled	integration / high	TEST_ENROLLMENT_WRITE_ LED and verified free TEST _ENROLLMENT_FREE_COURS	Expected: 201 pending enrollment membership true and created record deleted | Actual: Not executed in this run | Status: Not Run
TC-TEST-010-02	Student reads own active course IDs	Own course list	EnrollmentAuthenticationIT#shouldAllowStudentToReadOwnActiveCourseIds	integration / medium	Enrollment read fixtures and student credentials	Expected: 200 with data | Actual: Not executed in this run | Status: Not Run
TC-TEST-010-03	Enrollment membership checks	Existing membership across roles	EnrollmentAuthenticationIT#shouldAllowStudentTeacherAndAdminToCheckMembership	integration / medium	Enrollment read fixtures	Expected: Configured existing membership is true for student teacher admin and super admin | Actual: Not executed in this run | Status: Not Run
TC-TEST-010-04	Admin creates enrollment with selected status	Admin enrollment lifecycle	EnrollmentAuthenticationIT#shouldSupportAdminEnrollmentManagementWhenWriteFixturesAreExplicitlyEnabled	integration / high	TEST_ENROLLMENT_WRITE_ LED and unused alternate course	Expected: 201 pending enrollment and cleanup | Actual: Not executed in this run | Status: Not Run
TC-TEST-010-05	Admin lists active enrollments	Admin active enrollment list	EnrollmentAuthenticationIT#shouldAllowAdminToListActiveAndUnenrolledStudents	integration / medium	Enrollment Service and admin credentials	Expected: 200 with data | Actual: Not executed in this run | Status: Not Run
TC-TEST-010-06	Admin updates enrollment status	Pending approved rejected transitions	EnrollmentAuthenticationIT#shouldSupportAdminEnrollmentManagementWhenWriteFixturesAreExplicitlyEnabled	integration / high	TEST_ENROLLMENT_WRITE_ LED and unused alternate course	Expected: 200 for each requested status | Actual: Not executed in this run | Status: Not Run
TC-TEST-010-07	Teacher reads course student and enrollment lists	Teacher course enrollment reads	EnrollmentAuthenticationIT#shouldAllowTeacherToListStudentsByCourse	integration / high	Owned course fixture and teacher credentials	Expected: 200 student IDs and active enrollment records | Actual: Not executed in this run | Status: Not Run
TC-TEST-010-08	Admin lists unenrolled students	Unenrolled student listing	EnrollmentAuthenticationIT#shouldAllowAdminToListActiveAndUnenrolledStudents	integration / medium	Enrollment Service and admin credentials	Expected: 200 with data | Actual: Not executed in this run | Status: Not Run
TC-TEST-010-09	Enrollment management role restrictions	Deny student and teacher management actions	EnrollmentAuthenticationIT#shouldRejectStudentsAndTeachersFromManagingEnrollments	security / high	Enrollment Service and student/teacher/admin credentials	Expected: Restricted management operations return documented 403/404 responses | Actual: Not executed in this run | Status: Not Run
TC-TEST-010-10	Enrollment routes require authentication	Protected endpoint matrix	EnrollmentAuthenticationIT#shouldRejectProtectedEndpointsWithoutAuthentication	security / high	Enrollment Service	Expected: 401 for protected endpoints | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-010-11	Enrollment input validation	Invalid create update and self-enroll payloads	EnrollmentAuthenticationIT#shouldRejectInvalidEnrollmentRequests	validation / medium	Enrollment Service and admin/student credentials	Expected: 400 client errors | Actual: Not executed in this run | Status: Not Run
TC-TEST-010-12	Unknown enrollment update and deletion	Unknown enrollment identifiers	EnrollmentAuthenticationIT#shouldReturnNotFoundForUnknownEnrollmentUpdate	integration / medium	Enrollment Service and admin credentials	Expected: Not found or client error without 5xx | Actual: Not executed in this run | Status: Not Run
TC-TEST-010-13	Unknown membership returns false	Unknown student course pair	EnrollmentAuthenticationIT#shouldReturnFalseForUnknownMembershipCombination	integration / low	Enrollment Service and admin credentials	Expected: 200 false | Actual: Not executed in this run | Status: Not Run
TC-TEST-010-14	Enrollment API client rejects invalid IDs	Client path validation	EnrollmentClientTest#shouldRejectBlankPathIdentifiers	unit / low	None	Expected: IllegalArgumentException for blank identifiers | Actual: Passed; see JUnit report | Status: Passed
TC-TEST-010-15	Enrollment service URL configuration is parsed	Parse enrollment service configuration	TestEnvironmentConfigurationTest#shouldParseEnrolmentServiceConfigurationWhenEnrollmentTestingIsEnabled	unit / low	None	Expected: Required services and enrollment URL are parsed | Actual: Passed; see JUnit report | Status: Passed
TC-TEST-010-16	Admin soft-deletes created enrollment	Delete and membership removal	EnrollmentAuthenticationIT#shouldSupportAdminEnrollmentManagementWhenWriteFixturesAreExplicitlyEnabled	integration / high	TEST_ENROLLMENT_WRITE_LED and unused alternate course	Expected: 200 and no active membership or course-list entry | Actual: Not executed in this run | Status: Not Run
TC-TEST-010-17	Admin updates enrollment status and course list	Pending approved rejected and course transitions	EnrollmentAuthenticationIT#shouldSupportAdminEnrollmentManagementWhenWriteFixturesAreExplicitlyEnabled	integration / high	TEST_ENROLLMENT_WRITE_LED and two unused alternate courses	Expected: 200 statuses and membership follows changed course list | Actual: Not executed in this run | Status: Not Run
TC-TEST-010-18	Admin deletes enrollment and removes active membership	Soft delete created enrollment	EnrollmentAuthenticationIT#shouldSupportAdminEnrollmentManagementWhenWriteFixturesAreExplicitlyEnabled	integration / high	TEST_ENROLLMENT_WRITE_LED and alternate courses without active communication groups	Expected: Membership false and created ID absent from both active course lists | Actual: Not executed in this run | Status: Not Run
TC-TEST-010-19	Enrollment write workflow cleans up its created records	Admin-created enrollment is removed by test cleanup	EnrollmentAuthenticationIT#shouldSupportAdminEnrollmentManagementWhenWriteFixturesAreExplicitlyEnabled	integration / high	Opt-in non-production fixtures and Communication Service	Expected: Only test-created enrollment IDs are soft-deleted | Actual: Not executed in this run | Status: Not Run
TC-TEST-010-20	Enrollment role matrix includes super admin	Check membership access for configured roles	EnrollmentAuthenticationIT#shouldAllowStudentTeacherAndAdminToCheckMembership	security / high	Student teacher admin and super admin credentials	Expected: Configured roles receive 200 for allowed membership read | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-010-21	Teacher enrollment creation enforces course ownership	Deny creation for another teacher's course	EnrollmentAuthenticationIT#shouldDenyTeacherEnrollmentCreationForAnotherTeachersCourse	integration / high	TEST_ENROLLMENT_UNOWNURSE_ID and student fixture	Expected: Creation in other teacher's course returns 403 | Actual: Not executed in this run | Status: Not Run
TC-TEST-010-22	Students are denied teacher and administrator enrollment queries	Probe course directory and unenrolled-student queries with restricted roles	EnrollmentAuthenticationIT#shouldRejectStudentsFromTeacherAndAdminQueries	security / high	Enrollment Service and student/teacher credentials	Expected: Student receives 403 for teacher/admin queries and teacher receives 403 for unenrolled-student query | Actual: Not executed in this run | Status: Not Run
TC-TEST-010-23	Teacher enrollment update and delete enforce course ownership	Deny update and delete for another teacher's course	EnrollmentAuthenticationIT#shouldDenyTeacherUpdatingOrDeletingEnrollmentForAnotherTeachersCourse	security / high	TEST_ENROLLMENT_ID identifying an enrollment in another teacher's course	Expected: Update and delete both return 403 without changing the fixture | Actual: Not executed in this run | Status: Not Run
TC-TEST-010-24	Teacher can manage enrollment in owned course	Create update and delete enrollment in teacher-owned course	EnrollmentAuthenticationIT#shouldAllowTeacherToManageEnrollmentForOwnedCourseWhenWritesAreEnabled	integration / high	TEST_ENROLLMENT_WRITE_LED and owned-course/non-enrolled-student fixtures plus Communication Service	Expected: Created enrollment is approved then removed and membership is cleared | Actual: Not executed in this run | Status: Not Run
TC-COURSE-040	Teacher portal enforces course ownership	Teacher portal wrong-course denial	TeacherPortalDataIT#shouldDenyTeacherAnalyticsAndGradebookForAnotherTeachersCourse	integration / high	TEST_TEACHER_PORTAL_UND_COURSE_ID for another teacher's course	Expected: 403 Forbidden for each teacher portal course read | Actual: Not executed in this run | Status: Not Run
TC-COURSE-041	Teacher analytics validates period and date range	Analytics input boundary validation	TeacherPortalDataIT#shouldValidateAnalyticsPeriodsAndDateRanges	integration / medium	Teacher-owned TEST_TEACHE_PORTAL_COURSE_ID	Expected: Valid range returns 200; invalid period and inverted range return 400 | Actual: Not executed in this run | Status: Not Run
TC-COURSE-042	Teacher cannot mutate another teacher's course content	Reject cross-owner module lesson and learning-material mutations	TeacherCourseContentOwnershipIT#teacherCannotMutateAnotherTeachersCourseContent	integration / high	TEST_COURSE_CONTENT_OWHIP_WRITE_ENABLED plus valid unowned course module lesson and learning-material IDs	Expected: All create update and delete operations return 403 without mutating foreign records | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-26	User can read and submit no-op self profile update	Read and no-op update self profile	UserSelfServiceIT#shouldAllowUsersToReadAndNoOpUpdateTheirOwnProfile	integration / medium	User Service and student credentials	Expected: 200 with own profile and successful no-op update | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-27	Teacher-only profile and guidance routes reject students	Teacher profile/guidance authorization	UserSelfServiceIT#shouldAllowTeacherToSubmitNoOpProfileUpdateAndRestrictTeacherOnlyRoutes	security / high	User Service and teacher/student credentials	Expected: Teacher no-op update succeeds student denied invalid note 400 | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-021-28	User lookup routes validate missing records and inputs	Teacher and student lookup behavior	UserSelfServiceIT#shouldReturnNotFoundForUnknownTeacherAndEmptyForUnknownStudentLookup	integration / medium	User Service and teacher credentials	Expected: Unknown teacher 404 unknown student list empty | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-29	Protected user routes require authentication	Unauthenticated protected route matrix	UserSelfServiceIT#shouldRejectUnauthenticatedSelfServiceAndProtectedLookupCalls	security / high	User Service	Expected: 401 Unauthorized | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-30	Password change and student registration validate payloads	Reject empty password and registration requests	UserSelfServiceIT#shouldValidatePasswordChangeAndRequireAuthenticationForRegistration	validation / high	User Service and student credentials	Expected: 400 Bad Request | Actual: Not executed in this run | Status: Not Run
TEST-013-31	AI source list requires teacher or administrator role	Read AI sources by course	MaterialAiSourcesIT#shouldAllowTeacherToReadAiSourcesForConfiguredCourse	integration / high	Material Service and TEST_MATERIAL_COURSE_ID	Expected: 200 with paginated data for teacher; student/anonymous denied | Actual: Not executed in this run | Status: Not Run
TEST-013-32	Material list filters reflect lifecycle state	Filter by purpose and status	MaterialLifecycleIT#teacherCanUploadRetrieveListDownloadUpdateArchiveAndDeleteMaterial	integration / high	Material Service with TEST_MATERIAL_WRITE_ENABLED and approved course fixture	Expected: Uploaded item appears with LEARNING_MATERIAL and UPLOADED filters and later with ARCHIVED filter | Actual: Not executed in this run | Status: Not Run
TEST-013-33	Material signed URL returns uploaded bytes	Download uploaded bytes using presigned URL	MaterialLifecycleIT#teacherCanUploadRetrieveListDownloadUpdateArchiveAndDeleteMaterial	integration / high	TEST_MATERIAL_WRITE_ENAD plus TEST_MATERIAL_DOWOAD_BYTES_ENABLED and exact approved TEST_MATERIAL_DOWNLOAD_ALLOWED_H	Expected: HTTPS URL returns 200 and bytes exactly match uploaded synthetic file | Actual: Not executed in this run | Status: Not Run
TEST-013-34	Material signed URL host safety	Restrict URL byte probe to approved HTTPS host	MaterialDownloadUrlPolicyTest#acceptsOnlyHttpsOnTheExactConfiguredHost	unit / high	No external services	Expected: Accepts exact HTTPS host and rejects alternate host scheme credentials fragment and unsafe port | Actual: Passed; see JUnit report | Status: Passed
TEST-013-35	Material signed URL rejects alternate targets	Reject mismatched hosts and non-HTTPS URLs	MaterialDownloadUrlPolicyTest#rejectsUnconfiguredHostsAndNonHttpsSchemes	unit / high	No external services	Expected: Alternate host and HTTP scheme are rejected before network access | Actual: Passed; see JUnit report | Status: Passed
TEST-013-36	Material signed URL rejects unsafe URI components	Reject credentials fragments and nonstandard ports	MaterialDownloadUrlPolicyTest#rejectsCredentialsFragmentsAndUnapprovedPorts	unit / high	No external services	Expected: User info fragments and non-443 explicit ports are rejected | Actual: Passed; see JUnit report | Status: Passed
TEST-013-37	Material byte probe requires explicit URL configuration	Require URL and approved host configuration	MaterialDownloadUrlPolicyTest#requiresAnExplicitHostAndNonEmptyDownloadUrl	unit / medium	No external services	Expected: Missing or blank URL/host is rejected | Actual: Passed; see JUnit report | Status: Passed

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TEST-013-39	Material course membership controls student access	Reject an unenrolled student's material reads	MaterialLifecycleIT#rejectsUnenrolledSecondaryStudentFromMaterialReads	integration / high	TEST_MATERIAL_ID and TEST_MATERIAL_COURSE_ID plus distinct secondary student not enrolled in the course	Expected: Secondary student receives 403 for material retrieval and download | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-23	Payment verification requires authorized user	Admin verification and teacher/anonymous denial	PaymentAuthorizationIT#administratorCanVerifySuccessfulPaymentForUnknownStudentCoursePair	security / high	Payment Service and admin credentials	Expected: Admin gets false result for unknown pair teacher/anonymous denied | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-24	Invalid payment webhook signatures are rejected	Missing and malformed Stripe signature validation	PaymentValidationIT#webhookWithInvalidSignatureIsRejected	validation / high	Payment Service	Expected: 400 for malformed signature | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-25	Signed synthetic webhook replay is accepted	Valid signature and duplicate event handling	PaymentValidationIT#signedUnsupportedWebhookIsAcceptedAndSafeToReplay	integration / medium	TEST_STRIPE_WEBHOOK_TENABLED=true and TEST_STRIPE_WEBHOOK_SE must match isolated Payment Service	Expected: First delivery accepted and duplicate replay safely accepted without payment-state mutation | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-30	Signed payment success webhook transitions a disposable pending payment	Verify PaymentIntent success transition and idempotency	PaymentValidationIT#signedSuccessWebhookTransitionsDedicatedPendingPaymentAndIsIdempotent	integration / high	TEST_PAYMENT_WEBHOOK__ENABLED=true plus dedicated pending TEST_PAYMENT_WEBHOOK_SUCCESS_ and matching non-production Stripe webhook secret	Expected: Payment transitions to SUCCESS with paidAt and duplicate event delivery returns 200 without repeating transition | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-31	Signed payment failure webhook transitions a separate disposable pending payment	Verify PaymentIntent failure transition and idempotency	PaymentValidationIT#signedFailureWebhookTransitionsDedicatedPendingPaymentAndIsIdempotent	integration / high	TEST_PAYMENT_WEBHOOK__ENABLED=true plus dedicated pending TEST_PAYMENT_WEBHOOK_FAILURE_I and matching non-production Stripe webhook secret	Expected: Payment transitions to FAILED without paidAt and duplicate event delivery returns 200 without repeating transition | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-32	Unknown course payment initiation is not found	Reject a valid but unknown course ID before payment creation	PaymentValidationIT#invalidPaymentRequestsReturnExactClientErrorsOrNotFoundResponses	validation / medium	Payment Service and Course Service with student credentials	Expected: Valid unknown course UUID returns 404 without creating a payment or invoking Stripe | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-27	AI conversations are isolated to their owner	Reject cross-owner conversation reads and mutations	AiConversationIT#differentAdminOwnerCannotReadOrMutateTeachersConversation	integration / high	TEST_AI_WRITE_ENABLED=tr plus distinct teacher and admin credentials	Expected: Admin receives 404 for teacher-owned conversation read/messages/update/delete and owner cleanup succeeds | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-020-23	Deterministic AI chat and generation return success	Assert successful provider outputs	AiProviderIT#deterministicProviderSupportsSafeChatAndGenerationTests	integration / high	AI_TEST_MODE=true and TEST_AI_WRITE_ENABLED=tr	Expected: 200 with response data | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-24	Lesson summarization returns non-empty summary	Summarize configured lesson	AiContentGenerationIT#shouldSummarizeConfiguredLessonWithDeterministicProvider	integration / high	AI deterministic provider and TEST_AI_LESSON_ID	Expected: 200 with non-empty summary | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-25	Material summarization returns non-empty summary	Summarize configured material	AiContentGenerationIT#shouldSummarizeConfiguredMaterialWithDeterministicProvider	integration / high	AI deterministic provider and TEST_AI_MATERIAL_ID	Expected: 200 with non-empty summary | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-26	AI quiz generation creates the requested number of questions	Generate quiz draft with deterministic provider	AiContentGenerationIT#shouldGenerateQuizDraftWithDeterministicProvider	integration / high	AI deterministic provider and TEST_AI_COURSE_ID	Expected: 200 with three questions | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-16	Public multipart quiz-generation route validation	Verify unauthenticated route reaches multipart parameter validation	AiAuthorizationIT#quizGenerationIsPublicAccordingToEffectiveBackendSecurity	security / medium	AI Service only; malformed course UUID	Expected: Request reaches controller validation and returns 400 rather than 401 | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-17	AI quiz-generation semantic validation	Reject invalid count difficulty and question type before provider call	AiValidationIT#publicQuizGenerationRejectsInvalidSemanticParametersBeforeProviderCall	validation / high	AI Service only; no provider	Expected: Valid UUID with invalid semantic quiz-generation fields returns 400 without authentication or provider access | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-03	AI current-user identity and roles	Assert authenticated current-user response fields	AiAuthorizationIT#studentCannotUseChatOrConversationManagement	integration / medium	AI Service and student credentials	Expected: Current-user route returns a UUID userId and includes STUDENT in the roles list | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-25	User administration is restricted by role	Reject unauthorized user-management operations	UserAdministrationIT#shouldRestrictUserAdministrationToAllowedRoles	security / high	User Service and student/teacher/admin credentials	Expected: 403 Forbidden for disallowed roles | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-22	Student upcoming exam read requires student auth	Read upcoming exams	ExamAuthorizationIT#studentCannotUseManagementEndpointsButCanUseOwnResultsEndpoint	integration / medium	Exam Service and student credentials	Expected: 200 for student and 401 without token | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-23	Exam attachment reads require authentication	Reject anonymous attachment reads	ExamAuthorizationIT#unauthenticatedExamRequestsReturnUnauthorized	security / high	Exam Service	Expected: 401 Unauthorized | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-24	Students cannot delete exam attachments	Deny student attachment deletion	ExamAuthorizationIT#studentCannotUseManagementEndpointsButCanUseOwnResultsEndpoint	security / high	Exam Service and student credentials	Expected: 403 Forbidden | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-017-25	Exam attachment upload/list/download/delete lifecycle	Verify attachment lifecycle and metadata	ExamLifecycleIT#teacherCanRunTheExamLifecycleAndReadResults	integration / medium	TEST_EXAM_WRITE_ENABLE plus Course Enrollment Exam and Material services with approved course/student fixtures	Expected: Synthetic PDF upload/list/download metadata/delete succeed; optional byte comparison; attachment and test-created Material record are cleaned up | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-26	Student exam detail is visible only to enrolled students	Read scheduled exam as enrolled student	ExamLifecycleIT#teacherCanRunTheExamLifecycleAndReadResults	integration / high	TEST_EXAM_WRITE_ENABLE TEST_EXAM_COURSE_ID and TEST_EXAM_STUDENT_ID; configured Student login must represent that student	Expected: Student retrieves scheduled exam with 200; anonymous/non-student denied and unknown exam returns 404 | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-27	Exam attachment student visibility	Enforce visible and hidden attachment access	ExamLifecycleIT#teacherCanRunTheExamLifecycleAndReadResults	integration / high	TEST_EXAM_WRITE_ENABLE plus matching enrolled student and Course Enrollment Exam Material services	Expected: Student lists/downloads visible attachment but cannot list or download teacher-only attachment; test-created attachment and Material are cleaned up | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-28	Exam attachment PDF validation	Reject unsupported file and accept synthetic PDF	ExamLifecycleIT#teacherCanRunTheExamLifecycleAndReadResults	integration / high	TEST_EXAM_WRITE_ENABLE plus Course Enrollment Exam and Material services	Expected: TXT upload returns 400; PDF upload returns 201 and both local/remote synthetic artifacts are cleaned up | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-32	Message attachment operations require authentication	Reject anonymous attachment upload and download	CommunicationAttachmentAuthenticationIT#shouldRejectUnauthenticatedMessageAttachmentUploadAndDownload	security / high	Communication Service	Expected: 401 Unauthorized | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-33	Message attachment lifecycle and cleanup	Upload link download metadata and cleanup	CommunicationMessageWorkflowIT#shouldUploadAttachDownloadAndCleanUpMessageAttachment	integration / high	TEST_COMMUNICATION_WRI NABLED plus active teacher group membership and Material Service	Expected: Attachment is linked to message and metadata is returned; optional byte comparison; message and material cleaned up | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-34	Communication STOMP CONNECT authenticates teacher access token	Open direct WebSocket and authenticate a STOMP session	CommunicationWebSocketIT#teacherCanAuthenticateStompWebSocketConnection	integration / high	TEST_COMMUNICATION_WEB ET_ENABLED=true and teacher credential; Communication Service directly reachable	Expected: Authenticated teacher receives STOMP CONNECTED without creating or mutating data | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-29	AI STOMP CONNECT authenticates teacher access token	Open direct AI WebSocket and authenticate a STOMP session	AiWebSocketIT#teacherCanAuthenticateAiStompWebSocketConnection	integration / high	TEST_AI_WEBSOCKET_ENAB true and teacher credential; AI Service directly reachable	Expected: Authenticated teacher receives STOMP CONNECTED without sending chat or invoking provider | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-021-31	Student lookup results are restricted to the authenticated student	Enforce student lookup ownership	UserLookupOwnershipIT#studentCanLookupOnlyTheirOwnSummaryWhenAnotherStudentIsConfigured	integration / high	User Service plus distinct SECONDARY_STUDENT_TEST credentials	Expected: Own student summary returns 200 and another student's summary returns 403 from both lookup routes | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-32	Administrator list filters must return matching roles	Verify user list role filters	UserAdministrationIT#administratorCanListUsersAndReadAnalytics	integration / high	User Service and administrator plus student and teacher accounts	Expected: Each filtered list includes records and all returned roles match the requested role | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-33	Existence endpoints return false for unknown identities	Verify student and teacher existence booleans	UserValidationIT#invalidUserIdentifiersAndStatusPayloadsAreHandledSafely	integration / medium	User Service and administrator credentials	Expected: Unknown IDs return HTTP 200 with false and malformed student ID returns false | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-34	Unknown administration targets return not found	Verify unknown account lifecycle targets	UserAdministrationIT#administratorOperationsValidateUnknownUsersAndFields	integration / medium	User Service and administrator credentials	Expected: Valid status update invitation resend and delete requests for unknown user return 404 while malformed status payload returns 400 | Actual: Not executed in this run | Status: Not Run
TC-COURSE-002	Teacher retrieves assigned courses	Read teacher's assigned courses	CourseAuthenticationIT#shouldAllowTeacherToReadAssignedCourses	integration / high	Course Service and teacher credentials	Expected: 200 with success status and data | Actual: Not executed in this run | Status: Not Run
TC-COURSE-003	Student performs course lookup	Look up an unknown course as student	CourseAuthenticationIT#shouldAllowStudentToUseCourseLookup	security / medium	Course Service and student credentials	Expected: 404 for unknown course proves student lookup is allowed rather than role denied | Actual: Not executed in this run | Status: Not Run
TC-COURSE-004	Admin retrieves all courses	List courses as administrator	CourseAuthenticationIT#shouldAllowAdminToReadAllCourses	integration / high	Course Service and admin credentials	Expected: 200 with success status and data | Actual: Not executed in this run | Status: Not Run
TC-COURSE-005	Unauthenticated course list	Request course list without credentials	CourseAuthenticationIT#shouldRejectCourseListWithoutAuthentication	security / high	Course Service	Expected: 401 Unauthorized | Actual: Not executed in this run | Status: Not Run
TC-COURSE-006	Unauthenticated teacher courses	Request assigned courses without credentials	CourseAuthenticationIT#shouldRejectTeacherCoursesWithoutAuthentication	security / high	Course Service	Expected: 401 Unauthorized | Actual: Not executed in this run | Status: Not Run
TC-COURSE-007	Student reads all courses	Deny student access to full course list	CourseAuthenticationIT#shouldRejectStudentFromReadingAllCourses	security / high	Course Service and student credentials	Expected: 403 Forbidden | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-COURSE-008	Teacher reads all courses	Deny teacher access to full course list	CourseAuthenticationIT#shouldRejectTeacherFromReadingAllCourses	security / high	Course Service and teacher credentials	Expected: 403 Forbidden | Actual: Not executed in this run | Status: Not Run
TC-COURSE-009	Student creates a course	Deny student course creation	CourseAuthenticationIT#shouldRejectStudentFromCreatingACourse	security / high	Course Service and student credentials	Expected: 403 Forbidden before persistence | Actual: Not executed in this run | Status: Not Run
TC-COURSE-010	Teacher reaches course creation	Check backend teacher course-create permission without creating data	CourseAuthenticationIT#shouldAllowTeacherRoleToReachCourseCreation	security / medium	Course Service and teacher credentials	Expected: Request reaches validation/lookup and returns 404 for configured unknown grade rather than role-denied 403 | Actual: Not executed in this run | Status: Not Run
TC-COURSE-011	Student deletes a course	Deny student course deletion	CourseAuthenticationIT#shouldRejectStudentFromDeletingACourse	security / high	Course Service and student credentials	Expected: 403 Forbidden before persistence | Actual: Not executed in this run | Status: Not Run
TC-COURSE-012	Student performs bulk lookup	Deny student bulk course lookup	CourseAuthenticationIT#shouldRejectStudentFromBulkCourseLookup	security / high	Course Service and student credentials	Expected: 403 Forbidden | Actual: Not executed in this run | Status: Not Run
TC-COURSE-015	Admin creates a course	Create and clean up a synthetic course	CourseCrudIT#shouldCreateCourseAsAdmin	integration / high	Course Service; TEST_COURSE_GRADE_ID ar TEST_COURSE_TEACHER_ID	Expected: 201 with created course ID and test-created course deleted | Actual: Not executed in this run | Status: Not Run
TC-COURSE-016	Admin reads and updates a course	Create then read and update a synthetic course	CourseCrudIT#shouldReadAndUpdateCreatedCourseAsAdmin	integration / high	Course Service; TEST_COURSE_GRADE_ID ar TEST_COURSE_TEACHER_ID	Expected: Created course is readable and updated name persists; test-created course deleted | Actual: Not executed in this run | Status: Not Run
TC-COURSE-017	Student and teacher lookup	Read a created course as student and teacher	CourseCrudIT#shouldAllowStudentAndTeacherToLookupCreatedCourse	integration / high	Course Service; TEST_COURSE_GRADE_ID ar TEST_COURSE_TEACHER_ID	Expected: Both roles receive 200 and the matching course ID; course deleted by admin | Actual: Not executed in this run | Status: Not Run
TC-COURSE-018	Admin bulk lookup	Find created course by administrator bulk lookup	CourseCrudIT#shouldFindCreatedCourseByBulkLookupAsAdmin	integration / high	Course Service; TEST_COURSE_GRADE_ID ar TEST_COURSE_TEACHER_ID	Expected: 200 result contains created course ID; course deleted by admin | Actual: Not executed in this run | Status: Not Run
TC-COURSE-019	Admin deletes a course	Delete created course and verify it is not found	CourseCrudIT#shouldDeleteCreatedCourseAsAdmin	integration / high	Course Service; TEST_COURSE_GRADE_ID ar TEST_COURSE_TEACHER_ID	Expected: Delete returns 200 and subsequent read returns 404 | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-COURSE-020	Admin submits invalid course data	Reject empty course creation payload	CourseCrudIT#shouldRejectInvalidCoursePayloadAsAdmin	validation / medium	Course Service and admin credentials	Expected: 400 Bad Request | Actual: Not executed in this run | Status: Not Run
TC-COURSE-021	Admin reads an unknown course	Read unknown course ID	CourseCrudIT#shouldReturnNotFoundForUnknownCourse	validation / medium	Course Service and admin credentials	Expected: 404 Not Found | Actual: Not executed in this run | Status: Not Run
TC-COURSE-043	Announcement and progress role and validation boundaries	Exercise wrong-role and malformed announcement/progress requests	CourseAnnouncementsProgressIT#shouldRejectAnnouncementAndProgressOperationsForWrongRoles	security / high	Course Service with teacher and student credentials	Expected: Forbidden role operations return 403 and invalid request bodies return 400 | Actual: Not executed in this run | Status: Not Run
TC-COURSE-044	Announcement pagination client bounds	Reject invalid announcement page and size locally	CourseAnnouncementsProgressIT#shouldValidateAnnouncementPaginationBeforeCallingService	validation / medium	Student credentials	Expected: Negative page and size above the maximum throw IllegalArgumentException before an HTTP request | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-15	Assignment and submission routes require authentication	Probe protected assignment and submission routes anonymously	AssignmentAuthenticationIT#rejectsProtectedAssignmentAndSubmissionRequestsWithoutAuthentication	security / high	Assignment Service	Expected: Protected assignment/submission operations return 401 | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-17	Unknown assignment returns not found	Read assignment with unknown UUID	AssignmentWorkflowIT#unknownAssignmentIsNotReportedAsServerError	validation / medium	Assignment Service with read fixture configuration	Expected: Unknown assignment returns 404 | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-26	Student cannot manage assignments	Check student management and self-service routes	AssignmentAuthenticationIT#rejectsStudentAssignmentManagement	security / high	Assignment Service and student credentials	Expected: Management create/PATCH returns 403 while student assignment list returns 200 | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-27	Teacher cannot create student submissions or use student assignment listing	Check teacher is denied student-only actions	AssignmentAuthenticationIT#rejectsTeacherSubmissionCreation	security / high	Assignment Service and teacher credentials	Expected: Submission creation and student-only assignment list return 403 | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-28	Assignment due-date and attempt settings are persisted	Verify due date and allowed-attempt response values	AssignmentWorkflowIT#teacherCanCreateReadListUpdatePublishCloseAndDeleteAssignment	integration / medium	TEST_ASSIGNMENT_WRITE_ENABLED plus course and module fixtures	Expected: Created assignment returns configured due date and allowed-attempt count | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-29	Late submission is prohibited when assignment policy disables it	Try to submit a draft after its due date	SubmissionWorkflowIT#rejectsSubmittingADraftAfterItsDueDateWhenLateWorkIsDisabled	integration / high	TEST_ASSIGNMENT_LATE_SUBMISSION_ID is a draft on a published past-due assignment with late work disabled	Expected: 409 and draft remains unchanged | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-011-30	Submission attempt limit is enforced	Try to exceed the configured attempt limit	SubmissionWorkflowIT#rejectsAnAdditionalSubmissionAfterTheConfiguredAttemptLimit	integration / high	TEST_ASSIGNMENT_ATTEMPMIT_ID and configured student already at the limit	Expected: 409 and submission count remains unchanged | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-31	Rubric total cannot exceed assignment maximum	Reject rubric marks above maxMarks	AssignmentWorkflowIT#rejectsRubricMarksThatExceedAssignmentMaximum	integration / high	TEST_ASSIGNMENT_WRITE_LED plus teacher-owned course and module	Expected: 400 before assignment creation | Actual: Not executed in this run | Status: Not Run
TC-TEST-012-01	Unauthenticated assignment attachment routes reject access	Check anonymous attachment operations	AssignmentAttachmentAuthenticationIT#rejectsAttachmentAndFileRequestsWithoutAuthentication	security / high	Assignment Service	Expected: Unauthenticated assignment attachment list/create/download/delete return 401 | Actual: Not executed in this run | Status: Not Run
TC-TEST-012-02	Unauthenticated submission-file routes reject access	Check anonymous submission-file operations	AssignmentAttachmentAuthenticationIT#rejectsAttachmentAndFileRequestsWithoutAuthentication	security / high	Assignment Service	Expected: Unauthenticated submission-file list/create/download/delete return 401 | Actual: Not executed in this run | Status: Not Run
TC-TEST-012-03	Teacher links existing material to assignment	Create assignment attachment from configured material	AssignmentAttachmentWorkflowIT#teacherCanCreateListDownloadAndDeleteAssignmentMaterialAttachment	integration / high	TEST_ATTACHMENT_WRITE_LED with assignment/material fixtures	Expected: Attachment returns 201 and references configured Material ID | Actual: Not executed in this run | Status: Not Run
TC-TEST-012-04	Teacher lists and downloads assignment attachment	Inspect attachment collection and metadata as teacher	AssignmentAttachmentWorkflowIT#teacherCanCreateListDownloadAndDeleteAssignmentMaterialAttachment	integration / high	TEST_ATTACHMENT_WRITE_LED with assignment/material fixtures	Expected: List and download metadata return 200 and include the created attachment | Actual: Not executed in this run | Status: Not Run
TC-TEST-012-05	Student can read published assignment attachment	Check student attachment list and download access	AssignmentAttachmentWorkflowIT#teacherCanCreateListDownloadAndDeleteAssignmentMaterialAttachment	integration / high	TEST_ATTACHMENT_WRITE_LED with enrolled student and assignment/material fixtures	Expected: Student list and download return 200 | Actual: Not executed in this run | Status: Not Run
TC-TEST-012-06	Teacher deletes assignment attachment	Delete test-created assignment attachment	AssignmentAttachmentWorkflowIT#teacherCanCreateListDownloadAndDeleteAssignmentMaterialAttachment	integration / high	TEST_ATTACHMENT_WRITE_LED with assignment/material fixtures	Expected: Teacher delete returns 200 and test cleanup removes only its attachment | Actual: Not executed in this run | Status: Not Run
TC-TEST-012-07	Student cannot create assignment attachment	Check student attachment create denial	AssignmentAttachmentAuthenticationIT#rejectsInvalidAttachmentPayloadAndStudentManagement	security / high	Assignment Service and student credentials	Expected: Student attachment creation returns 403 | Actual: Not executed in this run | Status: Not Run
TC-TEST-012-08	Student links material to own draft submission	Create submission file association from configured material	SubmissionFileWorkflowIT#studentCanCreateListDownloadAndDeleteSubmissionMaterialFile	integration / high	TEST_ATTACHMENT_WRITE_LED with owned draft submission and material fixtures	Expected: Student create returns 201 and references configured material | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-012-09	Submission file owner and authorized teacher can read and owner deletes	Check submission file list download and deletion	SubmissionFileWorkflowIT#studentCanCreateListDownloadAndDeleteSubmissionMaterialFile	integration / high	TEST_ATTACHMENT_WRITE_LED with owned draft submission/material fixtures	Expected: Student and teacher read return 200 and student deletion returns 200 | Actual: Not executed in this run | Status: Not Run
TC-TEST-012-10	Teacher cannot create or delete submission files	Check teacher submission-file management denial	AssignmentAttachmentAuthenticationIT#rejectsTeacherSubmissionFileManagement	security / high	Assignment Service and teacher credentials	Expected: Teacher create and delete return 403 | Actual: Not executed in this run | Status: Not Run
TC-TEST-012-11	Invalid assignment attachment payload is rejected	Submit empty attachment request	AssignmentAttachmentAuthenticationIT#rejectsInvalidAttachmentPayloadAndStudentManagement	validation / medium	Assignment Service and teacher credentials	Expected: Teacher receives 400 for empty attachment request | Actual: Not executed in this run | Status: Not Run
TC-TEST-012-12	Unknown assignment attachment resources return not found	Read unknown assignment and attachment identifiers	AssignmentAttachmentWorkflowIT#rejectsUnknownAssignmentAttachmentResources	validation / medium	Assignment Service with attachment read fixtures	Expected: Unknown assignment and attachment return 404 | Actual: Not executed in this run | Status: Not Run
TC-TEST-012-17	Unknown submission-file resource returns not found	Download unknown file on configured submission	SubmissionFileWorkflowIT#rejectsUnknownSubmissionFileResources	validation / medium	Assignment Service with submission read fixture	Expected: Unknown submission-file ID returns 404 | Actual: Not executed in this run | Status: Not Run
TC-TEST-012-18	Assignment attachment cross-student ownership	Reject unenrolled student listing and downloading attachments	AssignmentAttachmentWorkflowIT#rejectsUnenrolledSecondaryStudentFromAssignmentAttachments	integration / high	TEST_ATTACHMENT_ASSIGN_ID with an existing attachment plus distinct secondary student not enrolled in its course	Expected: Secondary student receives 403 for list and download | Actual: Not executed in this run | Status: Not Run
TC-TEST-012-19	Assignment attachment download byte integrity	Compare signed URL bytes with uploaded synthetic attachment	AssignmentAttachmentWorkflowIT#teacherCanUploadAndCleanUpSyntheticAssignmentAttachment	integration / high	TEST_ATTACHMENT_WRITE_LED plus TEST_MATERIAL_DOWNLOAD_BYTES_ENABLED and exact approved storage host	Expected: Downloaded bytes equal the synthetic upload | Actual: Not executed in this run | Status: Not Run
TC-TEST-012-20	Submission file download byte integrity	Compare signed URL bytes with uploaded synthetic submission file	SubmissionFileWorkflowIT#studentCanUploadAndCleanUpSyntheticSubmissionFile	integration / high	TEST_ATTACHMENT_WRITE_LED plus TEST_MATERIAL_DOWNLOAD_BYTES_ENABLED and exact approved storage host	Expected: Downloaded bytes equal the synthetic upload | Actual: Not executed in this run | Status: Not Run
TC-TEST-012-21	Attachment cleanup failure handling	Run all cleanup actions and retain primary failure	CleanupScopeTest#runsRemainingActionsAndRetainsMultipleCleanupFailures	unit / medium	None	Expected: All cleanup actions run and secondary failures are retained as suppressed exceptions | Actual: Passed; see JUnit report | Status: Passed

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-011-16	Invalid assignment create payload is rejected	Submit an empty assignment create request	AssignmentAuthenticationIT#rejectsInvalidAssignmentPayloadBeforeCreation	validation / medium	Assignment Service and teacher credentials	Expected: Empty create payload returns 400 before persistence | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-08	Student creates a draft submission	Create a draft on configured assignment	SubmissionWorkflowIT#studentCanCreateUpdateAndSubmitDraftAndTeacherCanReviewIt	integration / high	TEST_ASSIGNMENT_WRITE_ENABLED with assigned course/assignment and student fixture	Expected: 201 draft response contains the created submission ID and DRAFT status | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-09	Student updates a draft submission	Change draft submission content	SubmissionWorkflowIT#studentCanCreateUpdateAndSubmitDraftAndTeacherCanReviewIt	integration / high	TEST_ASSIGNMENT_WRITE_ENABLED with assigned course/assignment and student fixture	Expected: 200 and updated content persists before submission | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-10	Student submits a draft submission	Submit draft and inspect status	SubmissionWorkflowIT#studentCanCreateUpdateAndSubmitDraftAndTeacherCanReviewIt	integration / high	TEST_ASSIGNMENT_WRITE_ENABLED with assigned course/assignment and student fixture	Expected: 200 and status is SUBMITTED | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-11	Student lists own submission	Verify submitted item appears in own submission list	SubmissionWorkflowIT#studentCanCreateUpdateAndSubmitDraftAndTeacherCanReviewIt	integration / high	TEST_ASSIGNMENT_WRITE_ENABLED with assigned course/assignment and student fixture	Expected: 200 list contains the created submission ID | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-12	Teacher lists submissions for assignment	Verify submitted item appears for teacher	SubmissionWorkflowIT#studentCanCreateUpdateAndSubmitDraftAndTeacherCanReviewIt	integration / high	TEST_ASSIGNMENT_WRITE_ENABLED with assigned course/assignment and teacher fixture	Expected: 200 list contains the created submission ID | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-13	Teacher grades a submission	Grade submitted work	SubmissionWorkflowIT#studentCanCreateUpdateAndSubmitDraftAndTeacherCanReviewIt	integration / high	TEST_ASSIGNMENT_WRITE_ENABLED with assigned course/assignment and teacher fixture	Expected: 200 and marks persist | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-14	Teacher returns graded submission	Return the graded submission	SubmissionWorkflowIT#studentCanCreateUpdateAndSubmitDraftAndTeacherCanReviewIt	integration / high	TEST_ASSIGNMENT_WRITE_ENABLED with assigned course/assignment and teacher fixture	Expected: 200 and status is RETURNED | Actual: Not executed in this run | Status: Not Run
TC-AUTH-CLIENT-001	Successful login creates an authenticated API client	Login and construct authenticated API client	AuthenticationClientIT#shouldLoginAndCreateAnAuthenticatedApiClient	integration / high	Authentication Service and Student credentials	Expected: Session is authenticated and authenticated API client is available | Actual: Not executed in this run | Status: Not Run
TC-AUTH-LOGIN-001	Valid Student credentials authenticate successfully	Login using valid Student credentials	AuthenticationLoginIT#shouldAuthenticateAStudentWithValidCredentials	integration / high	Authentication Service and Student credentials	Expected: Authenticated session contains a nonblank access token | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-AUTH-LOGIN-002	Invalid password is rejected without disclosing it	Reject invalid password	AuthenticationLoginIT#shouldRejectAnInvalidPassword	security / high	Authentication Service and Student credentials	Expected: 400 Bad Request and response does not echo password | Actual: Not executed in this run | Status: Not Run
TC-AUTH-LOGIN-003	Unknown account login is rejected	Reject unknown user	AuthenticationLoginIT#shouldRejectAnUnknownUser	security / high	Authentication Service	Expected: 400 Bad Request | Actual: Not executed in this run | Status: Not Run
TC-AUTH-LOGIN-004	Malformed login request is rejected	Reject invalid login request	AuthenticationLoginIT#shouldRejectAnInvalidLoginRequest	validation / high	Authentication Service	Expected: 400 Bad Request | Actual: Not executed in this run | Status: Not Run
TC-AUTH-REFRESH-001	Valid refresh token refreshes authenticated session	Refresh valid authentication session	AuthenticationRefreshIT#shouldRefreshAValidAuthenticationSession	integration / high	Authentication Service and refresh token included in login response	Expected: Refreshed session is authenticated and has nonblank access token | Actual: Not executed in this run | Status: Not Run
TC-AUTH-REFRESH-002	Invalid refresh token is rejected	Reject invalid refresh token	AuthenticationRefreshIT#shouldRejectAnInvalidRefreshToken	security / high	Authentication Service	Expected: 400 Bad Request | Actual: Not executed in this run | Status: Not Run
TC-AUTH-LOGOUT-001	Logout clears local authentication session	Clear session after successful logout	AuthenticationLogoutIT#shouldClearTheSessionAfterSuccessfulLogout	integration / high	Authentication Service and refresh token included in login response	Expected: Session no longer reports authenticated or refreshable | Actual: Not executed in this run | Status: Not Run
TC-AUTH-ACCESS-001	Public login route accepts unauthenticated requests	Submit invalid credentials without a bearer token	AuthorizationIT#shouldAllowPublicAccessWithoutAuthentication	security / high	API Gateway and Auth Service	Expected: Login route returns 400 for invalid credentials rather than rejecting the request as unauthenticated | Actual: Not executed in this run | Status: Not Run
TC-AUTH-ACCESS-002	Protected endpoint rejects unauthenticated request	Reject protected access without authentication	AuthorizationIT#shouldRejectProtectedAccessWithoutAuthentication	security / high	Authentication Service	Expected: 401 Unauthorized | Actual: Not executed in this run | Status: Not Run
TC-AUTH-ROLE-001	Student can access Student-only resource	Allow Student resource access	AuthorizationIT#shouldAllowAStudentToUseStudentResources	security / high	Authentication Service and Student credentials	Expected: 200 with success status | Actual: Not executed in this run | Status: Not Run
TC-AUTH-ROLE-002	Student cannot access administrator resource	Reject Student administrator access	AuthorizationIT#shouldRejectAStudentFromAdminResources	security / high	Authentication Service and Student credentials	Expected: 403 Forbidden | Actual: Not executed in this run | Status: Not Run
TC-AUTH-ROLE-003	Teacher can access authenticated resource but not administrator resource	Verify Teacher authorization	AuthorizationIT#shouldAllowATeacherToUseAuthenticatedResourcesButNotAdminResources	security / high	Authentication Service and Teacher credentials	Expected: Protected resource returns 200 and administrator resource returns 403 | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-AUTH-ROLE-004	Administrator can access administrator resource	Allow Admin resource access	AuthorizationIT#shouldAllowAnAdminToUseAdminResources	security / high	Authentication Service and Admin credentials	Expected: 200 with administrator authorization message | Actual: Not executed in this run | Status: Not Run
TC-AUTH-USER-001	Authenticated user can read own profile	Read authenticated user profile	AuthorizationIT#shouldAllowAnAuthenticatedUserToReadTheirProfile	security / high	Authentication Service and Student credentials	Expected: 200 response contains profile data | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-18	Protected AI endpoints reject anonymous requests	Unauthenticated current-user conversation and chat requests	AiAuthorizationIT#unauthenticatedProtectedAiRequestsReturnUnauthorized	security / high	AI Service and API Gateway	Expected: All protected probes return 401 Unauthorized | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-04	Teacher can create an AI conversation	Create disposable conversation	AiConversationIT#teacherCanUseTheConversationLifecycle	integration / high	TEST_AI_WRITE_ENABLED wi Teacher credentials and AI Service	Expected: 201 response contains conversation ID | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-05	Conversation list includes created owner conversation	List own conversations	AiConversationIT#teacherCanUseTheConversationLifecycle	integration / high	TEST_AI_WRITE_ENABLED wi Teacher credentials and AI Service	Expected: 200 list contains created conversation ID | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-06	Conversation owner can read conversation and messages	Read created conversation and messages	AiConversationIT#teacherCanUseTheConversationLifecycle	integration / high	TEST_AI_WRITE_ENABLED wi Teacher credentials and AI Service	Expected: 200 responses contain conversation and message data | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-07	Conversation owner can update title	Update created conversation title	AiConversationIT#teacherCanUseTheConversationLifecycle	integration / high	TEST_AI_WRITE_ENABLED wi Teacher credentials and AI Service	Expected: 200 response contains updated title | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-08	Conversation owner can delete conversation	Delete created conversation	AiConversationIT#teacherCanUseTheConversationLifecycle	integration / high	TEST_AI_WRITE_ENABLED wi Teacher credentials and AI Service	Expected: Delete succeeds and subsequent read returns 404 | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-02	AI health endpoint reports service readiness	Read AI health status	AiHealthIT#healthEndpointReportsAiServiceUp	integration / high	AI Service	Expected: 200 response reports ai-service and UP | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-11	Invalid AI request fields are rejected	Submit malformed conversation chat generation and summary inputs	AiValidationIT#invalidAiRequestsReturnClientErrorsOrNotFoundResponses	validation / high	AI Service and Teacher credentials	Expected: Invalid requests return documented 400 responses | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-19	Unknown conversation routes return not found	Read update and delete unknown conversation	AiValidationIT#unknownConversationReadUpdateAndDeleteRoutesReturnNotFound	validation / medium	AI Service and Teacher credentials	Expected: Unknown conversation operations return 404 | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-028-06	Gateway health becomes ready within bounded deadline	Wait for approved staging Gateway health	StagingSmokeIT#gatewayHealthIsReadyWithinBoundedDeadline	smoke / high	Manually approved staging target and protected workflow	Expected: HTTP 2xx within readiness deadline | Actual: Not executed in this run | Status: Not Run
TC-TEST-028-07	JWT signing key set is available	Validate published RSA RS256 JWT public signing key	StagingSmokeIT#jwtSigningKeysArePublished	smoke / high	Manually approved staging target and protected workflow	Expected: HTTP 200 JSON contains RSA signing key fields kid n and e | Actual: Not executed in this run | Status: Not Run
TC-TEST-028-08	Auth Service health is ready	Verify Auth Service health endpoint	StagingSmokeIT#authServiceHealthIsReadyWithinBoundedDeadline	smoke / high	Manually approved staging target and protected workflow	Expected: HTTP 2xx within readiness deadline | Actual: Not executed in this run | Status: Not Run
TC-TEST-028-09	Anonymous protected Gateway probe is rejected	Probe protected endpoint without token	StagingSmokeIT#protectedProbeRejectsAnonymousRequest	security / high	Manually approved staging target and protected workflow	Expected: HTTP 401 or 403 without sending a token | Actual: Not executed in this run | Status: Not Run
TC-TEST-024-01	Protected resources reject missing invalid and malformed tokens	Probe token authentication boundaries	SecurityRegressionIT#protectedEndpointsRejectMissingInvalidAndMalformedTokens	security / high	API Gateway and User Service with test environment enabled	Expected: All protected requests return 401 | Actual: Not executed in this run | Status: Not Run
TC-TEST-024-02	Student and Teacher cannot cross administrator boundary	Check administrator endpoint denies non-admin roles	SecurityRegressionIT#studentAndTeacherTokensCannotCrossAdministratorBoundary	security / high	API Gateway and distinct Student and Teacher credentials	Expected: Both requests return 403 | Actual: Not executed in this run | Status: Not Run
TC-TEST-024-03	Authentication errors do not leak secrets	Inspect protected error response for sensitive values	SecurityRegressionIT#protectedErrorResponsesDoNotExposeSensitiveValues	security / high	API Gateway and User Service	Expected: 401 response omits password token bearer and stack-trace values | Actual: Not executed in this run | Status: Not Run
TC-TEST-024-04	Expired signed JWT is rejected	Reject an expired access token with a valid signature	SecurityRegressionIT#expiredSignedAccessTokenIsRejected	security / high	API Gateway User Service student credentials and local AUTH_JWT_PRIVATE_KE _BASE64	Expected: Protected request returns 401 for an expired signed token | Actual: Not executed in this run | Status: Not Run
TC-TRACE-001	SRS-NOT-EXPLICIT (Gateway health probe; does not measure uptime)	shouldReadGatewayHealthAfterReadinessPasses	ApiClientIT#shouldReadGatewayHealthAfterReadinessPasses	integration / high	See ApiClientIT test configuration	Expected: Gateway health endpoint returns HTTP 200 after required services pass readiness | Actual: Not executed in this run | Status: Not Run
TC-TRACE-002	SRS-NOT-EXPLICIT; NFR-SE003 (role-scoped attendance access)	adminCanReadStudentSummary	AttendanceAdminIT#adminCanReadStudentSummary	integration / high	See AttendanceAdminIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TRACE-003	SRS-NOT-EXPLICIT (admin attendance session lifecycle)	adminCanCreateAndRetrieveAttendanceSessionWhenWritesAreEnabled	AttendanceAdminIT#adminCanCreateAndRetrieveAttendanceSessionWhenWritesAreEnabled	integration / high	See AttendanceAdminIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-004	FR-S033; FR-T038; NFR-SE001	protectedAttendanceRequestsRejectMissingAuthentication	AttendanceAuthenticationIT#protectedAttendanceRequestsRejectMissingAuthentication	security / high	See AttendanceAuthenticationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-005	FR-S033; FR-T038; NFR-SE003	studentCannotManageSessionsOrReadAnotherStudentSummary	AttendanceAuthenticationIT#studentCannotManageSessionsOrReadAnotherStudentSummary	security / high	See AttendanceAuthenticationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-006	FR-S033; FR-T038 (attendance record validation)	rejectsInvalidSessionAndRecordRequests	AttendanceValidationIT#rejectsInvalidSessionAndRecordRequests	integration / high	See AttendanceValidationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-007	FR-S033; FR-T038 (attendance resource lookup)	unknownAttendanceResourcesReturnNotFound	AttendanceValidationIT#unknownAttendanceResourcesReturnNotFound	integration / high	See AttendanceValidationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-008	SRS-NOT-EXPLICIT; NFR-SE001 (calendar authentication)	shouldRejectUpcomingCalendarRequestsWithoutAuthentication	CalendarAuthenticationIT#shouldRejectUpcomingCalendarRequestsWithoutAuthentication	security / high	See CalendarAuthenticationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-009	SRS-NOT-EXPLICIT; NFR-SE001 (calendar authentication)	shouldRejectEventCreationWithoutAuthentication	CalendarAuthenticationIT#shouldRejectEventCreationWithoutAuthentication	security / high	See CalendarAuthenticationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TRACE-010	SRS-NOT-EXPLICIT (student calendar event query)	shouldAllowAnAuthenticatedStudentToQueryUpcomingEvents	CalendarAuthenticationIT#shouldAllowAnAuthenticatedStudentToQueryUpcomingEvents	security / high	See CalendarAuthenticationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-011	SRS-NOT-EXPLICIT (custom calendar event creation)	shouldAllowAdminToCreateAnEvent	CalendarEventWorkflowIT#shouldAllowAdminToCreateAnEvent	integration / high	See CalendarEventWorkflowIT test configuration	Expected: Admin creates a custom event and receives HTTP 201; the test cleans it up | Actual: Not executed in this run | Status: Not Run
TC-TRACE-012	SRS-NOT-EXPLICIT (custom calendar event cancellation)	shouldAllowTeacherToCancelAnEvent	CalendarEventWorkflowIT#shouldAllowTeacherToCancelAnEvent	integration / high	See CalendarEventWorkflowIT test configuration	Expected: Teacher cancels a created custom event successfully | Actual: Not executed in this run | Status: Not Run
TC-TRACE-013	NFR-SE003 (student denied teacher event creation)	shouldRejectStudentEventCreation	CalendarEventWorkflowIT#shouldRejectStudentEventCreation	integration / high	See CalendarEventWorkflowIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-014	NFR-SE003 (student denied event updates)	shouldRejectStudentEventUpdate	CalendarEventWorkflowIT#shouldRejectStudentEventUpdate	integration / high	See CalendarEventWorkflowIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-015	NFR-SE003 (student denied event cancellation)	shouldRejectStudentEventCancellation	CalendarEventWorkflowIT#shouldRejectStudentEventCancellation	integration / high	See CalendarEventWorkflowIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-016	SRS-NOT-EXPLICIT (calendar input validation)	shouldRejectInvalidEventRequest	CalendarEventWorkflowIT#shouldRejectInvalidEventRequest	integration / high	See CalendarEventWorkflowIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-017	SRS-NOT-EXPLICIT (calendar date-range validation)	shouldRejectEventWithInvalidDateRange	CalendarEventWorkflowIT#shouldRejectEventWithInvalidDateRange	integration / high	See CalendarEventWorkflowIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TRACE-018	SRS-NOT-EXPLICIT; NFR-SE001 (venue authentication)	shouldRequireAuthenticationToListVenues	CalendarVenueAuthorizationIT#shouldRequireAuthenticationToListVenues	security / high	See CalendarVenueAuthorizationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-019	FR-S026; FR-S027; FR-T030; NFR-SE001	protectedCommunicationRequestsRejectMissingAuthentication	CommunicationAuthenticationIT#protectedCommunicationRequestsRejectMissingAuthentication	security / high	See CommunicationAuthenticationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-020	FR-S026; BR-008; NFR-SE003	studentCannotManageGroups	CommunicationAuthenticationIT#studentCannotManageGroups	security / high	See CommunicationAuthenticationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-021	FR-T030; BR-008; NFR-SE003	teacherCanCreateInspectManageAndArchiveAGroup	CommunicationGroupWorkflowIT#teacherCanCreateInspectManageAndArchiveAGroup	integration / high	See CommunicationGroupWorkflowIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-022	FR-S009; FR-T008; NFR-SE003	courseMembershipUsesTheCourseEndpoint	CommunicationGroupWorkflowIT#courseMembershipUsesTheCourseEndpoint	integration / high	See CommunicationGroupWorkflowIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-023	FR-S027; FR-T030; BR-008	teacherCanSendPageUpdateAndDeleteMessages	CommunicationMessageWorkflowIT#teacherCanSendPageUpdateAndDeleteMessages	integration / high	See CommunicationMessageWorkflowIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-024	SRS-NOT-EXPLICIT (communication preferences/settings)	authenticatedUserCanReadAndUpdateOwnSettings	CommunicationSettingsIT#authenticatedUserCanReadAndUpdateOwnSettings	integration / high	See CommunicationSettingsIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TRACE-025	FR-S026; FR-S027; FR-T030 (validation of communication requests)	invalidRequestsReturnClientErrors	CommunicationValidationIT#invalidRequestsReturnClientErrors	integration / high	See CommunicationValidationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-026	FR-S026; FR-S027; FR-T030 (unknown communication-resource handling)	unknownCommunicationIdsNeverProduceServerErrors	CommunicationValidationIT#unknownCommunicationIdsNeverProduceServerErrors	integration / high	See CommunicationValidationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-027	FR-S007; FR-S008; FR-S009; BR-001	studentCourseAndEnrollmentIdentifiersRemainConsistentThroughGateway	CourseEnrollmentJourneyIT#studentCourseAndEnrollmentIdentifiersRemainConsistentThroughGateway	e2e / high	See CourseEnrollmentJourneyIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-028	FR-A002; BR-005; NFR-SE003	gatewayPreservesRoleRestrictionsAcrossUserAndEnrollmentServices	CrossServiceAuthorizationIT#gatewayPreservesRoleRestrictionsAcrossUserAndEnrollmentServices	security / high	See CrossServiceAuthorizationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-029	FR-S002; FR-T002; NFR-SE001	authenticatedUserTravelsThroughGatewayToUserService	GatewaySmokeIT#authenticatedUserTravelsThroughGatewayToUserService	e2e / high	See GatewaySmokeIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-030	FR-S002; FR-T002; NFR-SE001	missingGatewayTokenIsRejected	GatewaySmokeIT#missingGatewayTokenIsRejected	e2e / high	See GatewaySmokeIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-031	SRS-NOT-EXPLICIT (provider journeys; payment is not specified and the notification journey is not exercised by this fixture guard)	paymentNotificationAndLearningProviderJourneysRequireExplicitFixtures	ProviderJourneyIT#paymentNotificationAndLearningProviderJourneysRequireExplicitFixtures	e2e / high	See ProviderJourneyIT test configuration	Expected: The test skips unless approved provider fixtures are enabled; it does not assert provider-journey behavior | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TRACE-032	FR-S009; BR-001; NFR-SE003	shouldRejectStudentFromAnotherStudentsCourseList	EnrollmentAuthenticationIT#shouldRejectStudentFromAnotherStudentsCourseList	security / high	See EnrollmentAuthenticationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-033	SRS-NOT-EXPLICIT (test-environment readiness prerequisite; does not measure uptime or recovery)	shouldKeepRequiredBackendServicesReady	BackendReadinessIT#shouldKeepRequiredBackendServicesReady	integration / high	See BackendReadinessIT test configuration	Expected: Required backend services must pass readiness before this test suite proceeds | Actual: Not executed in this run | Status: Not Run
TC-TRACE-034	SRS-NOT-EXPLICIT (exam-specific detail route); NFR-SE003	nonStudentsCannotUseStudentExamDetailRoute	ExamAuthorizationIT#nonStudentsCannotUseStudentExamDetailRoute	security / high	See ExamAuthorizationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-035	SRS-NOT-EXPLICIT (exam-specific student result visibility is not stated by FR-S024)	enrolledStudentCanRetrieveOwnExamResults	ExamStudentResultIT#enrolledStudentCanRetrieveOwnExamResults	integration / high	See ExamStudentResultIT test configuration	Expected: An enrolled student retrieves their own exam results; cross-student visibility is denied | Actual: Not executed in this run | Status: Not Run
TC-TRACE-036	SRS-NOT-EXPLICIT (exam/marks API validation)	invalidExamAndMarksPayloadsReturnClientErrors	ExamValidationIT#invalidExamAndMarksPayloadsReturnClientErrors	integration / high	See ExamValidationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-037	SRS-NOT-EXPLICIT (unknown exam resource handling)	unknownExamIdentifiersNeverProduceServerErrors	ExamValidationIT#unknownExamIdentifiersNeverProduceServerErrors	integration / high	See ExamValidationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-038	FR-S013; FR-AI002; NFR-SE003	shouldRejectStudentAndUnauthenticatedAiSourceReads	MaterialAiSourcesIT#shouldRejectStudentAndUnauthenticatedAiSourceReads	integration / high	See MaterialAiSourcesIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-039	FR-S013; FR-T013; NFR-SE001	rejectsProtectedMaterialRequestsWithoutAuthentication	MaterialAuthenticationIT#rejectsProtectedMaterialRequestsWithoutAuthentication	security / high	See MaterialAuthenticationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TRACE-040	FR-S013; FR-T013; NFR-SE003; BR-005	studentCannotManageMaterialsOrRunMaintenance	MaterialAuthenticationIT#studentCannotManageMaterialsOrRunMaintenance	security / high	See MaterialAuthenticationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-041	FR-T013; NFR-SE003	teacherCannotRunMaintenance	MaterialAuthenticationIT#teacherCannotRunMaintenance	security / high	See MaterialAuthenticationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-042	FR-S021; FR-T013 (submission attachment/material lifecycle)	enrolledStudentCanUploadSubmissionAttachmentMaterial	MaterialLifecycleIT#enrolledStudentCanUploadSubmissionAttachmentMaterial	integration / high	See MaterialLifecycleIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-043	FR-T013; FR-S021 (upload/update validation)	rejectsInvalidMaterialUploadsAndUpdates	MaterialLifecycleIT#rejectsInvalidMaterialUploadsAndUpdates	integration / high	See MaterialLifecycleIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-044	FR-T013; FR-S021 (unknown material operation handling)	unknownMaterialOperationsAreNotReportedAsServerErrors	MaterialLifecycleIT#unknownMaterialOperationsAreNotReportedAsServerErrors	integration / high	See MaterialLifecycleIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-045	SRS-NOT-EXPLICIT (material storage audit/cleanup); NFR-SE003 (admin-only access)	adminCanAuditAndRunNonDestructiveCleanup	MaterialLifecycleIT#adminCanAuditAndRunNonDestructiveCleanup	integration / high	See MaterialLifecycleIT test configuration	Expected: Admin audit and cleanup dry-run routes return success and confirm dryRun=true | Actual: Not executed in this run | Status: Not Run
TC-TRACE-046	FR-S030; FR-T035; BR-009; NFR-SE001	unauthenticatedNotificationRequestsReturnUnauthorized	NotificationAuthorizationIT#unauthenticatedNotificationRequestsReturnUnauthorized	security / high	See NotificationAuthorizationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TRACE-047	FR-S030; BR-009; NFR-SE003	notificationAccessIsAvailableToAuthenticatedUsersOnly	NotificationAuthorizationIT#notificationAccessIsAvailableToAuthenticatedUsersOnly	security / high	See NotificationAuthorizationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-048	FR-S030; BR-009; NFR-SE003 (notification ownership)	twoAuthenticatedUsersReceiveOnlyTheirOwnNotificationCollections	NotificationOwnershipIT#twoAuthenticatedUsersReceiveOnlyTheirOwnNotificationCollections	integration / high	See NotificationOwnershipIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-049	SRS-NOT-EXPLICIT (payment authorization); NFR-SE001	unauthenticatedPaymentRequestsReturnUnauthorized	PaymentAuthorizationIT#unauthenticatedPaymentRequestsReturnUnauthorized	security / high	See PaymentAuthorizationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-050	SRS-NOT-EXPLICIT (payment roles); NFR-SE003	studentCanUseOwnEndpointsButCannotUseAdministratorEndpoints	PaymentAuthorizationIT#studentCanUseOwnEndpointsButCannotUseAdministratorEndpoints	security / high	See PaymentAuthorizationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-051	SRS-NOT-EXPLICIT (payment roles); NFR-SE003	teacherCannotUseStudentOrAdministratorPaymentEndpoints	PaymentAuthorizationIT#teacherCannotUseStudentOrAdministratorPaymentEndpoints	security / high	See PaymentAuthorizationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-052	SRS-NOT-EXPLICIT (payment webhook signature validation); NFR-SE004	webhookWithoutSignatureIsRejected	PaymentValidationIT#webhookWithoutSignatureIsRejected	integration / high	See PaymentValidationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-053	FR-S022; FR-T024; NFR-SE003	teacherCannotStartOrSubmitStudentAttempt	QuizAuthenticationIT#teacherCannotStartOrSubmitStudentAttempt	security / high	See QuizAuthenticationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TRACE-054	FR-T024; FR-T026; BR-007	teacherManagesQuestionsAndPublishesQuiz	QuizManagementIT#teacherManagesQuestionsAndPublishesQuiz	integration / high	See QuizManagementIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-055	FR-T024 (quiz/question resource lookup)	unknownQuizAndQuestionResourcesReturnNotFound	QuizManagementIT#unknownQuizAndQuestionResourcesReturnNotFound	integration / high	See QuizManagementIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-056	FR-T024 (quiz request validation)	rejectsInvalidQuizRequests	QuizValidationIT#rejectsInvalidQuizRequests	integration / high	See QuizValidationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-057	FR-T024 (question/answer validation)	rejectsInvalidQuestionAndAnswerRequests	QuizValidationIT#rejectsInvalidQuestionAndAnswerRequests	integration / high	See QuizValidationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-058	SRS-NOT-EXPLICIT (fault-injection guard; downstream recovery behavior is not exercised)	controlledDownstreamFailureRequiresApprovedFaultInjection	DownstreamFailureIT#controlledDownstreamFailureRequiresApprovedFaultInjection	integration / high	See DownstreamFailureIT test configuration	Expected: Test requires configured fault injection; the current guard asserts no downstream failure or recovery behavior | Actual: Not executed in this run | Status: Not Run
TC-TRACE-059	SRS-NOT-EXPLICIT; NFR-SE001 (controlled provider failure)	controlledProviderFailureReturnsSanitizedServerError	AiProviderFailureIT#controlledProviderFailureReturnsSanitizedServerError	integration / high	See AiProviderFailureIT test configuration	Expected: Controlled provider failure returns 5xx without exposing its sentinel when provider test mode is enabled | Actual: Not executed in this run | Status: Not Run
TC-TRACE-060	FR-S001; FR-S002; FR-T001; FR-T002; NFR-SE001	publicAndProtectedAuthenticationProbesFollowTheirSecurityRules	AuthenticationIT#publicAndProtectedAuthenticationProbesFollowTheirSecurityRules	security / high	See AuthenticationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TRACE-061	FR-S002; FR-T002; FR-A002; NFR-SE003	roleProbesMatchStudentAndAdministratorTokens	AuthenticationIT#roleProbesMatchStudentAndAdministratorTokens	security / high	See AuthenticationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-062	FR-S001; FR-S002; NFR-SE001; NFR-SE004	invalidAuthenticationPayloadsAreRejectedWithoutSensitiveOutput	AuthenticationIT#invalidAuthenticationPayloadsAreRejectedWithoutSensitiveOutput	security / high	See AuthenticationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-063	FR-A001; FR-A002; FR-A003; BR-005; NFR-SE003	studentsAndTeachersCannotPerformAdministratorOperations	UserAuthorizationIT#studentsAndTeachersCannotPerformAdministratorOperations	security / high	See UserAuthorizationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-064	FR-A001; FR-A002; FR-A003; BR-005; NFR-SE001	protectedUserEndpointsRejectMissingAuthentication	UserAuthorizationIT#protectedUserEndpointsRejectMissingAuthentication	security / high	See UserAuthorizationIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TRACE-065	FR-S003; FR-T003; NFR-SE002	authenticatedUserCanRetrieveOwnProfileWithoutSensitiveFields	UserProfileIT#authenticatedUserCanRetrieveOwnProfileWithoutSensitiveFields	integration / high	See UserProfileIT test configuration	Expected: The scenario assertions pass; execution status is reported from JUnit, never inferred from this mapping | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-32	SRS-NOT-EXPLICIT; NFR-SE003	Delete a newly created open attendance session	AttendanceSessionWorkflowIT#teacherCanDeleteAnOpenAttendanceSession	integration / medium	TEST_ATTENDANCE_WRITE_LED with TEST_ATTENDANCE_COURSE and TEST_ATTENDANCE_STUT_ID; non-production fixtures	Expected: Teacher receives success deleting the newly created session and subsequent read returns 404 | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-35	FR-S026; FR-S027; FR-T030; NFR-SE003	Update a communication group and verify persisted fields	CommunicationGroupWorkflowIT#teacherCanCreateInspectManageAndArchiveAGroup	integration / high	TEST_COMMUNICATION_WRINABLED and teacher-owned course fixture	Expected: Update returns 200 and persisted group name and description match the request | Actual: Not executed in this run | Status: Not Run
TC-COURSE-045	SRS-NOT-EXPLICIT	Read the course service root greeting	CourseServiceRootIT#serviceRootReturnsCourseServiceGreeting	integration / low	Course Service directly reachable	Expected: Service root returns the expected greeting | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-COURSE-046	SRS-NOT-EXPLICIT	List public marketplace materials	MarketplaceProductIT#anonymousUserCanListMarketplaceProducts	integration / medium	Course Service and API Gateway	Expected: Anonymous request returns a successful paginated marketplace response | Actual: Not executed in this run | Status: Not Run
TC-COURSE-047	SRS-NOT-EXPLICIT	Read a published marketplace product	MarketplaceProductIT#anonymousUserCanReadAConfiguredPublishedProduct	integration / medium	TEST_MARKETPLACE_PRODUD points to a published non-production product	Expected: Anonymous request returns the selected product details | Actual: Not executed in this run | Status: Not Run
TC-COURSE-048	NFR-SE003	Administrator lists marketplace products	MarketplaceProductIT#administratorCanListMarketplaceProducts	security / medium	Course Service and admin credentials	Expected: Administrator receives a paginated product list | Actual: Not executed in this run | Status: Not Run
TC-COURSE-049	SRS-NOT-EXPLICIT	Create update publish and archive a marketplace product	MarketplaceProductIT#administratorCanCreateUpdatePublishAndArchiveMarketplaceProduct	integration / high	TEST_MARKETPLACE_WRITEBLED=true; non-production Course and Material services with upload storage available	Expected: Admin creates a product from synthetic uploaded material; update and status transitions succeed; product is archived and material cleanup is attempted | Actual: Not executed in this run | Status: Not Run
TC-COURSE-050	Course lifecycle and teacher ownership	Teacher creates reads and updates an owned course	CourseCrudIT#shouldAllowTeacherToCreateReadAndUpdateOwnedCourse	integration / high	TEST_COURSE_GRADE_ID plu teacher/admin credentials	Expected: Teacher-owned synthetic course is created read updated and deleted by admin cleanup | Actual: Not executed in this run | Status: Not Run
TC-COURSE-051	Course authorization and ownership	Reject teacher reads and updates on another teacher's course	CourseCrudIT#shouldDenyTeacherFromReadingOrUpdatingAnotherTeachersCourse	security / high	TEST_TEACHER_PORTAL_UND_COURSE_ID plus teacher/admin credentials	Expected: GET and PUT return 403 and admin confirms the foreign course is unchanged | Actual: Not executed in this run | Status: Not Run
TC-TEST-010-25	Duplicate enrollment prevention	Reject self-enrollment into an already active course	EnrollmentAuthenticationIT#shouldSupportStudentSelfEnrollmentWhenWriteFixturesAreExplicitlyEnabled	integration / high	TEST_ENROLLMENT_WRITE_LED and verified free TEST _ENROLLMENT_FREE_COURS plus Course and Communication services	Expected: First enrollment succeeds; duplicate returns 409; original membership remains and the created record is deleted | Actual: Not executed in this run | Status: Not Run
TC-TEST-013-38	SRS-NOT-EXPLICIT; NFR-SE003	Admin uploads material for marketplace listing	MarketplaceProductIT#administratorCanCreateUpdatePublishAndArchiveMarketplaceProduct	integration / high	TEST_MARKETPLACE_WRITEBLED=true; approved non-production Material Service and upload storage	Expected: Admin multipart upload returns a material ID used for a marketplace product and cleanup is attempted | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-ANALYTICS-001	SRS-NOT-EXPLICIT; NFR-SE003	Read aggregated admin analytics overview	AdminAnalyticsIT#administratorCanReadAdminAnalyticsOverview	integration / high	API Gateway and authenticated administrator; analytics dependencies available	Expected: Admin receives overview fields and aggregation partial-status indicator | Actual: Not executed in this run | Status: Not Run
TC-ANALYTICS-002	SRS-NOT-EXPLICIT; NFR-SE003	Restrict admin overview to admin roles	AdminAnalyticsIT#nonAdminCannotReadAdminAnalyticsOverview	security / high	API Gateway and student credentials plus anonymous request	Expected: Student receives 403 and unauthenticated request receives 401 | Actual: Not executed in this run | Status: Not Run
TC-TEST-013-39	SRS-NOT-EXPLICIT; NFR-SE001	Profile-image upload rejects a missing file	MaterialAuthenticationIT#profileImageUploadRequiresAnImagePart	validation / medium	Material Service and student credentials	Expected: Multipart upload without the required file part returns 400 and creates no object | Actual: Not executed in this run | Status: Not Run
TC-TEST-013-40	NFR-SE003	Student cannot read another users profile image	MaterialAuthenticationIT#studentCannotDownloadAnotherUsersProfileImage	security / high	Material Service and student credentials	Expected: Foreign profile-image key is rejected with 403 before storage access | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-29	SRS-NOT-EXPLICIT; NFR-SE003	Create and update an exam through multipart instruction routes	ExamLifecycleIT#teacherCanCreateAndUpdateExamUsingMultipartInstructionRoutes	integration / high	TEST_EXAM_WRITE_ENABLE and approved non-production course fixture	Expected: Multipart JSON exam part is accepted for create and update; resulting exam is cancelled during cleanup | Actual: Not executed in this run | Status: Not Run
TC-ANALYTICS-003	SRS-NOT-EXPLICIT; NFR-SE003	Read internal service analytics over service-discovery Gateway routes	InternalAnalyticsIT#administratorCanReadInternalAnalyticsOverviewsThroughTheirGatewayRoutes	integration / high	API Gateway and all analytics source services with administrator credentials	Expected: Eleven internal aggregate endpoints return successful responses through explicit or service-discovery Gateway ingress paths | Actual: Not executed in this run | Status: Not Run
TC-ANALYTICS-005	SRS-NOT-EXPLICIT; NFR-SE003	Reject unauthenticated requests to internal service routes	InternalRouteAuthorizationIT#unauthenticatedRequestsAreRejectedAcrossInternalRoutes	integration / high	TEST_INTERNAL_ROUTE_AUTNABLED=true; API Gateway and source services available; no credentials or mutation fixtures required	Expected: All 26 previously unmatched internal routes reject requests without required credentials with HTTP 401 or 403 | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-36	NFR-SE001; NFR-SE003	Reject unauthenticated STOMP connections through service and Gateway	CommunicationWebSocketIT#unauthenticatedStompConnectIsRejectedByServiceAndGateway	integration / high	TEST_COMMUNICATION_WEBET_ENABLED=true; Communication Service and API Gateway	Expected: STOMP CONNECT without Authorization does not receive CONNECTED and is rejected by ERROR or connection close | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-016-37	FR-T030; NFR-SE003	Reject group SEND and SUBSCRIBE when user is not an active member	CommunicationWebSocketIT#teacherCannotSendOrSubscribeToGroupWithoutMembership	integration / high	TEST_COMMUNICATION_WEBSOCKET_ENABLED=true; teacher credentials; random non-existent group UUID; Communication Service and API Gateway	Expected: Both membership-protected destinations reject before any group message can be persisted or received | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-38	SRS-NOT-EXPLICIT (WebSocket test echo route); NFR-SE001	Verify authenticated STOMP message delivery directly and through Gateway	CommunicationWebSocketIT#authenticatedTeacherCanSendAndReceiveTestBroadcastDirectlyAndThroughGateway	integration / medium	TEST_COMMUNICATION_WEBSOCKET_ENABLED=true; teacher credentials; Communication Service and API Gateway	Expected: An authenticated test message sent to /app/test is received on /topic/test through both ingress paths | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-39	FR-S026; FR-S027; FR-T030; NFR-SE003	Send and receive a persisted group message over STOMP and clean it up	CommunicationWebSocketIT#teacherCanSendPersistedGroupMessageOverGatewayAndCleanItUp	integration / high	TEST_COMMUNICATION_WEBSOCKET_ENABLED=true; TEST_COMMUNICATION_WRITE_ENABLED=true; TEST_COMMUNICATION_GROUP_ID is a non-production group where the teacher is an active member; API Gateway and Communication Service	Expected: Gateway STOMP message is broadcast with persisted message ID and the test deletes that message for everyone | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-30	NFR-SE001; NFR-SE003	Reject unauthenticated AI STOMP connections through service and Gateway	AiWebSocketIT#unauthenticatedAiStompConnectionsAreRejectedByServiceAndGateway	integration / high	TEST_AI_WEBSOCKET_ENABLED=true; AI Service and API Gateway	Expected: STOMP CONNECT without Authorization is rejected through both ingress paths | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-31	NFR-SE003	Reject student role at AI STOMP CONNECT	AiWebSocketIT#studentCannotAuthenticateToAiStompEndpoint	integration / high	TEST_AI_WEBSOCKET_ENABLED=true; student credential; AI Service	Expected: Student JWT does not receive STOMP CONNECTED because only teacher/admin roles are authorized | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-32	NFR-SE003	Reject unapproved AI STOMP destinations	AiWebSocketIT#teacherCannotUseUnapprovedAiStompDestinations	integration / high	TEST_AI_WEBSOCKET_ENABLED=true; teacher credential; AI Service and API Gateway	Expected: Unapproved SUBSCRIBE and SEND destinations are rejected by the STOMP interceptor | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-33	NFR-SE001	Deliver safe unknown-conversation error over AI STOMP user queue	AiWebSocketIT#teacherReceivesSafeErrorOnUserQueueForUnknownConversationWithoutProviderCall	integration / medium	TEST_AI_WEBSOCKET_ENABLED=true; teacher credential; AI Service and API Gateway; synthetic random conversation UUID	Expected: The user queue receives PROCESSING then a safe AI conversation not found error; provider is not invoked | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-E2E-001	SRS-NOT-EXPLICIT; NFR-SE001; NFR-SE003	Complete student teacher and admin platform journey through the API Gateway	CompleteWorkflowIT#studentTeacherAndAdminCompleteCrossServiceWorkflowsThroughGateway	e2e / high	TEST_E2E_ENABLED=true; TEST_E2E_WRITE_ENABLED=true TEST_E2E_COURSE_GRADE_ student teacher and admin credentials; Gateway plus Course Enrollment Assignment Material Payment Notification services	Expected: Teacher creates a course lesson assignment and quiz and uploads material; student enrolls accesses lesson downloads material submits assignment and quiz; teacher grades assignment and reads quiz result; admin reads user course payment and notification data; temporary course/enrollment/module/lesson/material/quiz are deleted; assignment is archived and its graded submission remains because no delete API exists | Actual: Not executed in this run | Status: Not Run
TC-UNIT-E2E-002	Repository test harness behavior; SRS requirement not applicable	Keep write journeys disabled by default	EndToEndClientTest#writeJourneysAreDisabledUnlessExplicitlyEnabled	unit / low	None; deterministic configuration test	Expected: Write and provider-backed E2E workflows remain disabled when their opt-in environment variables are absent | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-E2E-003	Repository test harness behavior; SRS requirement not applicable	Load explicit write journey configuration	EndToEndClientTest#loadsExplicitWriteJourneyConfiguration	unit / low	None; deterministic configuration test	Expected: Explicit E2E write opt-in and grade fixture settings are loaded from configuration | Actual: Passed; see JUnit report | Status: Passed
TC-TRACE-066	FR-T024; NFR-SE003 (quiz owner authorization)	anotherTeacherCannotReadOrManageQuizOwnedByTeacher	QuizManagementIT#anotherTeacherCannotReadOrManaQuizOwnedByTeacher	security / high	Distinct secondary teacher; synthetic quiz and approved course fixture	Expected: Another teacher receives 403 for quiz read, update, and delete; owner deletes the synthetic quiz | Actual: Not executed in this run | Status: Not Run
TC-TRACE-067	FR-S033; FR-T038; NFR-SE003 (attendance course ownership)	teacherCannotListSessionsForAnotherTeachersCourse	AttendanceAuthenticationIT#teacherCannotListSessionsForAnotherTeachersCourse	security / high	Distinct secondary teacher and course assigned only to primary teacher	Expected: Secondary teacher receives 403 when listing sessions for the unassigned course | Actual: Not executed in this run | Status: Not Run
TC-TRACE-068	NFR-SE003 (calendar course ownership)	anotherTeacherCannotUpdateOrCancelAnUnassignedCourseEvent	CalendarEventWorkflowIT#anotherTeacherCannotUpdateOrCancelAnUnassignedCourseEvent	security / high	Distinct secondary teacher; course assigned only to primary teacher	Expected: Secondary teacher receives 404 for event read and 403 for update/cancellation; owner cleans up the event | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-016-04	NFR-SE001	Reject unauthenticated group operations	CommunicationAuthenticationIT#protectedCommunicationRequestsRejectMissingAuthentication	security / high	Communication Service	Expected: Protected group operations return 401 | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-05	NFR-SE001	Reject unauthenticated message operations	CommunicationAuthenticationIT#protectedCommunicationRequestsRejectMissingAuthentication	security / high	Communication Service	Expected: Protected message operations return 401 | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-06	NFR-SE001	Reject unauthenticated settings operations	CommunicationAuthenticationIT#protectedCommunicationRequestsRejectMissingAuthentication	security / high	Communication Service	Expected: Settings operations return 401 | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-07	FR-S026; FR-S027; FR-T030	Create and manage a teacher group	CommunicationGroupWorkflowIT#teacherCanCreateInspectManageAndArchiveAGroup	integration / high	TEST_COMMUNICATION_WRINABLED and teacher-owned course fixture	Expected: Teacher creates inspects updates and archives/unarchives a group | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-08	NFR-SE003	Student cannot create or manage groups	CommunicationAuthenticationIT#studentCannotManageGroups	security / high	Communication Service and student credentials	Expected: Student group-management requests return 403 | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-09	NFR-SE001	Reject invalid group and message requests	CommunicationValidationIT#invalidRequestsReturnClientErrors	validation / high	Communication Service and teacher credentials	Expected: Invalid group and message payloads return 400 | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-10	FR-S026; FR-S027	List current user's groups	CommunicationGroupWorkflowIT#teacherCanCreateInspectManageAndArchiveAGroup	integration / medium	TEST_COMMUNICATION_WRINABLED and teacher-owned course fixture	Expected: Teacher receives the created group in the list response | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-11	FR-S026; FR-S027	Retrieve a created group	CommunicationGroupWorkflowIT#teacherCanCreateInspectManageAndArchiveAGroup	integration / medium	TEST_COMMUNICATION_WRINABLED and teacher-owned course fixture	Expected: Teacher retrieves the created group | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-12	FR-S026; FR-S027	List active group members	CommunicationGroupWorkflowIT#teacherCanCreateInspectManageAndArchiveAGroup	integration / medium	TEST_COMMUNICATION_WRINABLED and teacher-owned course fixture	Expected: Teacher receives group member list | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-13	FR-S026; FR-S027	Add a student to a group	CommunicationGroupWorkflowIT#teacherCanCreateInspectManageAndArchiveAGroup	integration / medium	TEST_COMMUNICATION_WRINABLED with course and student fixtures	Expected: Teacher successfully adds the configured student | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-14	FR-S026; FR-S027	Use the course endpoint for course membership	CommunicationGroupWorkflowIT#courseMembershipUsesTheCourseEndpoint	integration / medium	TEST_COMMUNICATION_WRINABLED and course/student fixtures	Expected: Student is added through course membership API | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-016-15	FR-S026; FR-S027	Archive a group	CommunicationGroupWorkflowIT#teacherCanCreateInspectManageAndArchiveAGroup	integration / medium	TEST_COMMUNICATION_WRINABLED and teacher-owned course fixture	Expected: Teacher archives the created group | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-16	FR-S026; FR-S027	Unarchive a group	CommunicationGroupWorkflowIT#teacherCanCreateInspectManageAndArchiveAGroup	integration / medium	TEST_COMMUNICATION_WRINABLED and teacher-owned course fixture	Expected: Teacher unarchives the created group | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-17	NFR-SE001	Unknown group identifiers return not found	CommunicationValidationIT#unknownCommunicationIdsNeverProduceServerErrors	validation / medium	Communication Service and teacher credentials	Expected: Unknown group reads return 404 | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-18	FR-S026; FR-S027	Send a message to a group	CommunicationMessageWorkflowIT#teacherCanSendPagUpdateAndDeleteMessages	integration / high	TEST_COMMUNICATION_WRINABLED and active teacher group fixture	Expected: Teacher creates a message and receives its identifier | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-19	NFR-SE001	Reject invalid message content	CommunicationValidationIT#invalidRequestsReturnClientErrors	validation / high	Communication Service and teacher credentials	Expected: Blank or missing message content returns 400 | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-20	FR-S026; FR-S027	Read paginated message history	CommunicationMessageWorkflowIT#teacherCanSendPagUpdateAndDeleteMessages	integration / medium	TEST_COMMUNICATION_WRINABLED and active teacher group fixture	Expected: History response includes page size and metadata | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-21	NFR-SE001	Reject invalid local message pagination	CommunicationClientTest#clientRejectsInvalidIdentifiersAndPagination	unit / medium	None	Expected: Invalid page and size values are rejected before HTTP request | Actual: Passed; see JUnit report | Status: Passed
TC-TEST-016-22	FR-S026; FR-S027	Update a sent message	CommunicationMessageWorkflowIT#teacherCanSendPagUpdateAndDeleteMessages	integration / medium	TEST_COMMUNICATION_WRINABLED and active teacher group fixture	Expected: Sender updates message content successfully | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-24	FR-S026; FR-S027	Delete a sent message for everyone	CommunicationMessageWorkflowIT#teacherCanSendPagUpdateAndDeleteMessages	integration / medium	TEST_COMMUNICATION_WRINABLED and active teacher group fixture	Expected: Sender deletes the synthetic message for everyone | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-25	NFR-SE001	Unknown message and group identifiers return not found	CommunicationValidationIT#unknownCommunicationIdsNeverProduceServerErrors	validation / medium	Communication Service and teacher credentials	Expected: Unknown group/message reads return 404 | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-29	NFR-SE003	Restrict group management by role	CommunicationAuthenticationIT#studentCannotManageGroups	security / high	Communication Service and student credentials	Expected: Student receives 403 for group management | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-016-40	NFR-SE001	Verify uploaded communication attachment bytes	CommunicationMessageWorkflowIT#shouldUploadAttachDownloadAndCleanUpMessageAttachment	integration / high	TEST_COMMUNICATION_WRINABLED and TEST_MATERIALOWNLOAD_BYTES_ENABLED approved storage host	Expected: Downloaded object bytes match the synthetic message attachment upload | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-02	SRS-NOT-EXPLICIT; NFR-SE003	Create a valid exam	ExamLifecycleIT#teacherCanRunTheExamLifecycleAndReadResults	integration / high	TEST_EXAM_WRITE_ENABLE and approved course/student fixtures	Expected: Teacher creates an exam and receives its identifier | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-03	SRS-NOT-EXPLICIT; NFR-SE003	Retrieve the created exam	ExamLifecycleIT#teacherCanRunTheExamLifecycleAndReadResults	integration / medium	TEST_EXAM_WRITE_ENABLE and approved course/student fixtures	Expected: Teacher retrieves the created exam | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-04	SRS-NOT-EXPLICIT; NFR-SE003	List exams for a course	ExamLifecycleIT#teacherCanRunTheExamLifecycleAndReadResults	integration / medium	TEST_EXAM_WRITE_ENABLE and approved course/student fixtures	Expected: Course exam list returns the created exam | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-05	SRS-NOT-EXPLICIT; NFR-SE003	Update editable exam fields	ExamLifecycleIT#teacherCanRunTheExamLifecycleAndReadResults	integration / high	TEST_EXAM_WRITE_ENABLE and approved course/student fixtures	Expected: Exam update succeeds while preserving the original course association | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-06	NFR-SE003	Prevent update from changing owning course	ExamLifecycleIT#teacherCanRunTheExamLifecycleAndReadResults	integration / high	TEST_EXAM_WRITE_ENABLE and approved course/student fixtures	Expected: Attempted courseId replacement is ignored and original courseId remains | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-07	SRS-NOT-EXPLICIT; NFR-SE003	Schedule an eligible exam	ExamLifecycleIT#teacherCanRunTheExamLifecycleAndReadResults	integration / high	TEST_EXAM_WRITE_ENABLE and approved course/student fixtures	Expected: Exam transitions to SCHEDULED | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-08	SRS-NOT-EXPLICIT; NFR-SE003	Start exam marking	ExamLifecycleIT#teacherCanRunTheExamLifecycleAndReadResults	integration / high	TEST_EXAM_WRITE_ENABLE and approved course/student fixtures	Expected: Scheduled exam transitions to MARKING | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-09	SRS-NOT-EXPLICIT; NFR-SE003	Enter marks for an enrolled student	ExamLifecycleIT#teacherCanRunTheExamLifecycleAndReadResults	integration / high	TEST_EXAM_WRITE_ENABLE and enrolled student fixture	Expected: Teacher marks request succeeds for the enrolled student | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-10	SRS-NOT-EXPLICIT; NFR-SE003	Complete an exam after marking	ExamLifecycleIT#teacherCanRunTheExamLifecycleAndReadResults	integration / high	TEST_EXAM_WRITE_ENABLE and enrolled student fixture	Expected: Exam transitions to COMPLETED | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-017-11	SRS-NOT-EXPLICIT; NFR-SE003	Read the exam result sheet	ExamLifecycleIT#teacherCanRunTheExamLifecycleAndReadResults	integration / high	TEST_EXAM_WRITE_ENABLE and enrolled student fixture	Expected: Teacher receives the result sheet | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-12	SRS-NOT-EXPLICIT; NFR-SE003	Read exam summary aggregates	ExamLifecycleIT#teacherCanRunTheExamLifecycleAndReadResults	integration / medium	TEST_EXAM_WRITE_ENABLE and enrolled student fixture	Expected: Teacher receives exam summary | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-13	SRS-NOT-EXPLICIT; NFR-SE003	Cancel an eligible exam	ExamLifecycleIT#teacherCanCancelAnEligibleExam	integration / high	TEST_EXAM_WRITE_ENABLE and approved course fixture	Expected: Exam transitions to CANCELLED | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-14	SRS-NOT-EXPLICIT; NFR-SE003	Student retrieves own exam results	ExamStudentResultIT#enrolledStudentCanRetrieveOwnExamResults	integration / high	Exam Service and student credentials	Expected: Student receives own results response | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-15	NFR-SE001	Reject unauthenticated exam access	ExamAuthorizationIT#unauthenticatedExamRequestsReturnUnauthorized	security / high	Exam Service	Expected: Protected exam routes return 401 | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-16	NFR-SE003	Student cannot manage exams	ExamAuthorizationIT#studentCannotUseManagementEndpointsButCanUseOwnResultsEndpoint	security / high	Exam Service and student credentials	Expected: Student management requests return 403 and own-result routes remain accessible | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-17	NFR-SE001	Reject invalid exam requests	ExamValidationIT#invalidExamAndMarksPayloadsReturnClientErrors	validation / high	Exam Service and teacher credentials	Expected: Invalid exam payload returns 400 | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-18	NFR-SE001	Reject invalid marks requests	ExamValidationIT#invalidExamAndMarksPayloadsReturnClientErrors	validation / high	Exam Service and teacher credentials	Expected: Empty or malformed marks return 400 | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-19	NFR-SE001	Unknown exam and course identifiers return not found	ExamValidationIT#unknownExamIdentifiersNeverProduceServerErrors	validation / medium	Exam Service and teacher credentials	Expected: Unknown exam and course identifiers return 404 | Actual: Not executed in this run | Status: Not Run
TC-TEST-017-30	NFR-SE001; NFR-SE003	Verify uploaded exam attachment bytes	ExamLifecycleIT#teacherCanRunTheExamLifecycleAndReadResults	integration / high	TEST_EXAM_WRITE_ENABLE and TEST_MATERIAL_DOWNL_BYTES_ENABLED with approved storage host	Expected: Teacher and enrolled student download bytes match the synthetic exam attachment upload | Actual: Not executed in this run | Status: Not Run
TC-UNIT-AI-001	Repository test harness behavior; SRS requirement not applicable	client Can Be Constructed	com.mentify.tests.ai.AiClientTest#clientCanBeConstructed	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-UNIT-AI-002	Repository test harness behavior; SRS requirement not applicable	client Rejects Blank Resource Identifiers	com.mentify.tests.ai.AiClientTest#clientRejectsBlankResourceIdentifiers	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-AI-003	Repository test harness behavior; SRS requirement not applicable	client Requires Api Client	com.mentify.tests.ai.AiClientTest#clientRequiresApiClient	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-API-001	Repository test harness behavior; SRS requirement not applicable	should Build Client For Requested Service Given Service Configuration	com.mentify.tests.api.ApiClientConfigurationTest#shouldBuildClientForRequestedServiceGivenServiceConfiguration	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-API-002	Repository test harness behavior; SRS requirement not applicable	should Use Configured Gateway Url Given Custom Environment	com.mentify.tests.api.ApiClientConfigurationTest#shouldUseConfiguredGatewayUrlGivenCustomEnvironment	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-API-003	Repository test harness behavior; SRS requirement not applicable	should Use Gateway And Timeout Configuration Given Environment Defaults	com.mentify.tests.api.ApiClientConfigurationTest#shouldUseGatewayAndTimeoutConfigurationGivenEnvironmentDefaults	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-API-004	Repository test harness behavior; SRS requirement not applicable	should Create Json Request Specification Given Valid Configuration	com.mentify.tests.api.ApiRequestSpecificationFactoryTest#shouldCreateJsonRequestSpecificationGivenValidConfiguration	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-API-005	Repository test harness behavior; SRS requirement not applicable	should Reject Invalid Path Given Path Does Not Start With Slash	com.mentify.tests.api.ApiRequestSpecificationFactoryTest#shouldRejectInvalidPathGivenPathDoesNotStartWithSlash	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ASSIGNMENT-001	Repository test harness behavior; SRS requirement not applicable	rejects Missing Assignment Id	com.mentify.tests.assignment.AssignmentAttachmentClientTest#rejectsMissingAssignmentId	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ASSIGNMENT-002	Repository test harness behavior; SRS requirement not applicable	rejects Missing Attachment Id	com.mentify.tests.assignment.AssignmentAttachmentClientTest#rejectsMissingAttachmentId	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-UNIT-ASSIGNMENT-003	Repository test harness behavior; SRS requirement not applicable	rejects Null Api Client	com.mentify.tests.assignment.AssignmentAttachmentClientTest#rejectsNullApiClient	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ASSIGNMENT-004	Repository test harness behavior; SRS requirement not applicable	creates Assignment Client	com.mentify.tests.assignment.AssignmentClientTest#createsAssignmentClient	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ASSIGNMENT-005	Repository test harness behavior; SRS requirement not applicable	rejects Blank Assignment And Course Identifiers	com.mentify.tests.assignment.AssignmentClientTest#rejectsBlankAssignmentAndCourseIdentifiers	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ASSIGNMENT-006	Repository test harness behavior; SRS requirement not applicable	requires Api Client	com.mentify.tests.assignment.AssignmentClientTest#requiresApiClient	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ASSIGNMENT-007	Repository test harness behavior; SRS requirement not applicable	try With Resources Keeps The Primary Failure And Suppresses Cleanup Failure	com.mentify.tests.assignment.CleanupScopeTest#tryWithResourcesKeepsThePrimaryFailureAndSuppressesCleanupFailure	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ASSIGNMENT-008	Repository test harness behavior; SRS requirement not applicable	creates Submission Client	com.mentify.tests.assignment.SubmissionClientTest#createsSubmissionClient	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ASSIGNMENT-009	Repository test harness behavior; SRS requirement not applicable	rejects Blank Submission Identifiers	com.mentify.tests.assignment.SubmissionClientTest#rejectsBlankSubmissionIdentifiers	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ASSIGNMENT-010	Repository test harness behavior; SRS requirement not applicable	requires Api Client	com.mentify.tests.assignment.SubmissionClientTest#requiresApiClient	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ASSIGNMENT-011	Repository test harness behavior; SRS requirement not applicable	rejects Missing File Id	com.mentify.tests.assignment.SubmissionFileClientTest#rejectsMissingFileId	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-UNIT-ASSIGNMENT-012	Repository test harness behavior; SRS requirement not applicable	rejects Missing Submission Id	com.mentify.tests.assignment.SubmissionFileClientTest#rejectsMissingSubmissionId	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ASSIGNMENT-013	Repository test harness behavior; SRS requirement not applicable	rejects Null Api Client	com.mentify.tests.assignment.SubmissionFileClientTest#rejectsNullApiClient	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ATTENDANCE-001	Repository test harness behavior; SRS requirement not applicable	client Can Be Constructed	com.mentify.tests.attendance.AttendanceClientTest#clientCanBeConstructed	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ATTENDANCE-002	Repository test harness behavior; SRS requirement not applicable	client Rejects Blank Identifiers	com.mentify.tests.attendance.AttendanceClientTest#clientRejectsBlankIdentifiers	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ATTENDANCE-003	Repository test harness behavior; SRS requirement not applicable	client Requires Api Client	com.mentify.tests.attendance.AttendanceClientTest#clientRequiresApiClient	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-AUTH-001	Repository test harness behavior; SRS requirement not applicable	should Read Role Specific Credentials And Hide Passwords	com.mentify.tests.auth.AuthenticationConfigurationTest#shouldReadRoleSpecificCredentialsAndHidePasswords	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-AUTH-002	Repository test harness behavior; SRS requirement not applicable	should Read Super Admin Credentials Separately From Admin	com.mentify.tests.auth.AuthenticationConfigurationTest#shouldReadSuperAdminCredentialsSeparatelyFromAdmin	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-AUTH-003	Repository test harness behavior; SRS requirement not applicable	should Reject Incomplete Credential Pairs	com.mentify.tests.auth.AuthenticationConfigurationTest#shouldRejectIncompleteCredentialPairs	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-AUTH-004	Repository test harness behavior; SRS requirement not applicable	should Require The Generic Test Credentials When Login Is Requested	com.mentify.tests.auth.AuthenticationConfigurationTest#shouldRequireTheGenericTestCredentialsWhenLoginIsRequested	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-UNIT-AUTH-005	Repository test harness behavior; SRS requirement not applicable	should Support A Separate Second Student Test Account	com.mentify.tests.auth.AuthenticationConfiguratio nTest#shouldSupportASeparateSecondStudentTestAcco unt	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-AUTH-006	Repository test harness behavior; SRS requirement not applicable	should Expose Only Safe Error Details	com.mentify.tests.auth.AuthenticationResponsePars erTest#shouldExposeOnlySafeErrorDetails	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-AUTH-007	Repository test harness behavior; SRS requirement not applicable	should Parse The Backend Authentication Envelope	com.mentify.tests.auth.AuthenticationResponsePars erTest#shouldParseTheBackendAuthenticationEnvelop e	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-AUTH-008	Repository test harness behavior; SRS requirement not applicable	should Clear Tokens Without Exposing Them In Diagnostics	com.mentify.tests.auth.AuthenticationSessionTest# shouldClearTokensWithoutExposingThemInDiagnostics	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-CALENDAR-001	Repository test harness behavior; SRS requirement not applicable	should Create Client With Configured Api Client	com.mentify.tests.calendar.CalendarClientTest#sho uldCreateClientWithConfiguredApiClient	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-CALENDAR-002	Repository test harness behavior; SRS requirement not applicable	should Reject Blank Event Id For Update And Cancel	com.mentify.tests.calendar.CalendarClientTest#sho uldRejectBlankEventIdForUpdateAndCancel	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-CALENDAR-003	Repository test harness behavior; SRS requirement not applicable	should Reject Null Api Client	com.mentify.tests.calendar.CalendarClientTest#sho uldRejectNullApiClient	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-COMMUNICATION-0	Repository test harness behavior; SRS requirement not applicable	client Can Be Constructed	com.mentify.tests.communication.CommunicationClie ntTest#clientCanBeConstructed	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-COMMUNICATION-0	Repository test harness behavior; SRS requirement not applicable	client Requires Api Client	com.mentify.tests.communication.CommunicationClie ntTest#clientRequiresApiClient	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-UNIT-CONTRACTS-001	Repository test harness behavior; SRS requirement not applicable	contract Validation Is Disabled By Default	com.mentify.tests.contracts.ContractConfigurationTest#contractValidationIsDisabledByDefault	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-CONTRACTS-002	Repository test harness behavior; SRS requirement not applicable	strict Contract Options Are Read From External Configuration	com.mentify.tests.contracts.ContractConfigurationTest#strictContractOptionsAreReadFromExternalConfiguration	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-COURSE-001	Repository test harness behavior; SRS requirement not applicable	should Create Client With Configured Api Client	com.mentify.tests.course.CourseClientTest#shouldCreateClientWithConfiguredApiClient	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-COURSE-002	Repository test harness behavior; SRS requirement not applicable	should Reject Null Api Client	com.mentify.tests.course.CourseClientTest#shouldRejectNullApiClient	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-E2E-001	Repository test harness behavior; SRS requirement not applicable	end To End Configuration Can Be Loaded	com.mentify.tests.e2e.EndToEndClientTest#endToEndConfigurationCanBeLoaded	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ENROLLMENT-001	Repository test harness behavior; SRS requirement not applicable	should Create Client With Configured Api Client	com.mentify.tests.enrollment.EnrollmentClientTest#shouldCreateClientWithConfiguredApiClient	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ENROLLMENT-002	Repository test harness behavior; SRS requirement not applicable	should Reject Null Api Client	com.mentify.tests.enrollment.EnrollmentClientTest#shouldRejectNullApiClient	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ENVIRONMENT-001	Repository test harness behavior; SRS requirement not applicable	should Complete Given Required Service Returns Successful Status	com.mentify.tests.environment.BackendReadinessClientTest#shouldCompleteGivenRequiredServiceReturnsSuccessfulStatus	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ENVIRONMENT-002	Repository test harness behavior; SRS requirement not applicable	should Fail After Timeout Given Service Never Becomes Ready	com.mentify.tests.environment.BackendReadinessClientTest#shouldFailAfterTimeoutGivenServiceNeverBecomesReady	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-UNIT-ENVIRONMENT-003	Repository test harness behavior; SRS requirement not applicable	should Fail Clearly Given Environment Is Disabled	com.mentify.tests.environment.BackendReadinessClientTest#shouldFailClearlyGivenEnvironmentIsDisabled	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ENVIRONMENT-004	Repository test harness behavior; SRS requirement not applicable	should Probe Auth Service Health Endpoint Given Jwt Environment	com.mentify.tests.environment.BackendReadinessClientTest#shouldProbeAuthServiceHealthEndpointGivenJwtEnvironment	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ENVIRONMENT-005	Repository test harness behavior; SRS requirement not applicable	should Retry Given Service Temporarily Fails	com.mentify.tests.environment.BackendReadinessClientTest#shouldRetryGivenServiceTemporarilyFails	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ENVIRONMENT-006	Repository test harness behavior; SRS requirement not applicable	should Configure Jwt Auth Service Given Custom Base Url	com.mentify.tests.environment.TestEnvironmentConfigurationTest#shouldConfigureJwtAuthServiceGivenCustomBaseUrl	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ENVIRONMENT-007	Repository test harness behavior; SRS requirement not applicable	should Parse Assignment Service Configuration When Assignment Testing Is Enabled	com.mentify.tests.environment.TestEnvironmentConfigurationTest#shouldParseAssignmentServiceConfigurationWhenAssignmentTestingIsEnabled	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ENVIRONMENT-008	Repository test harness behavior; SRS requirement not applicable	should Parse Calendar Service Configuration When Calendar Testing Is Enabled	com.mentify.tests.environment.TestEnvironmentConfigurationTest#shouldParseCalendarServiceConfigurationWhenCalendarTestingIsEnabled	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ENVIRONMENT-009	Repository test harness behavior; SRS requirement not applicable	should Parse Configured Service Names Given Valid Defaults	com.mentify.tests.environment.TestEnvironmentConfigurationTest#shouldParseConfiguredServiceNamesGivenValidDefaults	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ENVIRONMENT-010	Repository test harness behavior; SRS requirement not applicable	should Parse Course Service Configuration When Course Testing Is Enabled	com.mentify.tests.environment.TestEnvironmentConfigurationTest#shouldParseCourseServiceConfigurationWhenCourseTestingIsEnabled	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-ENVIRONMENT-011	Repository test harness behavior; SRS requirement not applicable	should Use Legacy Backend Base Url As Gateway Fallback Given Gateway Url Is Absent	com.mentify.tests.environment.TestEnvironmentConfigurationTest#shouldUseLegacyBackendBaseUrlAsGatewayFallbackGivenGatewayUrlIsAbsent	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-UNIT-ENVIRONMENT-012	Repository test harness behavior; SRS requirement not applicable	should Use Safe Defaults Given Environment Is Not Configured	com.mentify.tests.environment.TestEnvironmentConfigurationTest#shouldUseSafeDefaultsGivenEnvironmentIsNotConfigured	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-EXAM-001	Repository test harness behavior; SRS requirement not applicable	client Can Be Constructed	com.mentify.tests.exam.ExamClientTest#clientCanBeConstructed	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-EXAM-002	Repository test harness behavior; SRS requirement not applicable	client Rejects Blank Identifiers	com.mentify.tests.exam.ExamClientTest#clientRejectsBlankIdentifiers	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-EXAM-003	Repository test harness behavior; SRS requirement not applicable	client Requires Api Client	com.mentify.tests.exam.ExamClientTest#clientRequiresApiClient	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-FOUNDATION-001	Repository test harness behavior; SRS requirement not applicable	should Run Foundation Test Given Configured J Unit	com.mentify.tests.foundation.TestRepositorySmokeTest#shouldRunFoundationTestGivenConfiguredJUnit	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-MATERIAL-001	Repository test harness behavior; SRS requirement not applicable	rejects Missing Material Id	com.mentify.tests.material.MaterialClientTest#rejectsMissingMaterialId	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-MATERIAL-002	Repository test harness behavior; SRS requirement not applicable	rejects Null Api Client	com.mentify.tests.material.MaterialClientTest#rejectsNullApiClient	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-MATERIAL-003	Repository test harness behavior; SRS requirement not applicable	allows Http Only For An Explicit Loopback Origin When Enabled	com.mentify.tests.material.MaterialDownloadUrlPolicyTest#allowsHttpOnlyForAnExplicitLoopbackOriginWhenEnabled	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-NOTIFICATION-001	Repository test harness behavior; SRS requirement not applicable	client Can Be Constructed	com.mentify.tests.notification.NotificationClientTest#clientCanBeConstructed	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-UNIT-NOTIFICATION-002	Repository test harness behavior; SRS requirement not applicable	client Rejects Invalid Pagination And Identifiers	com.mentify.tests.notification.NotificationClientTest#clientRejectsInvalidPaginationAndIdentifiers	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-NOTIFICATION-003	Repository test harness behavior; SRS requirement not applicable	client Requires Api Client	com.mentify.tests.notification.NotificationClientTest#clientRequiresApiClient	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-PAYMENT-001	Repository test harness behavior; SRS requirement not applicable	client Can Be Constructed	com.mentify.tests.payment.PaymentClientTest#clientCanBeConstructed	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-PAYMENT-002	Repository test harness behavior; SRS requirement not applicable	client Rejects Invalid Identifiers And Pagination	com.mentify.tests.payment.PaymentClientTest#clientRejectsInvalidIdentifiersAndPagination	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-PAYMENT-003	Repository test harness behavior; SRS requirement not applicable	client Requires Api Client	com.mentify.tests.payment.PaymentClientTest#clientRequiresApiClient	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-PERFORMANCE-001	Repository test harness behavior; SRS requirement not applicable	configured Thresholds And Bounds Are Read	com.mentify.tests.performance.PerformanceConfigurationTest#configuredThresholdsAndBoundsAreRead	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-PERFORMANCE-002	Repository test harness behavior; SRS requirement not applicable	invalid Thresholds Are Rejected	com.mentify.tests.performance.PerformanceConfigurationTest#invalidThresholdsAreRejected	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-PERFORMANCE-003	Repository test harness behavior; SRS requirement not applicable	performance Is Disabled And Safe By Default	com.mentify.tests.performance.PerformanceConfigurationTest#performanceIsDisabledAndSafeByDefault	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-QUIZ-001	Repository test harness behavior; SRS requirement not applicable	clients Can Be Constructed	com.mentify.tests.quiz.QuizClientTest#clientsCanBeConstructed	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-UNIT-QUIZ-002	Repository test harness behavior; SRS requirement not applicable	clients Reject Blank Resource Ids	com.mentify.tests.quiz.QuizClientTest#clientsRejectBlankResourceIds	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-QUIZ-003	Repository test harness behavior; SRS requirement not applicable	clients Require Api Client	com.mentify.tests.quiz.QuizClientTest#clientsRequireApiClient	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-SECURITY-001	Repository test harness behavior; SRS requirement not applicable	explicit Modes Are Read Without Exposing Values	com.mentify.tests.security.SecurityResilienceTestConfigurationTest#explicitModesAreReadWithoutExposingValues	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-SECURITY-002	Repository test harness behavior; SRS requirement not applicable	security And Fault Injection Modes Are Disabled By Default	com.mentify.tests.security.SecurityResilienceTestConfigurationTest#securityAndFaultInjectionModesAreDisabledByDefault	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-STAGING-001	Repository test harness behavior; SRS requirement not applicable	applies Request Timeout Without Reading Response Body	com.mentify.tests.staging.StagingHttpProbeTest#appliesRequestTimeoutWithoutReadingResponseBody	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-STAGING-002	Repository test harness behavior; SRS requirement not applicable	does Not Follow Redirects To Another Host	com.mentify.tests.staging.StagingHttpProbeTest#doesNotFollowRedirectsToAnotherHost	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-STAGING-003	Repository test harness behavior; SRS requirement not applicable	accepts Confirmed Https Host Within Configured Staging Suffix	com.mentify.tests.staging.StagingTargetPolicyTest#acceptsConfirmedHttpsHostWithinConfiguredStagingSuffix	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-STAGING-004	Repository test harness behavior; SRS requirement not applicable	rejects Http Embedded Credentials Query And Unapproved Hosts	com.mentify.tests.staging.StagingTargetPolicyTest#rejectsHttpEmbeddedCredentialsQueryAndUnapprovedHosts	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-STAGING-005	Repository test harness behavior; SRS requirement not applicable	rejects Invalid Deployment Revision	com.mentify.tests.staging.StagingTargetPolicyTest#rejectsInvalidDeploymentRevision	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-UNIT-STAGING-006	Repository test harness behavior; SRS requirement not applicable	rejects Ip Address As Staging Host Suffix	com.mentify.tests.staging.StagingTargetPolicyTest#rejectsIpAddressAsStagingHostSuffix	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-STAGING-007	Repository test harness behavior; SRS requirement not applicable	rejects Missing Or Incorrect Confirmation Before Any Request	com.mentify.tests.staging.StagingTargetPolicyTest#rejectsMissingOrIncorrectConfirmationBeforeAnyRequest	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-STAGING-008	Repository test harness behavior; SRS requirement not applicable	rejects Non Staging Environment	com.mentify.tests.staging.StagingTargetPolicyTest#rejectsNonStagingEnvironment	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-USER-001	Repository test harness behavior; SRS requirement not applicable	client Can Be Constructed	com.mentify.tests.user.UserClientTest#clientCanBeConstructed	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-USER-002	Repository test harness behavior; SRS requirement not applicable	client Rejects Blank User Identifiers	com.mentify.tests.user.UserClientTest#clientRejectsBlankUserIdentifiers	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-UNIT-USER-003	Repository test harness behavior; SRS requirement not applicable	client Requires Api Client	com.mentify.tests.user.UserClientTest#clientRequiresApiClient	unit / low	None; deterministic unit test	Expected: The assertions in the mapped JUnit method pass; execution status is taken from JUnit. | Actual: Passed; see JUnit report | Status: Passed
TC-TRACE-069	FR-S007; FR-S008; FR-S009; SRS-NOT-EXPLICIT (lesson-to-assignment-to-quiz and graded-result journey)	studentCanFollowEnrolledCourseThroughLessonAssignmentQuizAndGradedResult	StudentLearningJourneyIT#studentCanFollowEnrolledCourseThroughLessonAssignmentQuizAndGradedResult	e2e / high	TEST_E2E_ENABLED=true; student credential; course /module/lesson/assignment/ published quiz/graded submission fixtures; Gateway, Course, Enrollment, Assignment services	Expected: Student identity and enrollment are verified across services; course lesson, assignment, quiz, and the same assignment's graded submission are readable through Gateway | Actual: Not executed in this run | Status: Not Run
TC-TRACE-070	SRS-NOT-EXPLICIT (teacher-owned course content and assignment grading E2E journey)	teacherCanFollowOwnedCourseFromLessonAndMaterialToAssignmentGrade	TeacherCourseJourneyIT#teacherCanFollowOwnedCourseFromLessonAndMaterialToAssignmentGrade	e2e / high	TEST_E2E_ENABLED=true; teacher credential; owned course/module/lesson/material/assignment/graded submission fixtures; Gateway, Course, Material, Assignment services	Expected: Teacher can follow owned course content through the persisted assignment grade without mutating fixture data | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TRACE-071	FR-A002; BR-005; NFR-SE003 (admin operations across Gateway services)	administratorCanReviewUsersCoursesPaymentsAndNotificationsThroughGateway	AdminOperationsJourneyIT#administratorCanReviewUsersCoursesPaymentsAndNotificationsThroughGateway	e2e / high	TEST_E2E_ENABLED=true; admin credential; Gateway, User, Course, Payment, Notification services	Expected: Administrator receives successful responses for user, course, payment, and notification operations through Gateway | Actual: Not executed in this run | Status: Not Run
TC-ANALYTICS-006	SRS-NOT-EXPLICIT; NFR-SE003	Read course analytics summaries through Gateway	InternalAnalyticsIT#administratorCanReadCourseAnalyticsSummariesForConfiguredCourse	integration / high	TEST_INTERNAL_ANALYTICS_URSE_ID plus API Gateway, seven analytics source services, and admin credentials	Expected: Six course-scoped analytics summary routes return HTTP 200 for an approved course | Actual: Not executed in this run | Status: Not Run
TC-TEST-038-01	SRS-NOT-EXPLICIT (test harness safe-default behavior)	Live integration tests are disabled by default	IntegrationTestExecutionConditionTest#shouldDisableLiveTestClassesByDefault	unit / high	None; deterministic environment map	Expected: Integration test classes receive a clear disabled reason when TEST_ENV_ENABLED is missing or false | Actual: Passed; see JUnit report | Status: Passed
TC-TEST-038-02	SRS-NOT-EXPLICIT (test harness safe-default behavior)	Unit tests remain enabled when live integration is disabled	IntegrationTestExecutionConditionTest#shouldLeaveUnitTestsEnabledWhenIntegrationIsDisabled	unit / high	None; deterministic environment map	Expected: Static unit test classes are not disabled by the live-integration condition | Actual: Passed; see JUnit report | Status: Passed
TC-TEST-038-03	SRS-NOT-EXPLICIT (test harness opt-in behavior)	Explicit integration enablement activates live tests	IntegrationTestExecutionConditionTest#shouldEnableLiveTestClassWhenEnvironmentIsExplicitlyEnabled	unit / high	None; deterministic environment map	Expected: TEST_ENV_ENABLED=true enables the integration class condition | Actual: Passed; see JUnit report | Status: Passed
TC-TEST-038-04	SRS-NOT-EXPLICIT (staging test target safeguard)	Staging smoke requires explicit staging target	IntegrationTestExecutionConditionTest#shouldRequireStagingTargetForStagingSmokeClass	unit / high	None; deterministic environment map	Expected: StagingSmokeIT is disabled unless STAGING_ENVIRONMENT=staging | Actual: Passed; see JUnit report | Status: Passed
TC-TEST-020-01	SRS-NOT-EXPLICIT	AI Service readiness	AiHealthIT#healthEndpointReportsAiServiceUp	integration / high	TEST_ENV_ENABLED=true; AI Service available	Expected: AI integration support verifies required services and AI Service readiness before the health test | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-09	SRS-NOT-EXPLICIT	Conversation ownership isolation	AiConversationIT#differentAdminOwnerCannotReadOrMutateTeachersConversation	integration / high	TEST_AI_WRITE_ENABLED; distinct Teacher and Admin credentials	Expected: Non-owner cannot read conversation/messages or update/delete; owner cleanup succeeds | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-020-10	SRS-NOT-EXPLICIT	Chat role authorization and allowed-role request	AiProviderIT#deterministicProviderSupportsSafeChatAndGenerationTests	integration / high	TEST_AI_WRITE_ENABLED; AI_TEST_MODE; Teacher and Admin credentials; configured conversation	Expected: Teacher chat and Admin generation return HTTP 200 with response data | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-12	SRS-NOT-EXPLICIT	Administrator deterministic content generation	AiProviderIT#deterministicProviderSupportsSafeChatAndGenerationTests	integration / high	TEST_AI_WRITE_ENABLED; AI_TEST_MODE; Admin credentials	Expected: Administrator generation returns HTTP 200 with response data | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-14	SRS-NOT-EXPLICIT	Configured lesson summary	AiContentGenerationIT#shouldSummarizeConfiguredLessonWithDeterministicProvider	integration / high	TEST_AI_WRITE_ENABLED; deterministic provider; TEST_AI_LESSON_ID	Expected: HTTP 200 includes the configured lesson ID and a non-empty summary | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-15	SRS-NOT-EXPLICIT	Configured material summary	AiContentGenerationIT#shouldSummarizeConfiguredMaterialWithDeterministicProvider	integration / high	TEST_AI_WRITE_ENABLED; deterministic provider; TEST_AI_MATERIAL_ID	Expected: HTTP 200 includes a non-empty summary and provider metadata | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-28	SRS-NOT-EXPLICIT	Public quiz-generation semantic validation	AiValidationIT#publicQuizGenerationRejectsInvalidSemanticParametersBeforeProviderCall	validation / high	AI Service only; no provider call	Expected: Invalid question count, difficulty, or question type returns HTTP 400 before provider invocation | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-13	SRS-NOT-EXPLICIT	Student and teacher are denied administrator-only general generation	AiAuthorizationIT#studentAndTeacherCannotUseAdministratorOnlyGeneration	security / high	AI Service; Student and Teacher credentials	Expected: Both non-admin roles receive HTTP 403 for general content generation | Actual: Not executed in this run | Status: Not Run
TC-TEST-020-20	SRS-NOT-EXPLICIT	Controlled AI provider failure is returned safely	AiProviderFailureIT#controlledProviderFailureReturnsSanitizedServerError	resilience / high	TEST_RESILIENCE_ENABLED ue; FAULT_INJECTION_ENABL D=true; TEST_PROVIDER_FAIL URE_MODE=true; AI Service backed by controlled failing provider; Admin credentials	Expected: HTTP 5xx response; controlled provider sentinel is not exposed | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-01	SRS-NOT-EXPLICIT	Teacher creates an assignment	AssignmentWorkflowIT#teacherCanCreateReadListUpdatePublishCloseAndDeleteAssignment	integration / high	TEST_ASSIGNMENT_WRITE_ LED; owned course/module; Teacher credentials	Expected: Created assignment returns an ID and the expected title, course, dates, attempts, and rubric | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-02	SRS-NOT-EXPLICIT	Teacher retrieves the created assignment	AssignmentWorkflowIT#teacherCanCreateReadListUpdatePublishCloseAndDeleteAssignment	integration / high	TEST_ASSIGNMENT_WRITE_ LED; owned course/module; Teacher credentials	Expected: Read response contains the created assignment ID | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-011-03	SRS-NOT-EXPLICIT	Teacher lists the created course assignment	AssignmentWorkflowIT#teacherCanCreateReadListUpdatePublishCloseAndDeleteAssignment	integration / high	TEST_ASSIGNMENT_WRITE_ENABLED; owned course/module; Teacher credentials	Expected: Course assignment list contains the created assignment ID | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-04	SRS-NOT-EXPLICIT	Teacher updates the assignment	AssignmentWorkflowIT#teacherCanCreateReadListUpdatePublishCloseAndDeleteAssignment	integration / high	TEST_ASSIGNMENT_WRITE_ENABLED; owned course/module; Teacher credentials	Expected: Update returns HTTP 200 and the changed title; omitted rubric data is preserved | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-05	SRS-NOT-EXPLICIT	Teacher publishes the assignment	AssignmentWorkflowIT#teacherCanCreateReadListUpdatePublishCloseAndDeleteAssignment	integration / high	TEST_ASSIGNMENT_WRITE_ENABLED; owned course/module; Teacher credentials	Expected: Publish returns HTTP 200 and status PUBLISHED | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-06	SRS-NOT-EXPLICIT	Teacher closes the assignment	AssignmentWorkflowIT#teacherCanCreateReadListUpdatePublishCloseAndDeleteAssignment	integration / high	TEST_ASSIGNMENT_WRITE_ENABLED; owned course/module; Teacher credentials	Expected: Close returns HTTP 200 and status CLOSED | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-07	SRS-NOT-EXPLICIT	Teacher deletes the test-created assignment	AssignmentWorkflowIT#teacherCanCreateReadListUpdatePublishCloseAndDeleteAssignment	integration / high	TEST_ASSIGNMENT_WRITE_ENABLED; owned course/module; Teacher credentials	Expected: Delete succeeds for the synthetic assignment created by the test | Actual: Not executed in this run | Status: Not Run
TC-TEST-011-18	SRS-NOT-EXPLICIT	Assignment client validates required path identifiers	AssignmentClientTest#rejectsBlankAssignmentAndCourseIdentifiers	unit / medium	No backend required	Expected: Blank assignment and course identifiers are rejected locally | Actual: Passed; see JUnit report | Status: Passed
TC-TEST-015-01	SRS-NOT-EXPLICIT	Teacher creates an open attendance session	AttendanceSessionWorkflowIT#teacherCanCreateListMarkAndCompleteSession	integration / high	TEST_ATTENDANCE_WRITE_ENABLED; approved course/student fixtures; Teacher credentials	Expected: HTTP 201; created session has status OPEN and includes the configured enrolled student | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-02	SRS-NOT-EXPLICIT	Admin creates and retrieves an attendance session	AttendanceAdminIT#adminCanCreateAndRetrieveAttendanceSessionWhenWritesAreEnabled	integration / high	TEST_ATTENDANCE_WRITE_ENABLED; approved course/student fixtures; Admin credentials	Expected: Admin creates a session, retrieves it, and cancels it | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-03	SRS-NOT-EXPLICIT	Teacher lists course attendance sessions	AttendanceSessionWorkflowIT#teacherCanCreateListMarkAndCompleteSession	integration / high	TEST_ATTENDANCE_WRITE_ENABLED; approved course/student fixtures; Teacher credentials	Expected: Session list includes the newly created session for the configured course | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-04	SRS-NOT-EXPLICIT	Teacher retrieves an attendance session	AttendanceSessionWorkflowIT#teacherCanCreateListMarkAndCompleteSession	integration / high	TEST_ATTENDANCE_WRITE_ENABLED; approved course/student fixtures; Teacher credentials	Expected: Session retrieval succeeds and completed status is persisted | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-015-05	SRS-NOT-EXPLICIT	Teacher marks one enrolled student	AttendanceSessionWorkflowIT#teacherCanCreateListMarkAndCompleteSession	integration / high	TEST_ATTENDANCE_WRITE_LED; configured student enrolled in the approved course	Expected: Single-record mark returns the configured student and LATE status | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-06	SRS-NOT-EXPLICIT	Teacher marks all enrolled students in a batch	AttendanceSessionWorkflowIT#teacherCanCreateListMarkAndCompleteSession	integration / high	TEST_ATTENDANCE_WRITE_LED; approved course with enrolled students	Expected: Batch mark returns records and all supplied statuses are PRESENT | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-07	SRS-NOT-EXPLICIT	Teacher completes a fully marked session	AttendanceSessionWorkflowIT#teacherCanCreateListMarkAndCompleteSession	integration / high	TEST_ATTENDANCE_WRITE_LED; approved course/student fixtures	Expected: Completion returns HTTP 200 and COMPLETED status; subsequent read preserves that status | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-08	SRS-NOT-EXPLICIT	Teacher cancels an open session	AttendanceSessionWorkflowIT#teacherCanCancelOpenSessionAndCancelledSessionCannotBeCompleted	integration / high	TEST_ATTENDANCE_WRITE_LED; approved course/student fixtures	Expected: Cancellation persists CANCELLED and later completion/mark attempts return 409 | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-17	SRS-NOT-EXPLICIT	Unauthenticated attendance requests are rejected	AttendanceAuthenticationIT#protectedAttendanceRequestsRejectMissingAuthentication	security / high	Attendance Service; no credentials	Expected: Protected session, marking, transition, and summary routes return HTTP 401 | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-18	SRS-NOT-EXPLICIT	Student cannot use session management routes	AttendanceAuthenticationIT#studentCannotManageSessionsOrReadAnotherStudentSummary	security / high	Attendance Service and Student credentials	Expected: Student receives HTTP 403 for session create/list/read/complete and summary access | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-20	SRS-NOT-EXPLICIT	Attendance session creation validates required fields	AttendanceValidationIT#rejectsInvalidSessionAndRecordRequests	validation / high	Attendance Service and Teacher credentials	Expected: Empty session body and invalid course/title fields return HTTP 400 | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-21	SRS-NOT-EXPLICIT	Attendance request rejects malformed UUID input	AttendanceValidationIT#rejectsInvalidSessionAndRecordRequests	validation / high	Attendance Service and Teacher credentials	Expected: Malformed course and student UUID inputs return HTTP 400 | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-22	SRS-NOT-EXPLICIT	Attendance marking rejects an empty record request	AttendanceValidationIT#rejectsInvalidSessionAndRecordRequests	validation / high	Attendance Service and Teacher credentials	Expected: Empty attendance records request returns HTTP 400 | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-015-24	SRS-NOT-EXPLICIT	Unknown session identifiers return not found	AttendanceValidationIT#unknownAttendanceResourcesReturnNotFound	integration / high	Attendance Service and Teacher credentials	Expected: Unknown session lookup returns HTTP 404 | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-25	SRS-NOT-EXPLICIT	Unknown course identifiers return not found	AttendanceValidationIT#unknownAttendanceResourcesReturnNotFound	integration / high	Attendance Service and Teacher credentials	Expected: Listing sessions for an unknown course returns HTTP 404 | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-28	SRS-NOT-EXPLICIT	Invalid attendance status is rejected	AttendanceValidationIT#rejectsInvalidSessionAndRecordRequests	validation / high	Attendance Service and Teacher credentials	Expected: Invalid attendance status returns HTTP 400 | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-29	SRS-NOT-EXPLICIT	Teacher cannot access another teacher course sessions	AttendanceAuthenticationIT#teacherCannotListSessionsForAnotherTeachersCourse	security / high	Attendance Service; primary-owned course; distinct secondary Teacher credentials	Expected: Secondary teacher receives HTTP 403 for the primary teacher course session list | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-19	SRS-NOT-EXPLICIT	Student cannot read another configured student attendance summary	AttendanceAuthenticationIT#studentCannotReadAnotherStudentsAttendanceSummary	security / high	Distinct TEST_ATTENDANCE_STUDEN and TEST_ATTENDANCE_OTH STUDENT_ID; matching Student and SECONDARY_STUDENT credentials; shared known course	Expected: Test verifies both authenticated identities match configured fixtures; Student receives HTTP 403 for the other student summary | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-23	SRS-NOT-EXPLICIT	Attendance marking rejects missing student ID or status	AttendanceValidationIT#rejectsInvalidSessionAndRecordRequests	validation / high	Attendance Service and Teacher credentials	Expected: Record payload missing studentId or status returns HTTP 400 | Actual: Not executed in this run | Status: Not Run
TC-CAL-001	SRS-NOT-EXPLICIT	Upcoming event route requires authentication	CalendarAuthenticationIT#shouldRejectUpcomingCalendarRequestsWithoutAuthentication	security / high	Calendar Service; no credentials	Expected: Upcoming events request returns HTTP 401 without authentication | Actual: Not executed in this run | Status: Not Run
TC-CAL-002	SRS-NOT-EXPLICIT	Event creation route requires authentication	CalendarAuthenticationIT#shouldRejectEventCreationWithoutAuthentication	security / high	Calendar Service; no credentials	Expected: Valid event creation request returns HTTP 401 without authentication | Actual: Not executed in this run | Status: Not Run
TC-CAL-003	SRS-NOT-EXPLICIT	Student reads upcoming calendar events	CalendarAuthenticationIT#shouldAllowAnAuthenticatedStudentToQueryUpcomingEvents	integration / high	Calendar Service and Student credentials	Expected: Authenticated upcoming request returns HTTP 200 and data | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-CAL-004	SRS-NOT-EXPLICIT	Calendar client constructs with a valid API client	CalendarClientTest#shouldCreateClientWithConfiguredApiClient	unit / medium	No backend required	Expected: Calendar client is created with configured API client | Actual: Passed; see JUnit report | Status: Passed
TC-CAL-005	SRS-NOT-EXPLICIT	Calendar Service readiness	CalendarAuthenticationIT#shouldAllowAnAuthenticatedStudentToQueryUpcomingEvents	integration / high	API Gateway, Auth Service, Calendar Service	Expected: Test setup waits for required services and Calendar Service readiness before the request | Actual: Not executed in this run | Status: Not Run
TC-CAL-006	SRS-NOT-EXPLICIT	Teacher creates a course calendar event	CalendarEventWorkflowIT#shouldAllowTeacherToCreateUpdateAndReadAnEvent	integration / high	Calendar Service; TEST_CALENDAR_COURSE_I assigned to Teacher	Expected: Teacher creates event successfully and test cleanup cancels it | Actual: Not executed in this run | Status: Not Run
TC-CAL-007	SRS-NOT-EXPLICIT	Admin creates a calendar event	CalendarEventWorkflowIT#shouldAllowAdminToCreateAnEvent	integration / high	Calendar Service and Admin credentials	Expected: Admin event creation returns documented success and cleanup cancels the event | Actual: Not executed in this run | Status: Not Run
TC-CAL-009	SRS-NOT-EXPLICIT	Teacher updates a created event	CalendarEventWorkflowIT#shouldAllowTeacherToCreateUpdateAndReadAnEvent	integration / high	Calendar Service; TEST_CALENDAR_COURSE_I assigned to Teacher	Expected: Update succeeds and a subsequent read returns the updated title | Actual: Not executed in this run | Status: Not Run
TC-CAL-010	SRS-NOT-EXPLICIT	Teacher cancels a created event	CalendarEventWorkflowIT#shouldAllowTeacherToCancelAnEvent	integration / high	Calendar Service; TEST_CALENDAR_COURSE_I assigned to Teacher	Expected: Cancellation succeeds and cleanup avoids retaining the synthetic event | Actual: Not executed in this run | Status: Not Run
TC-CAL-011	SRS-NOT-EXPLICIT	Student cannot create events	CalendarEventWorkflowIT#shouldRejectStudentEventCreation	security / high	Calendar Service and Student credentials	Expected: Student event creation returns HTTP 403 | Actual: Not executed in this run | Status: Not Run
TC-CAL-012	SRS-NOT-EXPLICIT	Student cannot update events	CalendarEventWorkflowIT#shouldRejectStudentEventUpdate	security / high	Calendar Service and Student credentials	Expected: Student event update returns HTTP 403 | Actual: Not executed in this run | Status: Not Run
TC-CAL-013	SRS-NOT-EXPLICIT	Student cannot cancel events	CalendarEventWorkflowIT#shouldRejectStudentEventCancellation	security / high	Calendar Service and Student credentials	Expected: Student event cancellation returns HTTP 403 | Actual: Not executed in this run | Status: Not Run
TC-CAL-014	SRS-NOT-EXPLICIT	Event creation validates required fields	CalendarEventWorkflowIT#shouldRejectInvalidEventRequest	validation / high	Calendar Service and Teacher credentials	Expected: Missing required event fields return HTTP 400 | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-CAL-015	SRS-NOT-EXPLICIT	Event creation validates chronological date range	CalendarEventWorkflowIT#shouldRejectEventWithInvalidDateRange	validation / high	Calendar Service and Teacher credentials	Expected: End time before start time returns HTTP 400 | Actual: Not executed in this run | Status: Not Run
TC-CAL-016	SRS-NOT-EXPLICIT	Unknown event identifiers return not found	CalendarEventWorkflowIT#shouldReturnNotFoundForUnknownEvent	integration / high	Calendar Service and Teacher credentials	Expected: Unknown event update and detail requests return HTTP 404 | Actual: Not executed in this run | Status: Not Run
TC-COURSE-013	SRS-NOT-EXPLICIT	Course client construction with configured API client	CourseClientTest#shouldCreateClientWithConfiguredApiClient	unit / medium	No backend required	Expected: Course client is instantiated with the valid configured API client | Actual: Passed; see JUnit report | Status: Passed
TC-COURSE-014	SRS-NOT-EXPLICIT	Course service environment URL configuration	TestEnvironmentConfigurationTest#shouldParseCourseServiceConfigurationWhenCourseTestingIsEnabled	unit / medium	Deterministic environment map; no backend required	Expected: Configured Course Service URL and required-service list are parsed correctly | Actual: Passed; see JUnit report | Status: Passed
TC-TEST-015-26	SRS-NOT-EXPLICIT	Student with no records in a known course receives a zero summary	AttendanceSummaryIT#studentWithNoAttendanceRecordsReceivesZeroSummary	integration / medium	Known TEST_ATTENDANCE_COURSE and TEST_ATTENDANCE_EMP SUMMARY_STUDENT_ID; Student credentials matching that ID	Expected: Response identifies the configured student/course and reports zero session, attendance, and percentage totals | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-27	SRS-NOT-EXPLICIT	Attendance rejects overlong session title or record remarks	AttendanceValidationIT#rejectsOverlongSessionTitleAndAttendanceRemarks	validation / medium	Valid TEST_ATTENDANCE_COURSE and Teacher credentials	Expected: Session title above 150 characters and remarks above 500 characters each return HTTP 400 | Actual: Not executed in this run | Status: Not Run
TC-TEST-015-33	SRS-NOT-EXPLICIT	Unknown course or student summary identifiers return not found	AttendanceValidationIT#unknownAttendanceResourcesReturnNotFound	integration / medium	Attendance Service and Teacher credentials	Expected: Unknown course and unknown student/course summary requests return HTTP 404 | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-01	SRS-NOT-EXPLICIT	Communication and required service readiness	CommunicationAuthenticationIT#protectedCommunicationRequestsRejectMissingAuthentication	integration / high	API Gateway, Auth Service, Communication Service	Expected: JUnit setup verifies required services and Communication Service readiness before the protected-route checks | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-02	SRS-NOT-EXPLICIT	Authenticated user identity is returned by /me	CommunicationAuthenticationIT#authenticatedUserCanReadTheirCommunicationIdentity	integration / high	Communication Service and Student credentials	Expected: Authenticated /me response returns a parseable userId | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-016-03	SRS-NOT-EXPLICIT	Communication /me requires authentication	CommunicationAuthenticationIT#protectedCommunicationRequestsRejectMissingAuthentication	security / high	Communication Service; no credentials	Expected: Unauthenticated /me request returns HTTP 401 | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-23	SRS-NOT-EXPLICIT	Sender deletes own message for self	CommunicationMessageWorkflowIT#teacherCanDeleteOwnMessageForSelfWithoutHidingItFromOthers	integration / high	TEST_COMMUNICATION_WRITE_ENABLED; active group membership; Teacher credentials	Expected: ME delete returns HTTP 200 and the message is absent from the sender history; test cleanup deletes it for everyone | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-26	SRS-NOT-EXPLICIT	Authenticated user reads own settings	CommunicationSettingsIT#authenticatedUserCanReadAndUpdateOwnSettings	integration / high	Communication Service and Student credentials	Expected: Settings response succeeds and the test restores original values after update | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-27	SRS-NOT-EXPLICIT	Authenticated user updates own settings	CommunicationSettingsIT#authenticatedUserCanReadAndUpdateOwnSettings	integration / high	Communication Service and Student credentials	Expected: Settings update succeeds and prior preferences are restored during cleanup | Actual: Not executed in this run | Status: Not Run
TC-TEST-016-28	SRS-NOT-EXPLICIT	Settings changes are isolated by authenticated user	CommunicationSettingsIT#settingsChangesAreIsolatedToTheAuthenticatedUser	integration / high	Distinct Student and SECONDARY_STUDENT credentials	Expected: Changing the primary student setting leaves the secondary student settings unchanged; primary setting is restored | Actual: Not executed in this run | Status: Not Run
TC-COURSE-001	SRS-NOT-EXPLICIT	Course Service readiness before integration request	CourseServiceRootIT#authenticatedUserCanReadCourseServiceRootGreeting	integration / high	API Gateway, Auth Service, Course Service, Teacher credentials	Expected: Course test setup verifies Course Service readiness before the authenticated root request | Actual: Not executed in this run | Status: Not Run
TEST-013-01	SRS-NOT-EXPLICIT	Enrolled student uploads a submission attachment	MaterialLifecycleIT#enrolledStudentCanUploadSubmissionAttachmentMaterial	integration / high	TEST_MATERIAL_WRITE_ENAD; enrolled Student, course, and material fixtures	Expected: Student upload returns HTTP 201 with SUBMISSION_ATTACHMENT purpose and a material ID | Actual: Not executed in this run | Status: Not Run
TEST-013-02	SRS-NOT-EXPLICIT	Teacher uploads learning material for an owned course	MaterialLifecycleIT#teacherCanUploadRetrieveListDownloadUpdateArchiveAndDeleteMaterial	integration / high	TEST_MATERIAL_WRITE_ENAD; Teacher-owned course fixture	Expected: Teacher upload returns HTTP 201 for LEARNING_MATERIAL | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TEST-013-03	SRS-NOT-EXPLICIT	Admin uploads and deletes synthetic material	MaterialLifecycleIT#administratorCanUploadAndDeleteSyntheticMaterial	integration / high	TEST_MATERIAL_WRITE_ENAD; approved course and Admin credentials	Expected: Admin creates a synthetic material and deletes it successfully | Actual: Not executed in this run | Status: Not Run
TEST-013-04	SRS-NOT-EXPLICIT	Material upload response includes file metadata	MaterialLifecycleIT#teacherCanUploadRetrieveListDownloadUpdateArchiveAndDeleteMaterial	integration / high	TEST_MATERIAL_WRITE_ENAD; Teacher-owned course fixture	Expected: Response includes material ID, original filename, content type, positive file size, purpose, and UPLOADED status | Actual: Not executed in this run | Status: Not Run
TEST-013-05	SRS-NOT-EXPLICIT	Authorized teacher retrieves uploaded material	MaterialLifecycleIT#teacherCanUploadRetrieveListDownloadUpdateArchiveAndDeleteMaterial	integration / high	TEST_MATERIAL_WRITE_ENAD; Teacher-owned course fixture	Expected: Retrieved material ID and title match the synthetic upload | Actual: Not executed in this run | Status: Not Run
TEST-013-06	SRS-NOT-EXPLICIT	Teacher lists uploaded material	MaterialLifecycleIT#teacherCanUploadRetrieveListDownloadUpdateArchiveAndDeleteMaterial	integration / high	TEST_MATERIAL_WRITE_ENAD; Teacher-owned course fixture	Expected: List response includes the uploaded material | Actual: Not executed in this run | Status: Not Run
TEST-013-07	SRS-NOT-EXPLICIT	Material list honors page size	MaterialLifecycleIT#teacherCanUploadRetrieveListDownloadUpdateArchiveAndDeleteMaterial	integration / medium	TEST_MATERIAL_WRITE_ENAD; Teacher-owned course fixture	Expected: List response reports requested page size and contains the uploaded result | Actual: Not executed in this run | Status: Not Run
TEST-013-08	SRS-NOT-EXPLICIT	Material title search returns uploaded material	MaterialLifecycleIT#teacherCanUploadRetrieveListDownloadUpdateArchiveAndDeleteMaterial	integration / medium	TEST_MATERIAL_WRITE_ENAD; Teacher-owned course fixture	Expected: Search by unique title returns the uploaded material ID | Actual: Not executed in this run | Status: Not Run
TEST-013-09	SRS-NOT-EXPLICIT	Material list filters by course	MaterialLifecycleIT#teacherCanUploadRetrieveListDownloadUpdateArchiveAndDeleteMaterial	integration / medium	TEST_MATERIAL_WRITE_ENAD; Teacher-owned course fixture	Expected: Course-filtered result contains uploaded material with matching course ID | Actual: Not executed in this run | Status: Not Run
TEST-013-10	SRS-NOT-EXPLICIT	Material download response provides URL and expiry	MaterialLifecycleIT#teacherCanUploadRetrieveListDownloadUpdateArchiveAndDeleteMaterial	integration / high	TEST_MATERIAL_WRITE_ENAD; configured Material storage	Expected: Download response includes material ID, download URL, and expiry metadata | Actual: Not executed in this run | Status: Not Run
TEST-013-11	SRS-NOT-EXPLICIT	Download response does not expose storage keys	MaterialLifecycleIT#teacherCanUploadRetrieveListDownloadUpdateArchiveAndDeleteMaterial	security / high	TEST_MATERIAL_WRITE_ENAD; configured Material storage	Expected: Download metadata has no storageKey or provider credentials | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TEST-013-12	SRS-NOT-EXPLICIT	Teacher updates managed material metadata	MaterialLifecycleIT#teacherCanUploadRetrieveListDownloadUpdateArchiveAndDeleteMaterial	integration / high	TEST_MATERIAL_WRITE_ENAD; Teacher-owned course fixture	Expected: Updated title and description are returned | Actual: Not executed in this run | Status: Not Run
TEST-013-13	SRS-NOT-EXPLICIT	Teacher archives managed material	MaterialLifecycleIT#teacherCanUploadRetrieveListDownloadUpdateArchiveAndDeleteMaterial	integration / high	TEST_MATERIAL_WRITE_ENAD; Teacher-owned course fixture	Expected: Archive returns status ARCHIVED and archived filter includes the material | Actual: Not executed in this run | Status: Not Run
TEST-013-14	SRS-NOT-EXPLICIT	Teacher deletes archived test material	MaterialLifecycleIT#teacherCanUploadRetrieveListDownloadUpdateArchiveAndDeleteMaterial	integration / high	TEST_MATERIAL_WRITE_ENAD; Teacher-owned course fixture	Expected: Archived synthetic material is deleted successfully | Actual: Not executed in this run | Status: Not Run
TEST-013-15	SRS-NOT-EXPLICIT	Admin performs a non-destructive maintenance audit	MaterialLifecycleIT#adminCanAuditAndRunNonDestructiveCleanup	integration / high	Admin credentials; Material Service	Expected: Audit request returns HTTP 200 and confirms dryRun=true | Actual: Not executed in this run | Status: Not Run
TEST-013-16	SRS-NOT-EXPLICIT	Admin runs cleanup in dry-run mode	MaterialLifecycleIT#adminCanAuditAndRunNonDestructiveCleanup	integration / high	Admin credentials; Material Service	Expected: Cleanup request returns HTTP 200 and confirms dryRun=true | Actual: Not executed in this run | Status: Not Run
TEST-013-18	SRS-NOT-EXPLICIT	Upload without a file is rejected	MaterialLifecycleIT#rejectsInvalidMaterialUploadsAndUpdates	validation / medium	TEST_MATERIAL_WRITE_ENAD; Teacher-owned course fixture	Expected: Multipart upload without a file returns HTTP 400 | Actual: Not executed in this run | Status: Not Run
TEST-013-19	SRS-NOT-EXPLICIT	Empty file upload is rejected	MaterialLifecycleIT#rejectsInvalidMaterialUploadsAndUpdates	validation / medium	TEST_MATERIAL_WRITE_ENAD; Teacher-owned course fixture	Expected: Empty synthetic file upload returns HTTP 400 | Actual: Not executed in this run | Status: Not Run
TEST-013-20	SRS-NOT-EXPLICIT	Unsupported file type is rejected	MaterialLifecycleIT#rejectsInvalidMaterialUploadsAndUpdates	validation / medium	TEST_MATERIAL_WRITE_ENAD; Teacher-owned course fixture	Expected: Unsupported synthetic .bin upload returns HTTP 400 | Actual: Not executed in this run | Status: Not Run
TEST-013-21	SRS-NOT-EXPLICIT	Upload without title is rejected	MaterialLifecycleIT#rejectsInvalidMaterialUploadsAndUpdates	validation / medium	TEST_MATERIAL_WRITE_ENAD; Teacher-owned course fixture	Expected: Upload missing title returns HTTP 400 | Actual: Not executed in this run | Status: Not Run
TEST-013-22	SRS-NOT-EXPLICIT	Upload without course ID is rejected	MaterialLifecycleIT#rejectsInvalidMaterialUploadsAndUpdates	validation / medium	TEST_MATERIAL_WRITE_ENAD; Teacher credentials	Expected: Upload missing course ID returns HTTP 400 | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TEST-013-23	SRS-NOT-EXPLICIT	Empty material update is rejected	MaterialLifecycleIT#teacherCanUploadRetrieveListDownloadUpdateArchiveAndDeleteMaterial	validation / medium	TEST_MATERIAL_WRITE_ENAD; Teacher-owned course fixture	Expected: Empty update payload for existing material returns HTTP 400 | Actual: Not executed in this run | Status: Not Run
TEST-013-24	SRS-NOT-EXPLICIT	Unknown material retrieval returns not found	MaterialLifecycleIT#unknownMaterialOperationsAreNotReportedAsServerErrors	integration / medium	TEST_MATERIAL_READ_FIXTS; Teacher credentials	Expected: Unknown material read returns HTTP 404 | Actual: Not executed in this run | Status: Not Run
TEST-013-25	SRS-NOT-EXPLICIT	Unknown material download returns not found	MaterialLifecycleIT#unknownMaterialOperationsAreNotReportedAsServerErrors	integration / medium	TEST_MATERIAL_READ_FIXTS; Teacher credentials	Expected: Unknown material download returns HTTP 404 | Actual: Not executed in this run | Status: Not Run
TEST-013-26	SRS-NOT-EXPLICIT	Unknown material update returns not found	MaterialLifecycleIT#unknownMaterialOperationsAreNotReportedAsServerErrors	integration / medium	TEST_MATERIAL_READ_FIXTS; Teacher credentials	Expected: Unknown material update returns HTTP 404 | Actual: Not executed in this run | Status: Not Run
TEST-013-27	SRS-NOT-EXPLICIT	Unknown archive and delete return not found	MaterialLifecycleIT#unknownMaterialOperationsAreNotReportedAsServerErrors	integration / medium	TEST_MATERIAL_READ_FIXTS; Teacher credentials	Expected: Unknown material archive and delete return HTTP 404 | Actual: Not executed in this run | Status: Not Run
TEST-013-28	SRS-NOT-EXPLICIT	Material protected routes require authentication	MaterialAuthenticationIT#rejectsProtectedMaterialRequestsWithoutAuthentication	security / high	Material Service; no credentials	Expected: Protected upload/list/read/download/update/archive/delete/maintenance requests return HTTP 401 | Actual: Not executed in this run | Status: Not Run
TEST-013-29	SRS-NOT-EXPLICIT	Student cannot manage materials or run maintenance	MaterialAuthenticationIT#studentCannotManageMaterialsOrRunMaintenance	security / high	Material Service and Student credentials	Expected: Student material management and maintenance requests return HTTP 403 | Actual: Not executed in this run | Status: Not Run
TEST-013-30	SRS-NOT-EXPLICIT	Temporary material files are cleaned up	MaterialLifecycleIT#teacherCanUploadRetrieveListDownloadUpdateArchiveAndDeleteMaterial	unit/integration / medium	Temporary filesystem access only	Expected: Try-with-resources closes the generated test file and deletes it after the upload lifecycle | Actual: Not executed in this run | Status: Not Run
TEST-013-40	SRS-NOT-EXPLICIT	Teacher cannot run material maintenance	MaterialAuthenticationIT#teacherCannotRunMaintenance	security / high	Material Service and Teacher credentials	Expected: Teacher audit and cleanup requests return HTTP 403 | Actual: Not executed in this run | Status: Not Run
TC-AUTH-ACCESS-003	NFR-SE001; NFR-SE003	Reject expired signed access token	SecurityRegressionIT#expiredSignedAccessTokenIsRejected	security / high	User Service; student credentials; AUTH_JWT_PRIVATE_KEY_BASE64 with synthetic key material	Expected: Expired signed access JWT returns 401 and session logout is attempted | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-017-01	SRS-NOT-EXPLICIT	Verify required services are ready before integration tests	BackendReadinessIT#shouldKeepRequiredBackendServicesReady	integration / high	TEST_ENV_ENABLED=true; configured API Gateway and required backend services	Expected: BeforeAll readiness check completes only when every configured required service is healthy | Actual: Not executed in this run | Status: Not Run
SEC-001	NFR-SE001	Missing bearer token is rejected	SecurityRegressionIT#protectedEndpointsRejectMissingInvalidAndMalformedTokens	security / high	API Gateway and Auth Service	Expected: Protected current-user endpoint returns 401 without a bearer token | Actual: Not executed in this run | Status: Not Run
SEC-002	NFR-SE001	Empty bearer token is rejected	SecurityRegressionIT#protectedEndpointsRejectMissingInvalidAndMalformedTokens	security / high	API Gateway and Auth Service	Expected: Protected current-user endpoint returns 401 for an empty bearer token | Actual: Not executed in this run | Status: Not Run
SEC-003	NFR-SE001	Invalid or malformed bearer token is rejected	SecurityRegressionIT#protectedEndpointsRejectMissingInvalidAndMalformedTokens	security / high	API Gateway and Auth Service	Expected: Protected endpoint returns 401 for malformed token values | Actual: Not executed in this run | Status: Not Run
SEC-004	NFR-SE003	Student cannot access administrator probe	SecurityRegressionIT#studentAndTeacherTokensCannotCrossAdministratorBoundary	security / high	Student credential and User Service	Expected: Administrator probe returns 403 to student | Actual: Not executed in this run | Status: Not Run
SEC-005	NFR-SE003	Teacher cannot access administrator probe	SecurityRegressionIT#studentAndTeacherTokensCannotCrossAdministratorBoundary	security / high	Teacher credential and User Service	Expected: Administrator probe returns 403 to teacher | Actual: Not executed in this run | Status: Not Run
SEC-006	NFR-SE001; NFR-SE004	Authentication error responses do not expose sensitive values	SecurityRegressionIT#protectedErrorResponsesDoNotExposeSensitiveValues	security / high	API Gateway	Expected: 401 response contains no password token bearer header or stack trace markers | Actual: Not executed in this run | Status: Not Run
RES-002	SRS-NOT-EXPLICIT; NFR-SE001	Controlled AI provider failure returns sanitized server error	AiProviderFailureIT#controlledProviderFailureReturnsSanitizedServerError	resilience / high	TEST_RESILIENCE_ENABLED=true; FAULT_INJECTION_ENABLED=true; TEST_PROVIDER_FAILURE_MODE=true; AI Service configured with a controlled failing provider	Expected: Request returns 5xx and does not expose the provider sentinel | Actual: Not executed in this run | Status: Not Run
RES-003	SRS-NOT-EXPLICIT; NFR-SE001	Fault injection modes are disabled by default	SecurityResilienceTestConfigurationTest#securityAndFaultInjectionModesAreDisabledByDefault	unit / medium	None	Expected: Security and fault-injection configuration default to disabled | Actual: Passed; see JUnit report | Status: Passed

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-019-01	SRS-NOT-EXPLICIT	Verify Gateway Auth and Notification services are ready	BackendReadinessIT#shouldKeepRequiredBackendServicesReady	integration / high	TEST_ENV_ENABLED=true; AP Gateway Auth and Notification service URLs configured	Expected: Test fails unless all configured required services respond healthy within readiness timeout | Actual: Not executed in this run | Status: Not Run
TC-TEST-019-03	SRS-NOT-EXPLICIT	Default notification pagination is correct over HTTP	NotificationWorkflowIT#userCanRetrievePaginatedAndUnreadNotifications	integration / high	Notification Service and student credentials	Expected: HTTP response reports page 0 and size 20 | Actual: Not executed in this run | Status: Not Run
TC-TEST-019-04	SRS-NOT-EXPLICIT	Later notification page respects requested bounds	NotificationWorkflowIT#userCanRetrievePaginatedAndUnreadNotifications	integration / medium	Notification Service and student credentials	Expected: HTTP response reports requested page 1 and no more than requested page size | Actual: Not executed in this run | Status: Not Run
TC-TEST-019-07	SRS-NOT-EXPLICIT	Unread count is non-negative and consistent with returned unread records	NotificationWorkflowIT#userCanRetrievePaginatedAndUnreadNotifications	integration / medium	Notification Service and student credentials	Expected: Unread count is non-negative and at least the returned unread-only page size | Actual: Not executed in this run | Status: Not Run
TC-TEST-019-09	SRS-NOT-EXPLICIT	Owner can mark all notifications read	NotificationWorkflowIT#ownerCanMarkReadMarkAllReadAndDeleteApprovedFixture	integration / medium	TEST_NOTIFICATION_WRITE_ABLED=true; approved disposable notification fixture	Expected: Mark-all-read returns 200 and a non-negative changed-record count | Actual: Not executed in this run | Status: Not Run
TC-TEST-019-10	SRS-NOT-EXPLICIT	Owner can delete approved notification	NotificationWorkflowIT#ownerCanMarkReadMarkAllReadAndDeleteApprovedFixture	integration / medium	TEST_NOTIFICATION_WRITE_ABLED=true; approved disposable notification fixture	Expected: Owner receives 200 when deleting the approved notification | Actual: Not executed in this run | Status: Not Run
TC-TEST-019-11	SRS-NOT-EXPLICIT; NFR-SE003	Authenticated users have separate notification collections	NotificationOwnershipIT#twoAuthenticatedUsersReceiveOnlyTheirOwnNotificationCollections	security / high	TEST_NOTIFICATION_ID owne by configured student; student and teacher credentials; Notification Service	Expected: Configured student-owned notification appears in the student's collection and is absent from the teacher's collection | Actual: Not executed in this run | Status: Not Run
TC-TEST-019-13	NFR-SE001	Notification endpoints reject unauthenticated access	NotificationAuthorizationIT#unauthenticatedNotificationRequestsReturnUnauthorized	security / high	Notification Service	Expected: All notification routes return 401 without authentication | Actual: Not executed in this run | Status: Not Run
TC-TEST-019-16	SRS-NOT-EXPLICIT	Notification endpoint rejects invalid pagination	NotificationValidationIT#httpEndpointRejectsInvalidPaginationRanges	validation / high	Notification Service and student credentials	Expected: HTTP endpoint returns 400 for negative page zero size and size above 100 | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-018-01	SRS-NOT-EXPLICIT	Verify Gateway Auth Course Enrollment and Payment services are ready	BackendReadinessIT#shouldKeepRequiredBackendServicesReady	integration / high	TEST_ENV_ENABLED=true; configured required service health URLs	Expected: The test waits for each configured service health endpoint and fails if any remains unavailable | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-09	SRS-NOT-EXPLICIT	Administrator payment pagination returns requested page metadata	PaymentAdminIT#administratorCanListFilterAndInspectPayments	integration / high	Payment Service and Admin credentials	Expected: Page zero default and page one with size one return matching page and size metadata with bounded content | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-10	SRS-NOT-EXPLICIT	Administrator can inspect a selected payment detail	PaymentAdminIT#administratorCanListFilterAndInspectPayments	integration / high	Payment Service; Admin credentials; TEST_PAYMENT_* fixtures	Expected: Admin payment reference filter returns the selected payment ID in the response | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-11	SRS-NOT-EXPLICIT	Administrator can verify a student course payment pair	PaymentAuthorizationIT#administratorCanVerifySuccessfulPaymentForUnknownStudentCoursePair	integration / high	Payment Service and Admin credentials	Expected: Unknown synthetic student/course pair returns 200 with successfulPaymentExists=false | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-13	NFR-SE003	Student and teacher cannot refund payments	PaymentAuthorizationIT#studentsAndTeachersCannotRefundPayments	security / high	Payment Service; Student and Teacher credentials	Expected: Student and teacher refund requests return 403 | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-15	NFR-SE001	Protected payment endpoints reject unauthenticated requests	PaymentAuthorizationIT#unauthenticatedPaymentRequestsReturnUnauthorized	security / high	Payment Service	Expected: Unauthenticated requests to protected payment routes return 401 | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-16	NFR-SE003	Student cannot access administrator payment operations	PaymentAuthorizationIT#studentCanUseOwnEndpointsButCannotUseAdministratorEndpoints	security / high	Payment Service and Student credentials	Expected: Administrator list detail refund and verification routes return 403 to student | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-17	NFR-SE003	Teacher cannot access student or administrator payment operations	PaymentAuthorizationIT#teacherCannotUseStudentOrAdministratorPaymentEndpoints	security / high	Payment Service and Teacher credentials	Expected: Student and administrator payment routes return 403 to teacher | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-018-19	NFR-SE001; NFR-SE004	Stripe webhook requires a valid signature	PaymentValidationIT#webhookWithoutSignatureIsRejected	integration / high	Payment Service	Expected: Missing Stripe signature is rejected with HTTP 400 | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-26	NFR-SE001	Invalid payment inputs and unknown student payment IDs are rejected	PaymentValidationIT#invalidPaymentRequestsReturnExactClientErrorsOrNotFoundResponses	validation / high	Payment Service and Student credentials	Expected: Malformed course and verification IDs return 400; unknown course and payment IDs return 404 | Actual: Not executed in this run | Status: Not Run
TC-TEST-018-28	SRS-NOT-EXPLICIT; NFR-SE003	Refund only uses the approved dedicated test payment fixture	PaymentAdminIT#administratorCanInspectAndSafelyRefundAnApprovedTestPayment	integration / high	TEST_PAYMENT_WRITE_ENA =true; one-time TEST_PAYMENT_REFUND_ID Admin credentials	Expected: Refund flow first verifies the configured payment ID belongs to the configured student and course and is SUCCESS before issuing refund | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-01	SRS-NOT-EXPLICIT	Teacher creates a quiz	QuizManagementIT#teacherManagesQuestionsAndPublishesQuiz	integration / high	TEST_QUIZ_WRITE_ENABLED ue; TEST_QUIZ_COURSE_ID; Teacher credentials	Expected: Quiz create returns 201 with an ID and requested values | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-02	SRS-NOT-EXPLICIT	Teacher retrieves a quiz	QuizManagementIT#teacherManagesQuestionsAndPublishesQuiz	integration / high	TEST_QUIZ_WRITE_ENABLED ue; TEST_QUIZ_COURSE_ID; Teacher credentials	Expected: Created quiz is retrieved with matching ID and DRAFT status | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-03	SRS-NOT-EXPLICIT	Teacher lists quizzes	QuizManagementIT#teacherManagesQuestionsAndPublishesQuiz	integration / medium	TEST_QUIZ_WRITE_ENABLED ue; TEST_QUIZ_COURSE_ID; Teacher credentials	Expected: Created quiz appears in teacher quiz list | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-04	SRS-NOT-EXPLICIT	Teacher updates a quiz	QuizManagementIT#teacherManagesQuestionsAndPublishesQuiz	integration / high	TEST_QUIZ_WRITE_ENABLED ue; TEST_QUIZ_COURSE_ID; Teacher credentials	Expected: Update returns 200 and persists modified duration and pass mark | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-05	SRS-NOT-EXPLICIT	Teacher creates a question	QuizManagementIT#teacherManagesQuestionsAndPublishesQuiz	integration / high	TEST_QUIZ_WRITE_ENABLED ue; TEST_QUIZ_COURSE_ID; Teacher credentials	Expected: Created question has identifier supported question type and four options | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-06	SRS-NOT-EXPLICIT	Teacher updates a question	QuizManagementIT#teacherManagesQuestionsAndPublishesQuiz	integration / high	TEST_QUIZ_WRITE_ENABLED ue; TEST_QUIZ_COURSE_ID; Teacher credentials	Expected: Question update returns 200 and updated question text | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-014-07	SRS-NOT-EXPLICIT	Teacher deletes a question	QuizManagementIT#teacherManagesQuestionsAndPublishesQuiz	integration / medium	TEST_QUIZ_WRITE_ENABLED ue; TEST_QUIZ_COURSE_ID; Teacher credentials	Expected: Created removable question is deleted successfully | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-08	SRS-NOT-EXPLICIT	Teacher publishes a quiz	QuizManagementIT#teacherManagesQuestionsAndPublishesQuiz	integration / high	TEST_QUIZ_WRITE_ENABLED ue; TEST_QUIZ_COURSE_ID; Teacher credentials	Expected: Quiz transitions to PUBLISHED | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-09	SRS-NOT-EXPLICIT	Enrolled student lists published course quizzes	QuizAttemptWorkflowIT#enrolledStudentCanAccessPublishedQuizAndCompleteAttempt	integration / high	TEST_QUIZ_WRITE_ENABLED ue; TEST_QUIZ_COURSE_ID; enrolled student credentials	Expected: Published quiz appears in the enrolled student's course list | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-12	SRS-NOT-EXPLICIT	Student saves an answer to an attempt	QuizAttemptWorkflowIT#enrolledStudentCanAccessPublishedQuizAndCompleteAttempt	integration / high	TEST_QUIZ_WRITE_ENABLED ue; TEST_QUIZ_COURSE_ID; enrolled student credentials	Expected: Answer submission is associated with the active attempt and selected option | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-15	NFR-SE003	Student cannot create quizzes	QuizAuthenticationIT#studentCannotUseTeacherQuizManagementEndpoints	security / high	TEST_QUIZ_WRITE_ENABLED ue; TEST_QUIZ_COURSE_ID; student and teacher credentials	Expected: Student quiz creation returns 403 | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-16	NFR-SE003	Student cannot create update or delete quiz questions	QuizAuthenticationIT#studentCannotUseTeacherQuizManagementEndpoints	security / high	TEST_QUIZ_WRITE_ENABLED ue; TEST_QUIZ_COURSE_ID; student and teacher credentials	Expected: Student question create update and delete requests return 403 | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-17	NFR-SE003	Teacher cannot start or submit student attempts	QuizAuthenticationIT#teacherCannotStartOrSubmitStudentAttempt	security / high	TEST_QUIZ_WRITE_ENABLED ue; TEST_QUIZ_COURSE_ID; teacher credentials	Expected: Teacher attempt start answer and submit requests return 403 | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-18	NFR-SE001	Quiz endpoints reject unauthenticated access	QuizAuthenticationIT#protectedQuizRequestsRejectMissingAuthentication	security / high	Quiz Service and API Gateway	Expected: Protected quiz question and attempt routes return 401 without authentication | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-19	NFR-SE001	Blank quiz title is rejected	QuizValidationIT#rejectsInvalidQuizRequests	validation / high	Quiz Service and Teacher credentials	Expected: Blank title request returns 400 | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-20	NFR-SE001	Invalid quiz duration is rejected	QuizValidationIT#rejectsInvalidQuizRequests	validation / high	Quiz Service and Teacher credentials	Expected: Zero duration request returns 400 | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-014-21	NFR-SE001	Invalid maximum attempts is rejected	QuizValidationIT#rejectsInvalidQuizRequests	validation / high	Quiz Service and Teacher credentials	Expected: Zero maximum attempts request returns 400 | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-22	NFR-SE001	Question without options is rejected	QuizValidationIT#rejectsInvalidQuestionAndAnswerRequests	validation / high	Quiz Service and Teacher credentials	Expected: Question request omitting options returns 400 | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-23	NFR-SE001	Question with too few options is rejected	QuizValidationIT#rejectsInvalidQuestionAndAnswerRequests	validation / high	Quiz Service and Teacher credentials	Expected: Single-option multiple-choice question returns 400 | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-24	NFR-SE001	Answer request with missing identifiers is rejected	QuizValidationIT#rejectsInvalidQuestionAndAnswerRequests	validation / high	Quiz Service and Student credentials	Expected: Answer request missing question or selected option identifiers returns 400 | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-25	NFR-SE001	Unknown quiz and question operations return not found	QuizManagementIT#unknownQuizAndQuestionResourceturnNotFound	validation / high	Quiz Service and Teacher credentials	Expected: Unknown quiz and question read update delete publish and create operations return 404 | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-26	NFR-SE001	Unknown attempt answer and submit operations return not found	QuizValidationIT#rejectsInvalidQuestionAndAnswerRequests	validation / high	Quiz Service and Student credentials	Expected: Valid answer and submit operations against unknown attempt IDs return 404 | Actual: Not executed in this run | Status: Not Run
TC-TEST-014-27	SRS-NOT-EXPLICIT	Live quiz integration tests remain disabled by default	IntegrationTestExecutionConditionTest#shouldDisableLiveTestClassesByDefault	unit / high	TEST_ENV_ENABLED omitted or false	Expected: Quiz and other live integration classes are disabled without an explicit environment opt-in | Actual: Passed; see JUnit report | Status: Passed
TC-TEST-014-41	SRS-NOT-EXPLICIT	Verify Gateway Auth Course Enrollment and Assignment dependencies before quiz tests	QuizManagementIT#requiredQuizServicesAreReady	integration / high	TEST_ENV_ENABLED=true; configured Gateway Auth Course Enrollment and Assignment health URLs	Expected: The explicit quiz readiness test fails unless every required dependency is healthy | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-01	SRS-NOT-EXPLICIT	Verify Gateway Auth and User services are ready	UserProfileIT#requiredUserServicesAreReady	integration / high	TEST_ENV_ENABLED=true; configured Gateway Auth and User Service health URLs	Expected: Readiness test fails unless every configured required service responds healthy within the configured timeout | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-021-02	SRS-NOT-EXPLICIT	Public authentication probe is available	AuthenticationIT#publicAndProtectedAuthenticationProbesFollowTheirSecurityRules	integration / medium	API Gateway and User Service	Expected: Public probe returns HTTP 200 and success response without authentication | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-03	NFR-SE001	Protected user endpoints reject missing authentication	UserAuthorizationIT#protectedUserEndpointsRejectMissingAuthentication	security / high	User Service	Expected: Protected profile listing and analytics requests return 401 without credentials | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-04	FR-S001; FR-S002; FR-A001; NFR-SE003	Student and administrator role probes follow access rules	AuthenticationIT#roleProbesMatchStudentAndAdministratorTokens	security / high	Student and Admin credentials; User Service	Expected: Student profile access succeeds student admin listing is denied and admin listing succeeds | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-05	NFR-SE001	Invalid login credentials and payloads are rejected	AuthenticationLoginIT#shouldRejectAnInvalidLoginRequest	security / high	Auth Service; synthetic invalid credentials	Expected: Invalid password login returns the documented client error | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-06	NFR-SE001	Invalid refresh token is rejected	AuthenticationRefreshIT#shouldRejectAnInvalidRefreshToken	security / high	Auth Service; synthetic invalid token	Expected: Invalid refresh token is rejected with the observed contract status | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-07	NFR-SE001	Invalid logout request is rejected safely	AuthenticationIT#invalidAuthenticationPayloadsAreRejectedWithoutSensitiveOutput	security / medium	Auth Service	Expected: Logout request without required data returns 400 | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-08	FR-S005; NFR-SE001; NFR-SE004	Invalid forgot-password email is rejected	AuthenticationIT#invalidAuthenticationPayloadsAreRejectedWithoutSensitiveOutput	validation / medium	Auth Service; malformed synthetic email	Expected: Malformed forgot-password email returns 400 | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-09	FR-S003; NFR-SE002	Authenticated student can retrieve their own profile	UserProfileIT#authenticatedUserCanRetrieveOwnProfileWithoutSensitiveFields	integration / high	User Service and Student credentials	Expected: Current-user request returns 200 for the authenticated profile | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-10	NFR-SE004	Profile response does not expose sensitive fields	UserProfileIT#authenticatedUserCanRetrieveOwnProfileWithoutSensitiveFields	security / high	User Service and Student credentials	Expected: Profile response contains no password refresh token or client secret fields | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-021-11	FR-A001; FR-A002; FR-A003	Administrator can list users by role	UserAdministrationIT#administratorCanListUsersAndReadAnalytics	integration / high	User Service and Admin credentials	Expected: Student and teacher lists contain only entries with the requested role | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-12	FR-A001; FR-A002	Administrator can list students with pagination	UserAdministrationIT#administratorCanListUsersAndReadAnalytics	integration / medium	User Service and Admin credentials	Expected: Student page returns records with STUDENT role | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-13	FR-A001; FR-A003	Administrator can list teachers with pagination	UserAdministrationIT#administratorCanListUsersAndReadAnalytics	integration / medium	User Service and Admin credentials	Expected: Teacher page returns records with TEACHER role | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-14	FR-A004	Administrator can read user registration analytics	UserAdministrationIT#administratorCanListUsersAndReadAnalytics	integration / medium	User Service and Admin credentials	Expected: Registration analytics endpoint returns HTTP 200 | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-15	NFR-SE001	Administrator user mutations validate fields and resource IDs	UserAdministrationIT#administratorOperationsValidateUnknownUsersAndFields	validation / high	User Service and Admin credentials	Expected: Malformed registration and status payloads return 400 and unknown user operations return 404 | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-16	NFR-SE003	Student cannot perform administrator user operations	UserAuthorizationIT#studentsAndTeachersCannotPerformAdministratorOperations	security / high	User Service and Student credentials	Expected: Student listing analytics status registration and delete attempts are denied | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-17	NFR-SE003	Teacher cannot perform administrator user operations	UserAuthorizationIT#studentsAndTeachersCannotPerformAdministratorOperations	security / high	User Service and Teacher credentials	Expected: Teacher listing registration and delete attempts are denied | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-22	SRS-NOT-EXPLICIT; NFR-SE004	Student and teacher existence checks return safe false values for unknown identities	UserValidationIT#invalidUserIdentifiersAndStatusPayloadsAreHandledSafely	validation / high	User Service and Admin credentials	Expected: Unknown student and teacher IDs and malformed student identity return explicit false without details | Actual: Not executed in this run | Status: Not Run
TC-TEST-028-01	SRS-NOT-EXPLICIT; NFR-SE001	Missing or incorrect staging confirmation fails before requests	StagingTargetPolicyTest#rejectsMissingOrIncorrectConfirmationBeforeAnyRequest	unit / high	No network target; policy-only test	Expected: Missing or incorrect confirmation throws configuration error before any HTTP request | Actual: Passed; see JUnit report | Status: Passed

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-028-02	NFR-SE001	Unsafe staging URL is rejected before network access	StagingTargetPolicyTest#rejectsHttpEmbeddedCredentialsQueryAndUnapprovedHosts	unit / high	No network target; policy-only test	Expected: HTTP embedded credentials query string non-443 port and unapproved host are rejected | Actual: Passed; see JUnit report | Status: Passed
TC-TEST-028-03	NFR-SE001	Malformed deployment revision is rejected	StagingTargetPolicyTest#rejectsInvalidDeploymentRevision	unit / high	No network target; policy-only test	Expected: Malformed deployment revision fails closed | Actual: Passed; see JUnit report | Status: Passed
TC-TEST-028-04	NFR-SE001	Staging HTTP probe does not follow cross-host redirects	StagingHttpProbeTest#doesNotFollowRedirectsToAnotherHost	unit / high	Loopback-only synthetic HTTP server	Expected: Probe returns redirect status and does not follow it to another host | Actual: Passed; see JUnit report | Status: Passed
TC-TEST-028-05	NFR-SE001	Staging HTTP probe enforces request timeout	StagingHttpProbeTest#appliesRequestTimeoutWithoutReadingResponseBody	unit / high	Loopback-only synthetic delayed HTTP server	Expected: Probe throws within the configured request timeout | Actual: Passed; see JUnit report | Status: Passed
TC-TEST-028-12	SRS-NOT-EXPLICIT	Default CI does not enable staging smoke tests	IntegrationTestExecutionConditionTest#shouldRequireStagingTargetForStagingSmokeClass	unit / high	STAGING_ENVIRONMENT and TEST_ENV_ENABLED absent	Expected: Staging integration test remains disabled unless explicit staging environment is configured | Actual: Passed; see JUnit report | Status: Passed
TC-TEST-028-13	SRS-NOT-EXPLICIT; NFR-SE001	Production environment is rejected by staging target policy	StagingTargetPolicyTest#rejectsNonStagingEnvironment	unit / high	No network target; policy-only test	Expected: Production environment setting fails closed before any request | Actual: Passed; see JUnit report | Status: Passed
TC-TEST-028-14	NFR-SE001	IP address is rejected as the staging host suffix	StagingTargetPolicyTest#rejectsIpAddressAsStagingHostSuffix	unit / medium	No network target; policy-only test	Expected: IP address host suffix is rejected by target policy | Actual: Passed; see JUnit report | Status: Passed
TC-TEST-021-18	FR-A001; FR-A002; NFR-SE003	Administrator registers a synthetic student account	UserRegistrationWorkflowIT#administratorCanRegisterSyntheticStudentAndCaptureInvitationAndResetEmails	integration / high	TEST_USER_WRITE_ENABLE ue; TEST_USER_MAIL_CAPTU _ENABLED=true; local Mailpit API; Admin credentials	Expected: Synthetic student registration returns 201 and created user is deleted during cleanup | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-19	FR-A003; NFR-SE003	Super Admin registers a synthetic administrator account	UserRegistrationWorkflowIT#superAdministratorCanRegisterSyntheticAdministratorAndCaptureInvitation	integration / high	TEST_USER_WRITE_ENABLE ue; TEST_USER_MAIL_CAPTU _ENABLED=true; TEST_SUPEF ADMIN_SUPPORTED=true; Super Admin credentials; local Mailpit API	Expected: Admin registration returns 201 with INVITED status and created account is deleted during cleanup | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-TEST-021-20	FR-A005; NFR-SE004	Invitation and password-reset emails are captured safely	UserRegistrationWorkflowIT#administratorCanRegisterSyntheticStudentAndCaptureInvitationAndResetEmails	integration / high	TEST_USER_WRITE_ENABLEue; TEST_USER_MAIL_CAPTU_ENABLED=true; backend SMTP pointed to local Mailpit; local Mailpit API	Expected: Invitation and reset messages reach only the local Mailpit inbox and are deleted during cleanup | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-21	FR-A004; NFR-SE003	Administrator deactivates and reactivates the dedicated test user	UserStatusLifecycleIT#administratorCanDeactivateAndReactivateDedicatedTestUser	integration / high	TEST_USER_WRITE_ENABLEue; TEST_USER_ID identifies a disposable user; Admin credentials	Expected: Status update endpoint accepts deactivation and reactivation and cleanup restores active state | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-35	SRS-NOT-EXPLICIT; NFR-SE001	Public JWKS endpoint is available through the Gateway	AuthenticationRecoveryIT#shouldExposePublicJwksDocument	integration / medium	API Gateway and Auth Service	Expected: Unauthenticated Gateway request returns HTTP 200 and a keys document | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-36	SRS-NOT-EXPLICIT; NFR-SE001	Student-only probe enforces its role restriction	AuthenticationIT#studentProbeRequiresTheStudentRole	security / high	API Gateway, User Service, Student and Teacher credentials	Expected: Student receives HTTP 200; missing authentication and teacher role receive 401 and 403 | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-37	FR-A004; NFR-SE003	Administrator deletes the synthetic registered user during cleanup	UserRegistrationWorkflowIT#administratorCanRegisterSyntheticStudentAndCaptureInvitationAndResetEmails	integration / high	TEST_USER_WRITE_ENABLEue; local Mailpit; disposable synthetic account	Expected: Delete endpoint returns 200 and the created account is removed | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-38	FR-A005; NFR-SE004	Administrator resends invitation for a synthetic invited student	UserRegistrationWorkflowIT#administratorCanRegisterSyntheticStudentAndCaptureInvitationAndResetEmails	integration / high	TEST_USER_WRITE_ENABLEue; TEST_USER_MAIL_CAPTU_ENABLED=true; backend SMTP points to local Mailpit	Expected: Resend endpoint returns 200 and the invitation message is captured and cleaned up | Actual: Not executed in this run | Status: Not Run
TC-TEST-021-39	FR-A001; FR-A002; NFR-SE004	Public registration creates a synthetic student and delivers credentials to local Mailpit	UserRegistrationWorkflowIT#publicStudentRegistrationCreatesAccountAndCapturesCredentialsEmail	integration / high	TEST_USER_WRITE_ENABLEue; TEST_USER_MAIL_CAPTU_ENABLED=true; local Mailpit; Admin credentials for cleanup	Expected: Public registration returns 201, Mailpit captures credentials email, and cleanup deletes the account/message | Actual: Not executed in this run | Status: Not Run
TC-AUTH-RECOVERY-003	NFR-SE004	Verify password reset OTP captured for a synthetic test account	UserRegistrationWorkflowIT#administratorCanRegisterSyntheticStudentAndCaptureInvitationAndResetEmails	integration / high	TEST_USER_WRITE_ENABLEue; TEST_USER_MAIL_CAPTU_ENABLED=true; backend SMTP points to local Mailpit	Expected: Captured six-digit OTP verifies with HTTP 200 and yields a nonblank reset token | Actual: Not executed in this run | Status: Not Run

Test Case Register

Requirement-linked cases and recorded execution status

Case ID	Requirement	Case / Input	Test file / method	Type / priority	Preconditions	Expected / actual / status
TC-AUTH-RECOVERY-004	NFR-SE004	Reset synthetic test account password with a valid reset token	UserRegistrationWorkflowIT#administratorCanRegisterSyntheticStudentAndCaptureInvitationAndResetEmails	integration / high	TEST_USER_WRITE_ENABLE ue; TEST_USER_MAIL_CAPTU _ENABLED=true; backend SMTP points to local Mailpit	Expected: Password reset returns HTTP 200 and invalidates the reset token | Actual: Not executed in this run | Status: Not Run
TC-TEST-019-17	Notification Service ingests supported Kafka events into the owner's notification list	Publish a synthetic QUIZ_CREATED event and retrieve it through the authenticated API	NotificationEventEmailWorkflowIT#supportedKafkaEventCreatesNotificationAndSendsEmailToLocalMailpit	integration / high	TEST_NOTIFICATION_EVENT_ ABLED=true; non-production Kafka broker; API Gateway, Auth and Notification Services	Expected: Synthetic event is stored with its event type and appears only in the authenticated recipient's notifications | Actual: Not executed in this run | Status: Not Run
TC-TEST-019-18	Notification Service delivers allowed notification email through the configured SMTP sink	Capture the synthetic notification email in local Mailpit	NotificationEventEmailWorkflowIT#supportedKafkaEventCreatesNotificationAndSendsEmailToLocalMailpit	integration / high	NOTIFICATION_EMAIL_TEST_ DE=true; backend SMTP points only to local Mailpit; synthetic .invalid recipient	Expected: Notification email status is SENT, Mailpit contains the message body, and created notification/message are cleaned up | Actual: Not executed in this run | Status: Not Run
TC-CAL-026	SRS-NOT-EXPLICIT	Administrator reads the Calendar venue list	CalendarVenueAuthorizationIT#administratorCanReadVenueList	integration / medium	API Gateway and Calendar Service; administrator credentials	Expected: Authenticated administrator receives HTTP 200 and a data field; read-only request creates no records | Actual: Not executed in this run | Status: Not Run
TC-CAL-025	SRS-NOT-EXPLICIT	Admin creates a uniquely named venue in a confirmed disposable Calendar database	CalendarVenueAuthorizationIT#administratorCanCreateVenueInConfirmedDisposableCalendarDatabase	integration / high	TEST_CALENDAR_VENUE_WI ENABLED=true; TEST_CALEND R_DISPOSABLE_DATABASE_ IRMED=true; Admin credentials	Expected: Venue creation returns 201 and appears in the authenticated listing; disposable database is discarded after the run | Actual: Not executed in this run | Status: Not Run
TEST-013-41	SRS-NOT-EXPLICIT; NFR-SE003	Student uploads a synthetic profile image to local test storage	MaterialAuthenticationIT#studentCanUploadAndDownloadOwnProfileImageFromLocalTestStorage	integration / high	TEST_PROFILE_IMAGE_WRIT NABLED=true; TEST_PROFILE IMAGE_STORAGE_CLEANUP_ LED=true; TEST_PROFILE_IMA GE_STORAGE_TARGET_CON D=true; local Gateway and S3-compatible test bucket; Student credentials	Expected: Upload returns 201 with image/png and an object key; test deletes the exact returned key during cleanup | Actual: Not executed in this run | Status: Not Run
TEST-013-42	SRS-NOT-EXPLICIT; NFR-SE003	Student downloads their own profile image bytes	MaterialAuthenticationIT#studentCanUploadAndDownloadOwnProfileImageFromLocalTestStorage	integration / high	Same local S3-compatible test bucket and synthetic upload as TEST-013-41	Expected: Download returns HTTP 200, image/png, and bytes identical to the synthetic PNG | Actual: Not executed in this run | Status: Not Run